An Analysis of the Performance of the Indonesian Treasuries Market

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Keywords and Abbreviations

- Accounting and Auditing Organization for Islamic Financial Institutions (AAOIFI)
- Additional Auction (Green Shoe Option)
- Auction Participants (APs)
- Bank Indonesia (BI)
- Bank Indonesia Electronic Trading Platform (BI-ETP)
- Bloomberg Professional Service Database (Bloomberg)
- Bond Auctions
- Conventional Treasuries
- Counterfactual Revenue Analysis
- Discriminatory-price Auction
- Efficiency Ranking
- Hypothetical uniform-price auction with truthful bidding (HUPATB)
- Indonesia Deposit Insurance Corporation (IDIC)
- Indonesian Treasuries
- Islamic Treasuries
- Primary Dealers (PDs)
- Revenue Equivalence Theorem
- Revenue Ranking
- Structural Approach
- The Ministry of Finance (MoF) of Republic of Indonesia
- Treasury Auctions
- Treasury Bills
- Treasury Bonds
- Uniform-price Auction

Abstract

In this thesis, I compare the performance of the discriminatory-price auction relative to the uniform-price auction in the Indonesian treasuries' market. An important debate in the empirical treasury auction literature concerns which auction mechanism best serves a government treasury market, with auction theory providing conflicting answers. Employing a counterfactual analysis, I derive optimal bidding conditions for both auction methods under the assumption that bidders have symmetric information and independent private value about the security. Using a non-parametric resampling strategy and recovering bidders' unknown true valuation corresponding to each observed bid point, I compute the underlying distribution of market clearing prices based on the structural model estimation approach. I then calculate the upper bound of the revenue to perform a counterfactual revenue comparison with the factual revenue.

For a large sample of Islamic (Sukuk) and conventional treasury bill auctions between 2003 and 2017 in Indonesia, I find switching from a discriminatory-price auction to a uniform-price auction improves both auction revenue and efficiency. This is consistent with the argument that bidders use their best effort to win an auction with the lowest price and avoid the winner's curse in discriminatory-price auctions. I find the opposite results for longer maturity treasury bonds which show that discriminatory-price auctions yield higher revenue than uniform-price auctions. This finding is consistent with the view that participants in uniform-price auctions can support a collusive Nash equilibrium and have more incentive to shade their bids; thus, bidders can steepen their schedules enough to neutralise the elasticity supply of sellers. Further, for treasury bonds,

I find discriminatory and uniform-price auctions yield roughly equal efficiency, consistent with the argument that in discriminatory-price auctions, bidders shade their bids according to the effectiveness of getting an allocation. In contrast, in uniform-price auctions, differential bid incentives occur when a winner has a favourable likelihood of influencing the market clearing price. I find no difference in results between conventional treasuries and Sukuk, which is perhaps not surprising since the information environment is identical.

Next, I run regressions to examine the factors that may explain both the revenue and efficiency gains when switching to a uniform-price auction. Consistent with auction theory, the level of competition is negatively associated with both revenue and efficiency gains. I find higher uncertainty is positively associated with both revenue and efficiency gains, suggesting that the discriminatory-price auction is favoured in the presence of information opacity. My findings support the argument that bidders in uniform-price auctions employ mixed strategies and that absorbing the uncertainty can provide economic benefits to them.

I contribute to the literature by providing new evidence on the ranking of revenue and efficiency gains between discriminatory and uniform-price auctions in emerging markets, thus providing further insights into the debate on whether switching to uniform-price auctions is beneficial in Indonesia. My contribution also lies in demonstrating how auction characteristics influence auction performance when switching to a uniform-price auction. My research has potential policy implications for the Indonesian treasuries market − switching to a uniform-price auction is unlikely to be significantly beneficial for treasury bond auctions given the low level of competition and the high level of dispersion of information in the Indonesian treasuries market. However, switching to a

uniform-price auction is likely to be more beneficial for treasury bills where the level of competition is high, and the level of uncertainty is low.

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Statement of Original Authorship

The work contained in this thesis has not been previously submitted to meet requirements for an award at this or any other higher education institution. To the best of my knowledge and belief, the thesis contains no material previously published or written by another person except where due reference is made.

Signature:

QUT Verified Signature

Wawan Sugiyarto Date: 22/10/2020

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"Allah has permitted trading and forbidden interest." QS Al Baqarah: 275

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Chapter 1

Introduction

1.1 Introduction

The increasing issuance of government securities to fund government projects[1] and control the money supply has resulted in a proliferation of writings on the revenue maximisation problem[2]—which method should governments use to sell their treasuries? Around the world, the most popular mechanism to sell government treasuries is through discriminatory-price (multiple-price) auctions and uniform-price (single-price) auctions. The primary difference between these two auction systems is the price paid by successful bidders (Becker, 2001). In uniform-price auctions, everyone pays the same (market clearing) price, equal to the lowest winning bid, regardless of their actual bid. In discriminatory-price auctions the price paid is the winning bid submitted. Thus, the auction price is the market-clearing price for uniform-price auctions and of the average winning price for discriminatory-price auctions.

The selection of auction format has been extensively debated in the multi-unit auction literature. The most controversial issue concerning market design in treasury auctions (Boolaky, 2010). There are several problems that auction design must address. Firstly, the winner's curse, the situation where the winning bidder pays more than the

[1] For example, in Indonesia, government treasuries contributed as much as 37 percent to the annual budget in 2019, with the total amount of finance obtained through issuing treasuries amounting to USD 65 billion (the Ministry of Finance (MoF) of Republic of Indonesia, 2019). These funds have been essential to finance infrastructure projects, including energy, telecommunications, transportation, agriculture, manufacturing, and housing.

[2] The goal of treasury issuances is to raise as much revenue as possible, which essentially means to sell a given amount of securities at the lowest 'discount' (or yield) possible.

shared average price of the item. Theory suggests that highly sophisticated and seasoned bidders involved in treasury bond auctions anticipate the winner's curse. Bidders' fear of the winner's curse makes them shade their bids, placing bids well below what they believe is the accurate value of the security (Engelbrecht-Wiggans & Kahn, 1998b; Swinkles, 1999). This tend to lower auction efficiency.[3] The second problem is information acquisition; bidders do not know the value of the common goods and must independently estimate it, which can be highly costly. To offset these costs, buyers underbid in an auction. The third problem is the cost of reselling. By increasing the importance of specialised knowledge in terms of market demand and information costs, discriminatory-price auctions encourage fewer bidders to participate in competitive bidding than do uniform-price auctions. Hence, bidders in these auctions may face bigger losses if they cannot sell their inventories. Tending to those inventories has a cost, which must also be offset by underbidding. The final problem with auctions is non-competitive bidding. Since there is an opportunity to influence market-clearing price, bidders have more incentive to shade their other bid units than their first bid unit in the uniform-price auction. This demand reduction results in an inefficient equilibrium with the winning bids not placed with those who value them the most (Nautz & Wolfstetter, 1997; Gordy, 1999). Unfortunately, auction theory does not provide a definite answer to the questions which auction method is revenue-optimal (Klemperer, 2003).

A survey by Brenner, Galai, and Sade (2009) reflects the ambiguity, where 24 of countries surveyed use discriminatory-prices, 9 countries use uniform-prices, and 9

[3] Efficiency occurs when the treasuries end up in the hands of the bidders who value them most (Ausubel et al., 2014).

countries use both auction mechanisms. Ausubel et al. (2014) note that there may be no one model that may optimally fit all contexts, given that auction revenue depends on many features of the different demand functions. Consistent with Ausubel et al. (2014), empirical studies show that the one that yields the best results for the issuer varies across countries. Using a natural experiment, Goldreich (2007) shows that in the US treasury bond market, the uniform-price auction yields higher revenue than the discriminatory-price auction. However, using a structural approach, Hortaçsu and McAdams (2010) find that for Turkey's treasury bills, the discriminatory-price auction yields higher revenue ex-post that the uniform-price auction, but cannot reject equivalence revenue ex-ante.

To increase their revenue, several countries have switched treasury auction design over time. For example, the United States, Mexico, and Korea switched from a discriminatory-price to a uniform-price auction method to encourage more aggressive bidding by auction participants. For the United States (Nyborg and Sundaresan, 1996; Binmore & Swierzbinski, 2000), the switch yielded no economically significant benefit in revenue (or efficiency), possibly because both auctions were highly competitive. For Mexico, auction revenue increased slightly (Umlauf, 1993). Nevertheless, the Mexican government returned to discriminatory-price auctions in 1995 to increase domestic participation in the auctions and to reduce sharp swings in the treasury bill rate.

Although receiving less attention in the literature, efficiency is also an important criterion for choosing an auction design, Market designers of the auction of any product with a private value should be concerned with whether the auction method allocates the product to the highest value buyer. In a multi-unit auction, there is no general rating of the efficiency of either method. The unequal rewards for shade bids on different good

units suggest that inefficient equilibria can be achieved in both uniform-price and discriminatory-price auctions (Engelbrecht-Wiggans & Kahn, 1998b). Extant empirical literature about which of the two auction designs is more efficient are inconclusive. Research suggests that both auction formats result in bid shading, the practice of a bidder submitting a bid that is well below what he believes is the a true value inhibiting efficiency (Engelbrecht-Wiggans & Kahn, 1998b; Kastl, 2011; Ausubel et al, 2014).

1.2 Research Aims and Questions

In this thesis I set out to determine which auction method is most beneficial for the Indonesian Treasury. In particular, I provide an empirical comparison of the performance of discriminatory-price auctions relative to uniform-price auctions for treasury bills and bonds as well as conventional and Islamic treasuries. Firstly, I examine whether switching from a discriminatory-price auction to a uniform-price auction improves auction revenue and efficiency. While the empirical literature is mixed about the relative performance of the two auction formats, I argue that in the case of the relatively illiquid Indonesian treasuries market, uniform-price auctions generate higher revenue and efficiency. Secondly, using multiple variable regressions, I examine the factors that drive the performance gains. Based on established theories, I argue that the level of competition amongst bidders and information uncertainty are key determinants of both revenue and efficiency gains.

The motivation for my research is simple: to maximise both the revenue and efficiency of the treasuries' auctions for the Indonesian government. Doing so will help the Indonesian government meet their financial obligations. With the essential role of government treasuries in financing infrastructure spending, government securities dominates the domestic bond market in Indonesia (OJK, 2019).

1.3 Research Design

I focus on a comparison of the performance of discriminatory-price auctions relative to uniform-price auctions for Indonesian treasuries. My primary data source is the Ministry of Finance of the Republic of Indonesia (henceforth, MoF), which has kindly provided highly detailed individual bid data for the Indonesian treasury bill and bond auctions. My sample period begins in 2003 and ends in 2017. Indonesia issues two types of treasuries: conventional treasuries and Islamic treasuries (Sukuk). They differ in the type of clientele, with Islamic treasuries focusing predominantly on Islamic investors. The final sample comprises 297 conventional treasury bill auctions, 632 conventional treasury bond auctions, 99 Islamic treasury bill auctions, and 331 Islamic treasury bond auctions.

The method of quantitative data analysis in this study is a counterfactual approach, also called an income estimation model. The reason for employing this approach is that, because Indonesia has not previously issued treasuries using both auction formats, it is not possible to carry out a natural experimental (which would have been the preferred option). Neither is a laboratory experimental method chosen because it requires many strong assumptions to change the complex treasury auction practice into a manageable process. The main limitation of the laboratory experiment method is that it does not adequately describe the actual conditions due to test and perception bias of experimental participants about reality. Therefore, the preferred method chosen, and used to compare the performance of discriminatory-price auctions with uniform-price auctions, is the revenue estimation model.

I apply the structural model of strategic bidding developed by Hortaçsu and McAdams (2010). The structural model results in a first-order crucial circumstance in

which the bidding price for a specific quantity is equivalent to the marginal value of the bidder. The contrary elasticity of residual supply function for every bidder will determine the marginal value of the bidder. Counterfactual estimation is conducted through converting mark-down order to diagnose unobserved marginal valuation. The resulting estimated price for the uniform-price auction is then compared with the actual observed price from the discriminatory-price auction. The benefit of using the structural approach is that the model allows researchers to simulate counterfactual estimations. The main assumption is that the ex-ante information of bidders is stable across the auctions. Moreover, the assessment of distributional aspects of a policy intervention can be estimated using data on an individual bidder.

1.4 Summary of Main Findings

Based on a large sample of Indonesian treasuries, I find that switching from a discriminatory auction to a uniform-price auction format significantly (at the 95 percent confidence level) improves auction revenue and efficiency for treasury bills, but not for treasury bonds. This finding supports the proposition that bidders in discriminatory-price auctions are induced to shade their bids downwards, whereas uniform-price auctions eliminate this problem (Friedman, 1959; Brimmer, 1962; Rieber, 1964; and Goldstein, 1962). The argument of theoretical literature cautions that uniform-price auctions generate lower revenue, e.g., through collusive-type equilibria (Back & Zender, 1993; Daripa, 2001; Goswami et al., 1996; Liu, 2015), does not seem to hold in the context of the Indonesian treasuries market.

Based on multiple variable regressions, I find both revenue and efficiency gains when switching to uniform-price auctions decrease when the level of competition amongst the bidders in the auction is more intense. Thus, increasing the level of competition reduces the performance difference between discriminatory and uniform-price auctions. Further, revenue gains are inversely related to the level of information uncertainty. Bidders may employ mixed strategies and understand that absorbing uncertainty will provide economic benefits to them. In sum, my findings support auction theory predictions that auction characteristics determine auction revenue and efficiency gains, such as the level of competition and uncertainty in the market (Bikhchandani & Huang, 1989; Engelbrecht-wiggans et al., 1983; French & Mccormick, 1984; Milgrom & Weber, 1982).

1.5 Contributions

My research contributes to recent developments in the treasury auction literature as follow. Firstly, I add to the literature on the performance equivalence between discriminatory and uniform-price auctions. I show that the relative performance of the discriminatory-price auction is not uniform across treasuries. Previous studies document that one auction format outperforms another, such as discriminatory-price auctions (Fevrier, Preget, & Visser, 2004; Kang & Puller, 2008; Hortaçsu & McAdams, 2010; and Marszalec, 2017) or uniform-price auctions (Kastl, 2011). Specifically, I show that for Indonesian treasury bills, switching to a uniform-price auction format may benefit the Indonesian government both in terms of revenue and efficiency. This result does not extend to treasury bonds, with the Indonesian government expected to suffer a slight loss in revenue if it were to switch to a uniform-price auction format.

Furthermore, I extend the literature on how auction characteristics influence auction performance. In response to Vickrey (1961), Mcafee and Mcmillan (1987), and Klemperer (2003) who propose that competition may have a significant impact on multi-unit auction performance, I find that increased competition amongst bidders in the auction reduces both revenue and efficiency gains achieved by uniform-price auctions. Finally, I find that uncertainty decreases revenue gains achieved by switching to a uniform-price auction, but has no impact on efficiency gains.

1.6 Thesis Layout

The remainder of this thesis is organised as follows. Chapter 2 provides the institutional background of the Indonesian treasuries market, where the scope, the type of securities, and the mechanisms of the Indonesian treasury auctions are discussed in detail. Chapter 3 presents a review of the relevant literature on multi-unit auctions for treasuries, including the underlying theory and empirical evidence on auction performance. This is followed by a discussion of the determinants of auction performance. Chapter 4 develops the theoretical framework. Chapter 5 discusses the data and research methods adopted to test my predictions. Chapter 6 presents a discussion of the empirical findings of the auction performance. Chapter 7 provides the regression results of the determinants of auction performance. A summary and conclusion are provided in Chapter 8, which also outlines the limitations and suggestions for possible areas for future research.

Chapter 2

Institutional Background of the Indonesian Treasuries' Market

2.1 Introduction

This chapter presents an overview of the Indonesian treasuries' market, where I discuss the history, issuance methods, types of treasury securities offered for sale, auction mechanisms and main participants in Indonesian auctions.

2.2 Scope

The first issuance of Indonesian treasuries was sparked by the 1997 Asian financial crisis. The crisis resulted in a slump in the local currency, soaring non-performing loans, and a systemic banking crisis. Bank Indonesia, the central bank of Indonesia, and the Ministry of Finance, bailed out the Indonesian banking sector by injecting liquidity and capital in the form of a USD 60 billion government recapitalisation bond issue.[4] Because the recap bonds had no tradable features, they appeared as an asset on the balance sheet of banks, which themselves could not execute. In other words, the banks' balance sheet became illiquid. To remedy this problem, the Ministry of Finance (MoF) converted the non-tradable recap bonds to tradable ones in 2000, and issued treasuries for the first time in 2002.

The government of Indonesia authorised the MoF to issue treasuries in 2002 following the enactment of *Law No. 24/2002 on Conventional Based* government

[4]The government of Indonesia issued bonds to restructure the banking sector in 1999 as a response to minimise effects of the 1997 Asian financial crisis. These bonds are used as payment for the shares bought by the government from ailing banks.

Securities.[5] In 2005, the Indonesian government for the first time financed its expenditures, in excess of tax receipts, through the issuance of treasuries. To diversify treasury instruments and widen the investor base, the MoF started to issue Islamic treasuries[6] in 2008, subsequent to the endorsement of *Law No. 19/2008 on Sovereign Sharia Securities*. At the end of December 2017, conventional treasuries comprised 70 percent of the Indonesian treasuries market, with the rest filled by Islamic treasuries (MoF, 2017).

In Indonesia, treasuries are issued via three methods: book building, private placements, and discriminatory-price auctions. Book building is a method to sell treasuries through a selling agent who collects purchase orders within a predetermined period of bidding, making up 30 percent of total issuance of treasuries. Private placements, a method to sell treasuries directly to investors based on private negotiations, makes up a maximum 5 percent of total issuance of treasuries, depending on investor demand. Finally, a discriminatory-price auction is a method of issuance and sale of treasuries through auction participants (bidders) offering (competitive) bids within a predetermined bidding period. Discriminatory-price auctions dominate treasury issuance because they provide high regularity and flexibility for the MoF to set up a treasury issuance program, while at the same time increasing transparency and reducing the costs of the issuance process.

[5] Law No. 24/2002 became the foundation for Ministry of Finance as the representative of the Government of Indonesia to issue government bonds which aim to finance the state budget, cover short term state budget cash mismatches, and manage a government bond portfolio. Before the law enactment, the purpose of bond issuance was only to restructure the national banking system.

[6] The treasuries are issued using Islamic principles which are deemed acceptable to Sharia requirement.

2.3 Types of Treasury Securities

Table 2.1 summarises the two main types of treasuries issued by the MoF: conventional and Islamic treasuries (Sukuk). Of the USD 250 billion in outstanding treasuries at the end of December 2017, about USD 200 billion (70 percent) was in the form of conventional treasuries. Conventional treasury bonds range in maturity from three months to 30 years, providing semi-annual coupon payments for investors. Most of the treasury bonds are in the form of nominal securities for which the principal payments and coupons are fixed, mainly in rupiah and less frequently in other denominations such as US dollars, Japanese Yen, and Euro. In contrast, conventional treasury bills have a maturity of three months or one year, are issued at a discount from their face value (par), and are non-coupon paying.

The MoF regularly issues Islamic treasuries. As discounted securities, Islamic treasury bills have a maturity of six months. In contrast, Islamic treasury bonds have maturities ranging from 2 to 35 years, providing semi-annual coupon payments to holders. As of 31 December 2017, Islamic treasury bonds issued are dominated in Rupiah and US dollars. Because of the prohibition of transferring legal ownership of a state-owned asset to other parties, Islamic treasury bonds issued are predominantly asset-based rather than asset-backed Islamic.[7] Table 2.2 presents a summary of key differences between asset-based and asset-backed Islamic treasuries. There is no true sale or transfer of the asset

[7] This is different from asset-backed Islamic bonds which refer to transactions involving the true sale and transfer of legal ownership of the asset from the originator to the SPV. The SPV issues the Islamic bond to the investors, who then provide financing to the SPV and obtain the Islamic bond certificate. In turn, the SPV acts as a trustee of the asset for the investors who, collectively, are the undivided proportionate true owners of the asset. For redemption of the Islamic bond certificates, the investors rely heavily on the underlying asset. In the case of periodic redemption, the holders will be paid from the revenue generated by the asset. In the case of default, the investors have recourse against the asset by selling the asset and applying the proceeds towards satisfaction of any amounts owed to them.

from the originator to the special purpose vehicle (SPV) in asset-based Islamic bonds. The asset remains on the balance sheet of the originator after issuance. Islamic bond investors will only attain beneficial (or constructive) ownership based on either an accounting or a legal perspective, not real ownership of the asset. The real asset that becomes an underlying asset is used to make the transaction structure to conform to the principles of Shariah and serves neither as a source of direct benefit payments nor a pledge. Principal and returns/coupon payments of the Islamic bond are guaranteed by the obligor/originator and not directly related to the income stream from the underlying asset. Thus, asset-based Islamic treasuries very much resemble the characteristics of conventional secured bonds more closely than those of asset securitisation.

The issuance of Islamic treasuries provides several benefits to the Indonesian government. Firstly, it allows financing instrument diversification for the state budget, which previously relied entirely on conventional bonds and foreign loans. Because of the prohibition of Islamic investors to invest in conventional treasury instruments (*riba* is forbidden), Islamic bonds provide an expanded investor base. Secondly, Islamic bonds encourage the utilisation of state-owned assets to generate potential revenue for the state budget. Thirdly, Islamic bonds, being essentially risk-free, are the benchmark instrument in the Islamic capital market. This encourages the private sector to issue similar securities, thus allowing for the development of the Islamic capital market in Indonesia. A summary of the main differences between conventional and Islamic treasuries issued in Indonesia is provided in Table 2.3.

2.4 Islamic Treasuries Structure

In this subsection, I shed some light on the concepts and the structures of the various types of Indonesian Islamic Treasuries. As an asset-based instrument, the legal

structures adopted are either Ijarah sale-and-lease-back, Wakala, Ijarah Al Khadamat and Ijarah asset-to-be-leased. According to DGBFRM MoF Report (2017), and based on data from 2008 to 2017, Ijarah sale-and-lease-back *and* Ijarah asset-to-be-leased are the two preferred structures by MoF, where Ijarah asset-to-be-leased in comprises 77.25% of the market in 2017. The following sections briefly explain these four Sukuk structures.

2.4.1 Ijarah Sale-and-Lease-Back

Sukuk Ijarah, based on the concept of Islamic leasing, is an extremely popular and relatively simple form of Sukuk. The key element of Sukuk Ijarah is its simplicity, and it is sometimes viewed as the classic Sukuk structure from which other forms have developed. The term Ijarah essentially means 'lease' – or the transfer of the rights of an asset in exchange for payment. Its literal meaning is 'to compensate'.

The process itself is straightforward. At the first step, the government sells the beneficial title of Stated-owned assets to the Sukuk Negara issuing company (SNIC) to be used as Sukuk assets (*bai'* contract). SNIC is a special purpose vehicle (SPV) to facilitate an asset securitization transaction that comprises the sale and lease back of the asset from the government to the SPV via the issuance of Sukuk to investors. The SPV acts as trustee and holds the asset in trust for the Sukuk holders.

SNIC issue Sukuk an equivalent amount to the purchase price as evidence of investor's participation or ownership on Sukuk Assets, with profit payments to investors based on the rental income from the original lease. Sukuk represents rights in the underlying asset. Thus, Sukuk holders have a proportionate interest in the assets, which are held in trust. Following the purchase, SPV leases it back to the government for daily operations in return for regular rental payments under a sale-and-leaseback (Ijarah)

agreement. The government then makes periodic lease payments to the SNIC. These payments are distributed to the Sukuk holders as a regular income distribution payment in proportion to their holdings. Upon the maturity of the lease period (which equals the maturity date of the Sukuk), the government and the SNIC exercise their purchase and sale undertakings, with the SNIC transferring the asset back to the government at the exercised price – the proceeds of which the SNIC then uses to redeem the Sukuk. The sale and purchase (bai) contract and lease (Ijarah) contract are carried out separately.

The State-owned assets used in Sukuk issuance take the form of land and buildings across various ministries and institutions throughout Indonesia. AAOIFI categories it as Sukuk (certificate) of ownership of leased asset) and mentioned in Sharia Standards Number 17 (3/1). Fatwa National Sharia Board - Indonesian Council of Ulama Number 71/DSN-MUI/VI/2008 states the contract allowed to be made. Fatwa DSN- MUI Number 72/DSN-MUI/VI/2008 provides the basis and guidelines for issuance of Sukuk Negara using these structures.

There are several benefits using Ijarah Sale and Lease Back structure. It can be issued at a premium price, at par, or at a discount. The coupon can be fixed coupon or floating coupon with tradable in the secondary market. The maturity of Sukuk can be a short-term, medium-term or long term. Also, it has a simple transaction flow; thus, it easy to understand and has legal document issuance efficiency.

Figure 2.1 shows the flow of Sukuk Negara Ijarah Sale and Lease Back transactions. The government as obligor or originator sale the beneficial title of state-owned asset s to the Sukuk Negara issuing company to be used as Sukuk Negara assets or the underlying asset for the issuance. Sukuk Negara issuing company issue Sukuk Negara as evidence of investor's participation or ownership on Sukuk Negara assets.

Investors deliver proceeds of Sukuk Negara issuance to the Sukuk Negara issuing company. Sukuk Negara issuing company transfer proceeds of Sukuk Negara issuance to the government. Sukuk Negara issuing company leased Sukuk Negara assets to the government to be utilized for the government's daily activity. The government pays the lease Sukuk Negara assets as a coupon of Sukuk Negara to the Sukuk Negara holders through the paying agent. The government purchases Sukuk Negara assets from the Sukuk Negara issuing company at maturity time. Payment for the purchase of Sukuk Negara assets shall be made by the government to the Sukuk Negara holders through the paying agent for Sukuk Negara redemption.

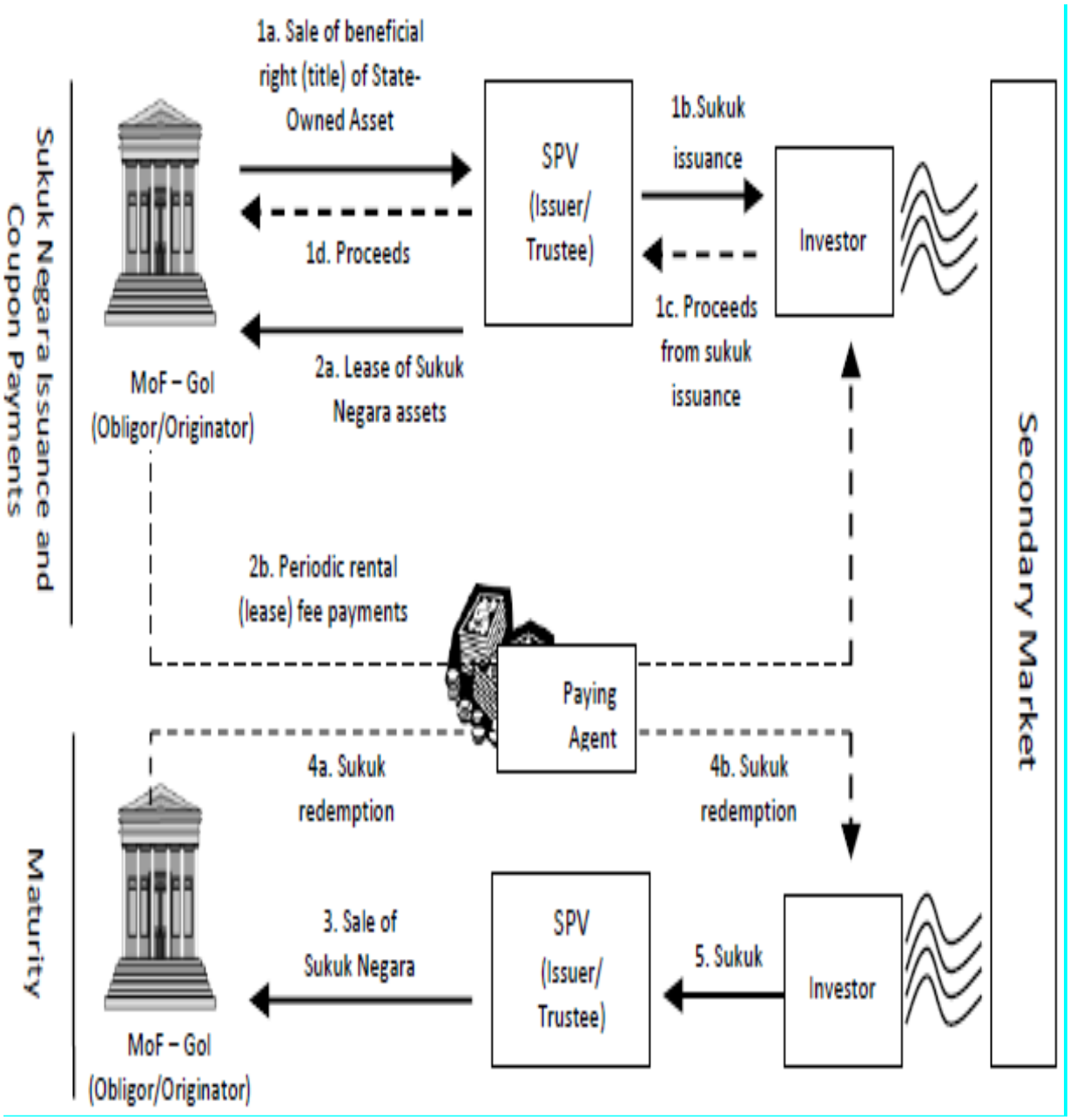

Figure 2.1: Structure and Flow of Transaction of Sukuk Negara Ijarah Sale-and-Lease-Back

2.4.2 Wakala

Sukuk Al-Wakala structure stems from the concept of a Wakala which, literally translated, means an arrangement whereby one party entrusts another party to act on its behalf. A Wakala is thereby akin to an agency arrangement. In the Fatwa National Sharia Board - Indonesian Council of Ulama number 95/DSN-MUI/VII/2014 on Sovereign Sharia Securities (SBSN) Wakala, Sukuk Negara Wakala defined as Sukuk Negara issued based on the principles of Shariah, as evidence of ownership of part of assets in investment activities managed by the Sukuk Negara issuing company as the agent of the Sukuk Negara holder.

Sukuk Wakala structure provides an alternative to Sukuk Negara structure that has flexibility in the use of contract and able to save the use of underlying assets in the form of State-owned assets, able to optimize a variety of potential underlying assets (such as procurement of goods and services). The structure combines at least two types of contract and assets, tradable in the secondary market, as well as widely accepted in domestic and international markets. Structure of Wakala was first used in Sukuk Negara issuance in foreign currencies in the international primary market in 2014, replacing the structure of Ijarah sale-and-lease-back used in the previous issuance.

Sukuk Negara Issuing company as an agent (wakeel) invests the funds provided by the investor into a pool of investments or assets and lends it expertise and manages those investments on behalf of the investor for a particular duration in order to generate an agreed-upon profit. As an agent, the Sukuk Negara issuing company conveys to potential investors about the planned use of the funds , among others which includes the types of planned activities, the composition of planned activities, estimated gains of each

activity, and estimated gain composition. Wakala contract to be used may be Wakala with a fee or Wakala without a fee.

Further, the Sukuk Negara issuing company issues Sukuk Negara to investors with a nominal value equivalent to planned activities to be carried out. The Sukuk Negara issuing company manages the proceeds to invest in a variety of activities that generate profits. In the case of Sukuk Negara issued with the aim to be tradable in the secondary market, the Sukuk Negara issuing company will keep the composition of at least 51% of activities in the form of tangible assets.

The Sukuk Negara issuing company generates profits from investment activities and delivered to the investor as a return or coupon. The coupon shall be periodically given to the Sukuk Negara holders during the term of Sukuk Negara and with a due date according to the agreement and type of gain generated. The government will buy back the assets of Sukuk Negara Wakala from the investors at an agreed price at maturity time through the sale-and-purchase contract. The purchase price of Sukuk Negara asset is equivalent to the nominal value of the Sukuk. The government will pay the proceeds to the investors as Sukuk redemption.

Figure 2.2 displays the flow transactions of Sukuk Negara Wakala. The Sukuk Negara issuing company shall declare that it acts as agent of the Sukuk Negara holder, to manage the proceeds of Sukuk Negara issuance to perform activities that generate gains or profits. The Sukuk Negara issuing company conveys to the potential investors about planned use of funds in various activities to be carried out. Wakala contract used: Wakala with fee (Wakala bil ujrah) or Wakala without fee (Wakala bi dunil ujrah). The Sukuk Negara issuing company informs investment activities undertaken, including: type of activity, composition of activities, calculation of gain/profits of each activity, and

calculation of composition. In case Sukuk Negara will be traded in the secondary market, the Sukuk Negara issuing company maintains the composition of activities of use of funds at least 51% in the form of tangible assets.

Then, the Sukuk Negara issuing company issue Sukuk Negara Wakala. The Sukuk Negara issuing company shall conduct various profitable activities, whether they are ijarah, tijarah, and other activities under the principles of Shariah with the government as counterparts. Any gain derived from these activities will be given to the Sukuk Negara holders as the coupons thereof. The coupons of Sukuk Negara shall be periodically given during the term of Sukuk Negara or on its maturity under the agreement.

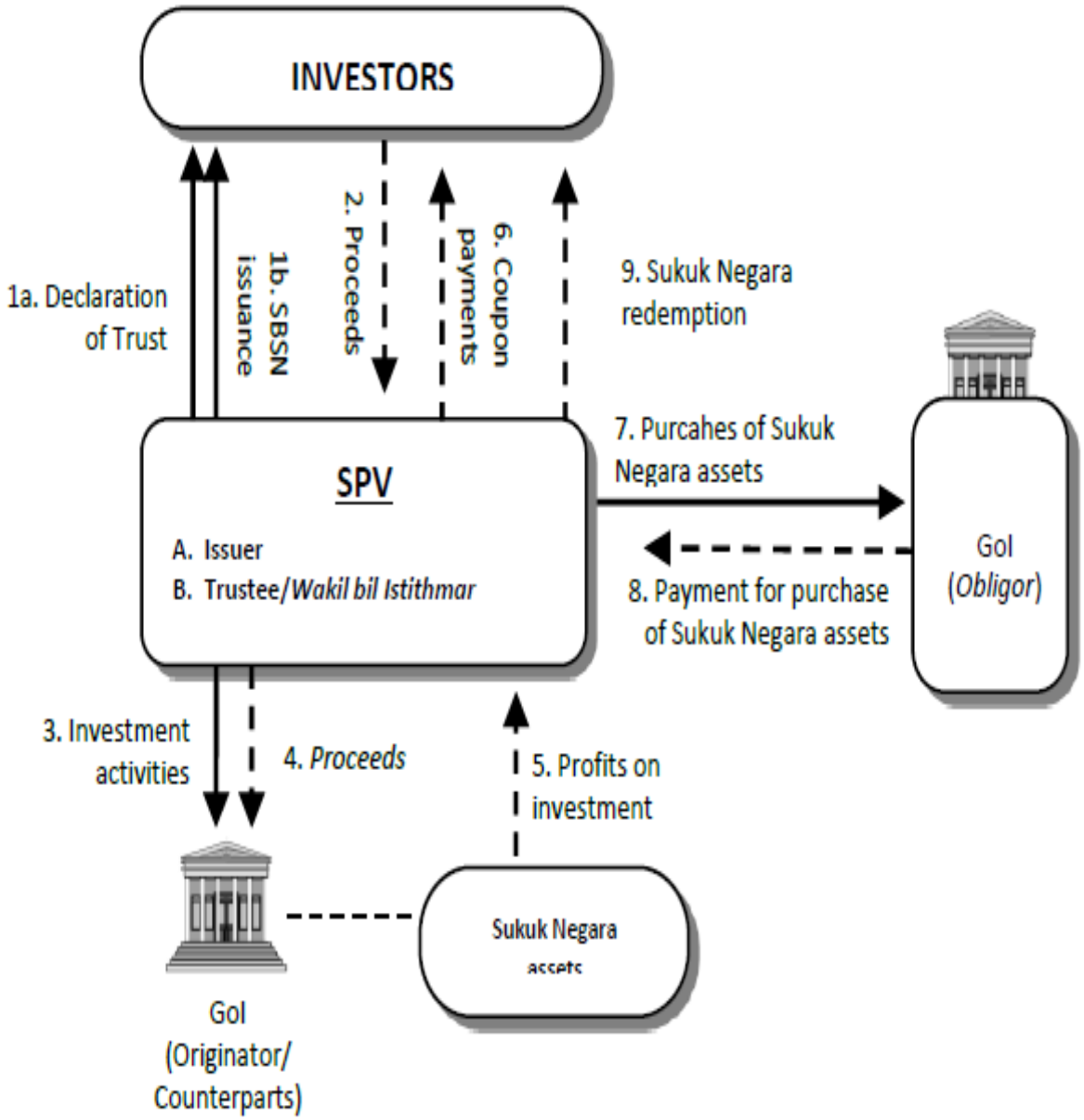

Figure 2.2: Structure and Flow of Transaction of Sukuk Negara Wakala

2.4.3 Ijarah Al Khadamat

The Ijarah Al-Khadamat contract itself can be understood as an ijarah contract for services. This is in accordance with the classification of objects of Ijarah which can be in the form of benefits for goods and rent (Ijarah al-ain). Examples are rental of houses and vehicles, or whose objects of the transaction are benefits for services and wages Ijarah Al-Khadamat), such as assistance and consultant services.

Referring AAOIFI Sharia Standards Number 17 (3/2/4), Sukuk Ijarah Al-Khadamat represents ownership of services available in the future (certificates of ownership of described future services). Ijarah Al-Khadamat uses an underlying asset in the form of services, namely hajj services, including flight services, catering services, and dormitory services. The features of Ijarah Al Khadamat structure are fixed coupons, do not use BMN as the underlying asset, but instead, use hajj services as the basis for issuance and can be issued in medium-long tenors.

Sukuk Negara Ijarah Al-Khadamat produces non-tradable features which represent ownership on services to be provided in the future as well as receivables of service revenues. According to the principles of Sharia, any Sukuk may be traded only if it represents ownership of tangible assets, whereas if it represents ownership of loans/receivable, the Sukuk is non-tradable and must follow Sharia provisions related to currency exchange and loans/receivable (sharf). These provisions also apply to Sukuk Negara Ijarah Al-Khadamat and thus according to Sharia principles, Sukuk Ijarah Al-Khadamat is not tradable. This is in line with Shariah provisions on Sukuk trading contained in AAOIFI Sharia Standards 17 (5/2/8), (5/2/9) and (5.2.10).

Figure 2.3 describes the flow of transactions of Sukuk Negara Ijarah *Al-Khadamat* structure. Firstly, the government makes the agreement on Hajj service delivery with a

Special Purpose Vehicle (SPV), namely the Sukuk Negara issuing company which is established specifically for Sukuk issuance.

The agreement shall be made under an Ijarah Al-Khadamat contract. Based on the contract, the government as the service user requests the SPV (service providers) to provide Hajj services in the forms of flight services, catering services and housing services during specified Hajj period. On the basis of the contract then SPV issues Sukuk with a nominal value equals to the value to procure the services. In this regard, Sukuk issued represent ownership over Sukuk Negara assets in the form Hajj services. SPV subsequently authorize the procurement of services to the government (as agent) through an authorization (wakala) agreement. The scope of this agreement includes the provision of services in the forms of Hajj flight, catering and housing services. The proceeds of Sukuk issuance shall be then given to the government as Hajj services providing costs. The government as the service user shall then pay a fee (ujrah) to the SPV (service providers) as a service fee payment.

Payment of coupon shall be made periodically to investors, which comes from the component of fee (ujrah) paid by the government (service user) to the SPV (service providers). To facilitate the payment mechanism, the fee payment is divided into two components. The first part is a fee component which is equivalent to total coupon paid to investors or so-called fee (ujrah) installment, while the second part is fee component which is equivalent to nominal value of the Sukuk, or referred to as residual fee (ujrah) payment. The first fee (ujrah) component is paid periodically to the investors as a form of coupon payment, while the residual fee (ujrah) payment will be paid in full at maturity as Sukuk redemption. Fee (ujrah) payment methods and mechanisms are set forth clearly in Ijarah Al-Khadamat Contract signed and agreed by both contracting parties.

After procurement process is carried out, the government (agent) will submit a report on procurement of services to SPV (Principal). With the completion of procurement process and report on procurement of services from the government (Agent), SPV (service users) and the government (service provider) will sign Official Minutes of Work Handover (BAST) as specified in Ijarah Al-Khadamat. On that basis, the government (service users) will settle the residual fee (ujrah) payment which is equivalent to nominal value of the Sukuk. Proceeds from this residual fee (ujrah) payment are paid in full to investors as Sukuk redemption.

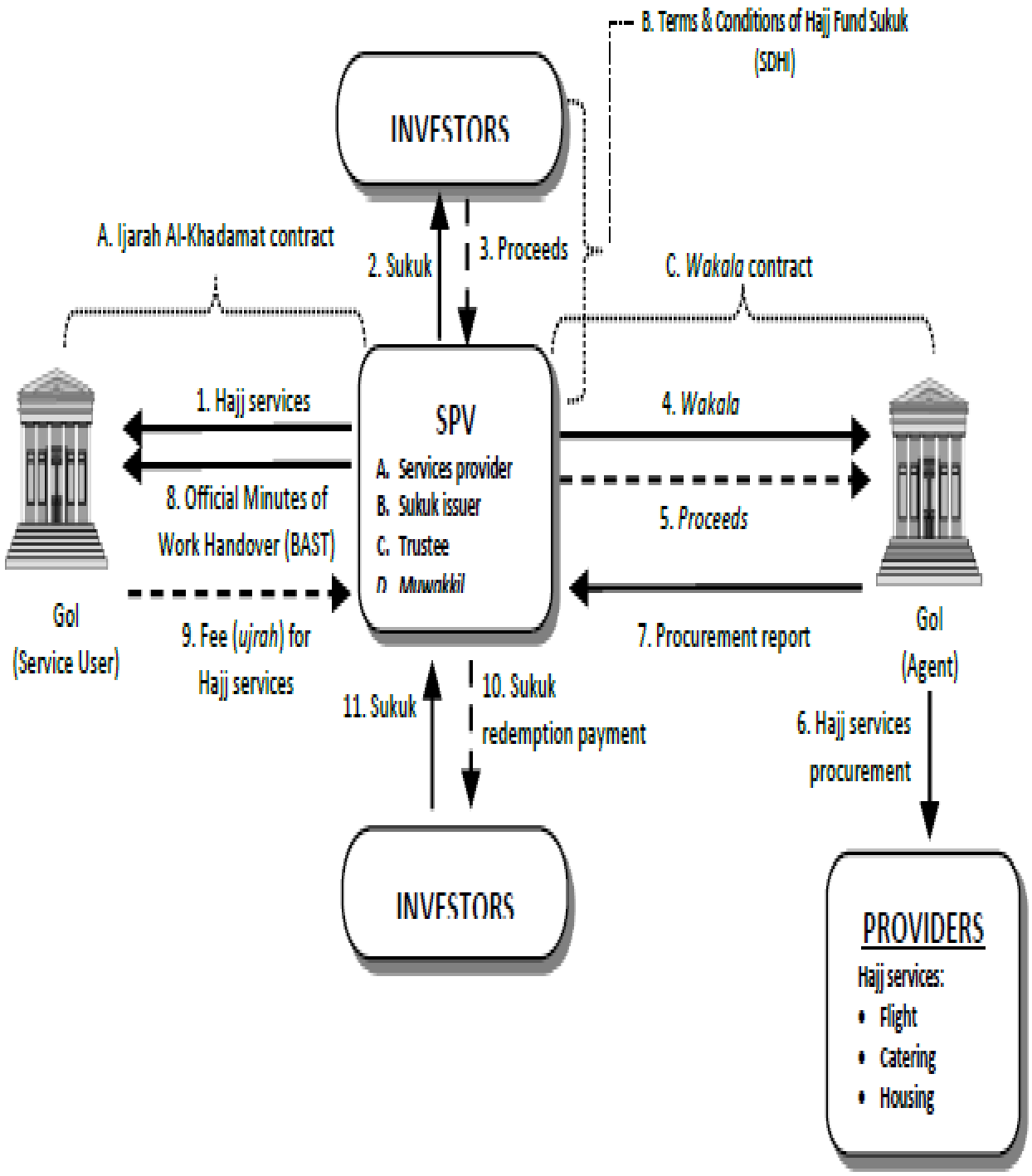

Figure 2.3: The Scheme of Sukuk Negara Ijarah Al-Khadamat Transaction

2.4.4 Ijarah Asset-to-be-Leased

Sukuk Negara Ijarah assets-to-be-leased is the Sukuk structure developed for Sukuk Negara issuance to finance the government projects, using Ijarah assets-to-be-leased contract. Based on AAOIFI Sharia Standards Number 17 (3/1), it represents ownership of tangible assets promised to be leased. Ijarah Asset To Be Leased contract is a contract which object specification has been determined, and some object has been existing when the contract is made, but the delivery of all object is made in the future as mentioned in the agreement. The contract has similarity characteristics with Ijarah Mawsufah fi Dhimmah (Forward Lease), yet more powerful because some objects are in existence well before the contract is made.

The Shariah basis of Sukuk Negara Ijarah Asset to be leased issuance is Fatwa National Sharia Board - Indonesian Council of Ulama Number 76/DSN-MUI/VI/2010 on Sukuk Negara Ijarah Asset to be leased. The government for the first time used this structure of Sukuk Negara Ijarah Asset to be leased when issuing Sukuk Negara series PBS in 2011. Furthermore, this structure has also been used in the issuance of retail Sukuk since 2012, replacing the structure of Ijarah Sale and Lease Back which had been used previously for the series.

The structure of Sukuk Negara Ijarah Asset to be leased has advantage features that suit the preferences of issuer and investor. These advantages include fixed or floating coupons, tradable, medium or long-term tenor, and issuable at a premium price, at par, or at discount. Besides, this structure also has simple transaction flow, so as to provide the efficiency of issuance documents used by the government. In accordance with the purpose of issuance, Sukuk Negara Ijarah asset-to-be-leased is issued to finance the development of projects. For that, underlying assets used are project or activities of Ministries or

government agencies which have been allocated in state budget. The use of the project as the basis for issuance of Sukuk Negara Ijarah asset to be leased surely must be complying with and not contrary to the principles of Shariah.

Figure 2.4 describe the transaction flow of Sukuk Negara Ijarah assets-to-be-leased. Figure 2.4 describes the transaction flow of Sukuk Negara Ijarah assets-to-be-leased. The government as the originator places an order to The Sukuk Negara issuing company (SPV) on assets in certain specifications to be leased through Ijarah Assets to be leased (Ijarah Object). Ijarah object itself is also the bases of Sukuk issuance (underlying asset) consisting of two types namely projects that will be built and tangible assets that may be the parts of projects that have been existing and state-owned asset s in the form of land or building. On the basis of such order and in order to provide Ijarah Object, The Sukuk Negara issuing company grant authority to the government (Agent) to supply the Ijarah Object through Wakala contract.

Based on such order and authority to supply Ijarah Object, the Sukuk Negara issuing company will issue Sukuk to the investors in nominal value equivalent to the price of assets to be used as Ijarah objects. The proceeds from the investors will be then used as payment to supply such assets to the government (as Agent). The issued Sukuk represents ownership of Sukuk Negara assets in the form of Ijarah Object, both already existing (state-owned asset and parts of projects already existing) or that will come into existence (projects).

The Sukuk Negara issuing company as trustee of the investors will lease Sukuk Negara assets to the government. The assets that have already existed shall be used directly, while assets under construction may be used after the project has been completed. Based on the lease of Sukuk Negara asset, the government as the lessee will

pay the lease fee (ujrah) periodically to the investors through the paying agent namely the central bank. The lease fee is a coupon received by the investors until Sukuk maturity. The Sukuk Negara further represents ownership of Sukuk Negara assets being leased to the government.

On maturity, through bai' (sale and purchase) contract, the government will buy back the Sukuk Negara assets from the investors at a price agreed upon in advance. The purchase shall be based on the purchase undertaking by the government and the sale undertaking by the Sukuk Negara issuing company. The purchase price of Sukuk Negara asset shall be equivalent to the nominal value of Sukuk. The proceeds from this purchase will be then paid to investors as Sukuk redemption.

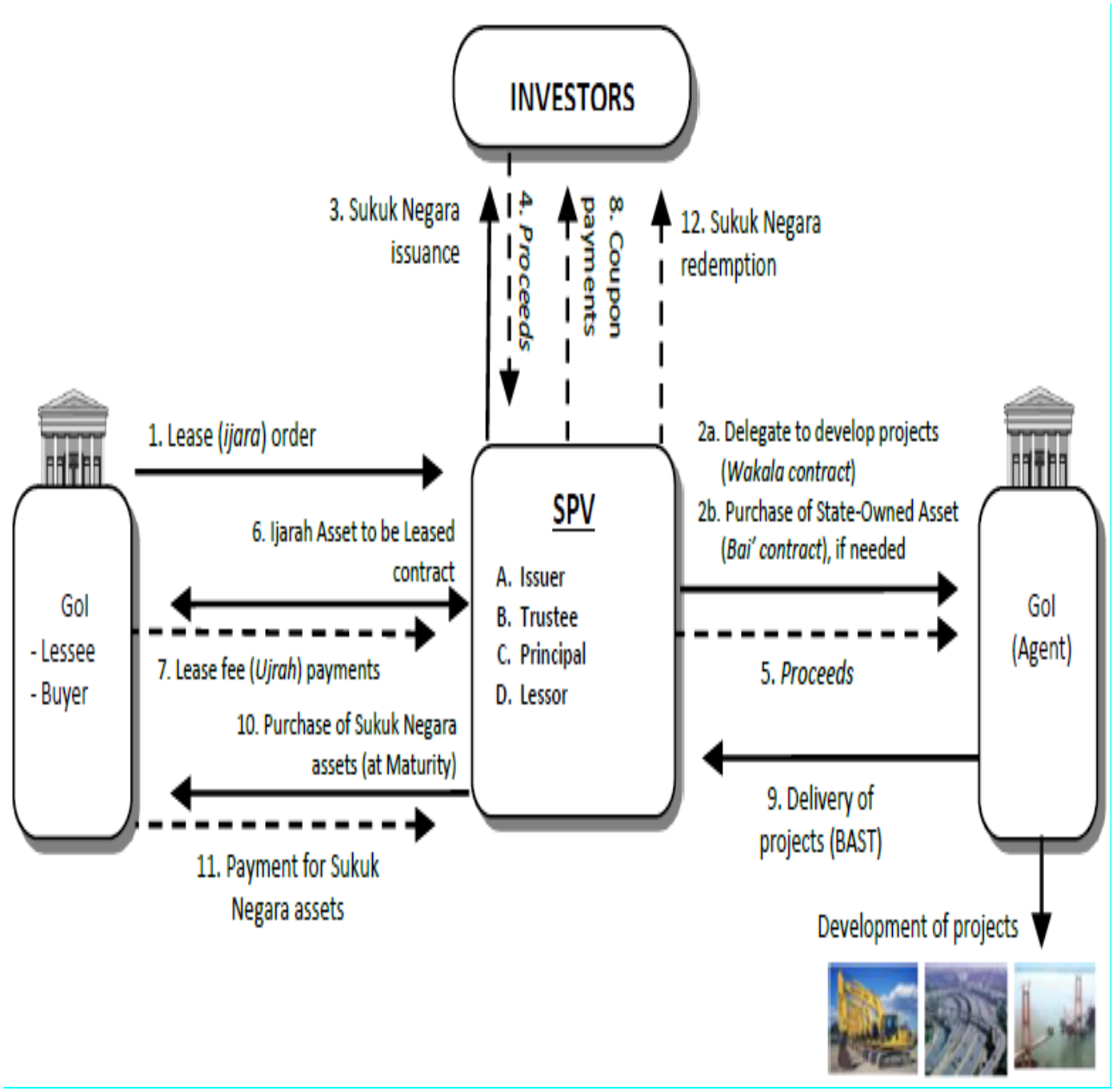

Figure 2.4: The Scheme of Sukuk Negara *Ijarah Asset-to-be-Leased* Transaction

Table 2.1: ***Treasuries Issued in Indonesia.***

Type	Maturity	Return	Minimum Denomination	Issuance Method	Tax Rate	Redemption
Islamic Treasury Bills	6 mths	Difference between redemption value and issue price	IDR1 Billion	Auction	15 percent	Single payment on maturity
Conventional Treasury Bills	3 and 12 mths	Difference between redemption value and issue price	IDR1 Billion	Auction	15 percent	Single payment on maturity
Conventional Treasury Bonds	> 1 year	Fixed semi-annual coupon, difference between redemption value and issue price	IDR1 Billion	Auction	15 percent	Single payment on maturity
Islamic Treasury Bonds	> 1 year	Fixed semi-annual coupon, difference between redemption value and issue price	IDR1 Billion	Auction	15 percent	Single payment on maturity
Savings Bonds (Retail)	2 years	Fixed semi-annual coupon, difference between redemption value and issue price	IDR1 Million	Bookbuilding	15 percent	Single payment on maturity
Indonesian Bonds (Retail)	3 years	Fixed semi-annual coupon, difference between redemption value and issue price	IDR1 Million	Bookbuilding	15 percent	Single payment on maturity
Islamic Saving Bonds (Retail)	2 years	Fixed semi-annual coupon, difference between redemption value and issue price	IDR1 Million	Bookbuilding	15 percent	Single payment on maturity
Islamic Bonds (Retail)	3 years	Fixed semi-annual coupon, difference between redemption value and issue price	IDR1 Million	Bookbuilding	15 percent	Single payment on maturity

Table 2.2: ***Comparison between Conventional and Islamic Treasuries Issued in Indonesia.***

Characteristics	Conventional Bonds	Islamic Bonds
Definition	Securities issued based on conventional (interest-based) principles.	Securities issued under the principles of Shariah, as evidence of ownership/participation of underlying assets.
Underlying Asset	Do not require underlying assets as the basis for issuance.	Require underlying assets as the basis for issuance.
Unit of Issuance	Each unit bond represents a share of debt.	Each unit Islamic bond represents a share of the underlying asset.
Proceeds Utilisation	Proceeds can be used to finance any state budget expenses.	Project-based Islamic bond (PBS) proceeds must be earmarked to specific projects conforming to underlying asset used.
Sharia Compliance	Do not require fatwa or Shariah endorsement.	Need fatwa or Shariah endorsement.
Auction Participants	20 primary dealers, comprising 4 local securities, 8 foreign banks, and 8 local banks plus Central Bank and the Indonesia Deposit Insurance Corporation (IDIC).	21 registered auction participants, consisting 3 Shariah banks, 4 local securities, 7 foreign banks, and 7 local banks plus Central Bank and the Indonesia Deposit Insurance Corporation (IDIC).
Auction Participants Liability in Primary Market	Absorb a minimum of 2 percent total issuance.	Submit bids in every auction without obligation to absorb.
Auction Participants Liability in Secondary Market	Must trade a minimum of 2 percent of benchmark bond trading volume. Release price quotation of benchmark bonds which can be executed at any time.	No liability.
Multiple Yield in Auction	1/100 or 0.01.	1/32 or 0.03125.

Table 2.3: ***Comparison between Asset-based and Asset-backed Islamic Treasuries.***

	Asset-based	**Asset-backed**
Description	Ownership of underlying asset is beneficial ownership, with the obligor/originator guaranteeing the coupon and nominal payments.	Ownership of underlying asset is true ownership, with coupon and nominal payments coming entirely from the income stream of the underlying asset.
Instrument Characteristic	Resemble characteristic of conventional bonds.	Resemble asset securitisation.
Underlying Asset Ownership	Asset transfer process to investors is based on the concept of "beneficial ownership". Legal title of the underlying asset remains with the originator.	Asset transfer process to investors is based on the concept of "true sale". Investors have full ownership (legal title) of the underlying assets.
Islamic Bond Rating	Based on Islamic bond originator rating, not on underlying asset.	Based on the value and performance of the underlying asset.
Nominal Payment	Principal payment is secured by obligor/originator through purchase undertaking.	Principal payment depends on the market price of the underlying asset. There is no purchase undertaking by obligor.
Return (Coupon) Payment	Coupon payment is secured and fully conducted by obligor/originator.	Coupon payment is entirely based on the income stream of the underlying asset.
Default	Instrument becomes unsecured when default because investors do not have direct ownership right of the underlying asset.	Instrument becomes secured when default because investors have direct ownership right of the underlying asset.

2.5 The Indonesian Treasuries Auction Market

Indonesia uses a treasuries discriminatory-price auction process with the following characteristics. The auction has private values where an unlimited number of bids can be submitted. The process of bidding is sequential, from high to low. The quantity does not decrease if the price rises (according to theory, quantity will drop if the price rises). At the start of the auction, the MoF announces the value of the indicative issuance (i.e., the expected total issues at each auction) and the fair yield for each type treasury. This announcement includes only the total revenue, while reasonable yields are not known by both the public and bidders. Each bid value (requested yield value) entered is sorted from low to high, with the value taken being the exact value below the fair yield. After the auction closes, a closed meeting will determine the auction winners, with the following notes: each auction is seen as one unit of all treasuries auctioned in a period. If the revenue target in an auction is not reached, it will be updated in the next period.

2.5.1 Conventional Treasuries

Bidders in Indonesian treasury auctions consist of three main groups: primary dealers (hereafter "PDs"), dealer's clients (investors or indirect bidders), and direct bidders. PDs are banks and securities companies which are appointed to participate in the government bond auction market. PDs submit bids on their own behalf and on behalf of their clients. Participation in conventional treasury auctions is limited to PDs so any parties wanting to purchase treasuries can only do so by using a PD. As of January 2019, there are 20 PDs: 16 banks and four securities companies (see Table 2.4).

PDs play a significant role in treasury auctions. They are required to participate meaningfully in both treasury markets and to provide policy-relevant market information to the MoF. In terms of participation, PDs must submit bids in every auction and absorb

a minimum 2 percent of the total issuance of treasury bond auctions within three months. In the secondary market, PDs must trade a minimum 2 percent of total trading volume of benchmark bonds within three months and release a price quotation of benchmark bonds to be executed at any time. Beside these obligations, PDs have several privileges. For example, they are the exclusive auction participants in treasury auctions and buyback treasury auctions. Additionally, PDs get securities lending facility,[8] and have privileged access to information relating state budget deficit financing policy and operational government bonds from the MoF.

In treasury auctions, there are two institutions which act as direct bidders: Indonesia Deposit Insurance Corporation (IDIC)[9] and Bank Indonesia (B1)[10]. These institutions can only submit bids on behalf of themselves via non-competitive bidding. Unlike IDIC, BI can submit bids both for treasury bills and treasury bonds. There is prohibition in Act No. 3/2004 concerning BI purchasing treasury bonds in the primary market on its own account. However, an exception is for treasury bills which BI uses in monetary market operation. BI and IDIC have the potency to influence the market condition since they have a lot of liquidity in their accounts.

Treasury auctions are carried out fortnightly on Tuesdays, starting at 10 am and finishing at 12 pm Jakarta time. At the end of each calendar year, the MoF announces the timetable for the issuance of treasuries, including treasury auction dates, tenure, and indicative targets of treasury auctions the following year. For a maximum of two days

[8] PDs are able to borrow Treasury Bonds and Treasury Bills through a securities lending facility provided by the Ministry of Finance (MoF) of the Republic of Indonesia. The lending facility allows PDs to borrow Treasury Bonds and Treasury Bills when they are not readily available from other sources in the market.
[9] IDIC is an independent institution which functions to insure the funds of depositors and is actively involved in maintaining the stability of the banking system in accordance with its authorised mandate.
[10] Bank Indonesia (BI) is the central bank of the Republic of Indonesia.

prior to the auction, the MoF releases planned issues, including treasury series, maturity date, auction date, percentage allocation for non-competitive bids, and indicative targets of the issuance. In addition, the MoF has the option to conduct an additional auction (green shoe option) one day after the regular auction has completed. Announcements for additional auctions are conducted simultaneously with announcements of the results of regular auctions.

Bidders can submit multiple offers for several units in sealed-bid auctions. PDs submit tenders electronically via the electronic auction system, the Bank Indonesia Electronic Trading Platform (BI ETP), prior to the specified auction deadline. PDs can submit either a competitive offer or a non-competitive offer. Bidders who submit a competitive offer must include the yield (price) and quantity, while bidders who submit a non-competitive bid only need to specify the quantity. For instance, if a competitive offer is to purchase IDR 10 billion at 8 percent yield, a non-competitive tender shall consist of only the amount of IDR 10 billion. Each auction participant must submit minimum bidding of IDR 1 billion, with the remaining amount being a multiple of IDR 100 million.

The bids submitted are ranked in ascending order according to yield. The allotment of treasuries is carried out based on the corresponding yield bid less than or equal to the benchmark yield set by the MoF. The MoF allots partial of the total bids when the total bid units at the cut-off yield exceeds the total winning bids. For bids at the cut-off yield, the MoF allocates amongst the bidder using weighted amount based on the quantity of each bid. Non-competitive bidders pay the corresponding weighted average of the winning competitive yields. The multiplication of the yield is 1/100 or 0.01.

2.5.2 Islamic Treasuries

Islamic treasury auctions consist of three types of bidders: Auction Participants (hereafter "APs"); APs clients (indirect bidders); and direct bidders. APs are banks and securities companies appointed by the MoF as registered auction participants in Islamic treasury auctions, not as primary dealers. Similar to PDs in conventional treasury auctions, APs can bid for themselves and for their clients. All parties, except direct bidders, willing to buy Islamic treasuries in the auction, must submit orders through APs. As of 31 January 2019, there are 21 APs: 14 conventional banks, three Shariah banks, and four securities companies. Table 2.5 lists them.

APs are different from PDs in a number of aspects. Although APs must submit bids in every auction, there is no obligation to win the auction. In line with the primary market, there are no obligations in the secondary market such as trading a minimum amount or release price quotations of Islamic treasuries. As a consequence of the absence of obligations, APs do not have the same privileges as PDs.

Although dealer participants may differ, there are a number of similarities between direct bidders in conventional and Islamic treasury auctions. BI and IDIC are direct bidders who can only submit non-competitive bids on their account and they both hold large cash reserves which can influence the market. However, only BI can submit bids for Islamic treasury bills. BI cannot buy treasury bonds in the primary market, except for Islamic treasury bills for monetary activities.

The Islamic treasury auction is carried out fortnightly on Tuesdays, starting at 10 am and finishing at 12 pm Jakarta time. The issuance calendar is published by the MoF at the end of the year for the year following. Bidders submit bids through APs to the auction agent by 12 pm on the auction date. Upon receipt of a bid summary from BI, the

MoF makes a decision on the accepted minimum yield. The MoF accepts bids with a lower yield in whole while bids presenting the accepted maximum yield can be absorbed with a reduction.

Since 2010, the MoF has conducted an additional auction or *green shoe option*. APs who submit bids at a regular auction have the right to purchase in this additional auction within the amount sold and at the same yield as the weighted average awarded yield at the regular auction. The amount sold during the additional auction is calculated as a product of the total nominal amount of all APs purchased by a given AP at the regular auction. Bidders submit bids at the additional auction within the time period specified in the announcement published after the regular auction. The total nominal value of all bids of a given AP set at the additional auction should not exceed the maximum amount of a given AP in the regular auction. APs can submit multiple offers in one sealed-bid auctions for multiple units. APs submit offers either competitive or non-competitive. Bidders who submit a competitive bid must include a yield and amount. For non-competitive offer, bidders merely submit the amount or quantity.

The MoF ranks submitted bids based on an ascending order according to yield. The allotment of treasuries is carried out based on the corresponding yield bid less than or equal to the benchmark yield set by the MoF. If the total bidding unit at the cut-off yield is greater than the total winning bid, the total bidding at the cut-off yield is only partially allotted. If there are multiple bidders at the cut-off yield, the remaining amount is shared among the bidders, weighted according to quantity bid. Non-competitive bidders pay the corresponding weighted average of the winning competitive yield. Multiplication of yields is 1/32.

2.5.3 Treasury Auction Mechanism

Table 2.4 illustrates the treasury auction allocation under discriminatory price auction on one day auction for one series, namely PBSXXX which has tenure of 5 years and coupon of 7% per annum. There were 18 bids submitted by 10 bidders, constituting a total bid value of IDR5.4 trillion or equivalent USD500 million. The MoF decide cut-off yield is 7.81250% or equivalent with the market-clearing price is 100.08149% and weighted average winning yield is 7.80083%. Based on the market-clearing price, the MoF absorbs IDR2.4 trillion or equivalent USD240 million. The total rejected bids are IDR3 trillion or equivalent US300 million which 55.56% from total incoming bids. From this auction, the MoF total cash proceed is IDR2.403 trillion or equivalent USD240 million.

Each participant will pay based on their bidding price. For example, bidder F submits bids IDR400 billion with yield 7.79700% in number 5, will pay as of IDR400,578,794,431 or equivalent with price 100.14470 for each unit. Another example is Bidder E which has two winning bids in this auction. For the first bid in the number 4, bidder E submits bids IDR200 billion with yield 7.79356. He pays PBSXXX with price 100.14878 for each unit or IDR200.297,556,675 in total. Then for the second bids, bidder E pays PBSXXX with price 100.09168 for each unit or IDR500,458,403,210 in total. In general, bidder E has weighted average winning price 100.10799 for PBSXXX.

Table 2.5 show the summary of incoming and winning bids of Indonesian treasury auctions over time. It allows us to assess the proportion and the success rate of bids each investor type. The incoming bids of conventional treasury auction dominated by the bank and foreign investors are 46.58% and 34.50%, respectively. For Islamic treasury auctions, bank and foreign investors also dominate incoming bids with the percentage of 71.54 and

12.85 respectively. In winning bids, bank and foreign investors dominate with a percentage of 42.25% and 34.17 respectively for conventional treasury auctions while for Islamic treasury auctions the percentages are 67.54 and 11.19 respectively.

Table 2.6 reports example the announcement of Islamic treasury auction results over time. The Islamic treasury auction on Tuesday, February 9, 2016, has total incoming bids IDR14.826 trillion or equivalent USD1.083 billion. Islamic T-Bills (SPN-S 10082016) has the highest incoming bids IDR5.291 trillion (USD386.5 million), followed by Islamic T Bonds (PBS009) IDR5.213 trillion (USD380.82 million), PBS006 IDR1.97 trillion (USD143 million), PBS011 IDR1.442 trillion (USD105.34 million) and the lowest PBS012 IDR0.910 trillion (USD 66 million). Based on the benchmark price for each series, the government absorb IDR5.25 trillion (USD383.52). PBS009 is the highest winning bids IDR2.92 trillion (USD213.31 million), followed by PBS006 IDR1.33 trillion (USD97.16 million), SPN-S 10082016 IDR0.5 trillion (USD36.53 million), PBS011 IDR0.34 trillion (USD24.84 million) and PBS012 IDR0.16 trillion (USD11.69 million).

Table 2.4: ***Illustration Treasury Auction Under Discriminatory Mechanism***

No	Dealer	Series	Yield	Weighted Yield	Price	Bids		Cash Proceeds	Cumulative Cash Proceeds
						Units	Cummulative Units		
1	A	PBSXXX	7.78500%	7.78500%	100.19367	100,000	100,000	100,193,668,801	100,193,668,801
2	C	PBSXXX	7.79000%	7.78750%	100.17326	100,000	200,000	100,173,260,991	200,366,929,792
3	D	PBSXXX	7.79500%	7.79286%	100.15286	500,000	700,000	500,764,291,353	701,131,221,145
4	E	PBSXXX	7.79600%	7.79356%	100.14878	200,000	900,000	200,297,556,675	901,428,777,820
5	F	PBSXXX	7.79700%	7.79462%	100.14470	400,000	1,300,000	400,578,794,431	#############
6	A	PBSXXX	7.80250%	7.79567%	100.12226	200,000	1,500,000	200,244,527,466	#############
7	C	PBSXXX	7.80500%	7.79676%	100.11207	200,000	1,700,000	200,224,136,196	#############
8	E	PBSXXX	7.81000%	7.79977%	100.09168	500,000	2,200,000	500,458,403,210	#############
9	G	PBSXXX	7.81250%	7.80083%	100.08149	200,000	2,400,000	200,162,977,642	#############
10	B	PBSXXX	7.82500%	7.80391%	100.03055	350,000	2,750,000	350,106,920,731	#############
11	D	PBSXXX	7.83250%	7.80903%	100.00000	600,000	3,350,000	600,000,000,000	#############
12	C	PBSXXX	7.81250%	7.80927%	100.08149	250,000	3,600,000	250,203,722,053	#############
13	G	PBSXXX	7.81250%	7.80973%	100.08149	600,000	4,200,000	600,488,932,927	#############
14	F	PBSXXX	7.81250%	7.80983%	100.08149	150,000	4,350,000	150,122,233,232	#############
15	E	PBSXXX	7.83250%	7.81129%	100.00000	300,000	4,650,000	300,000,000,000	#############
16	A	PBSXXX	7.83250%	7.81237%	100.00000	250,000	4,900,000	250,000,000,000	#############
17	B	PBSXXX	7.83250%	7.81316%	100.00000	200,000	5,100,000	200,000,000,000	#############
18	A	PBSXXX	7.83250%	7.81424%	100.00000	300,000	5,400,000	300,000,000,000	#############

Table 2.5: ***Summary of Incoming and Winning Bids by Investors Type in Indonesian Treasury Auction Market***

Investor Type	**% Incoming Bids**		**% Winning Bids**	
	Conventional	Islamic	Conventional	Islamic
Foreign	34.50%	12.85%	34.17%	11.19%
Insurance	1.29%	5.21%	1.82%	7.36%
Bank	46.58%	71.54%	42.25%	67.54%
Pension Fund	5.12%	4.51%	6.39%	6.43%
Others	4.37%	3.31%	6.75%	4.65%
Mutual Fund	2.43%	0.10%	2.65%	0.14%
Securities	5.70%	2.48%	5.96%	2.69%
Total	100%	100%	100%	100%

Table 2.6: ***Sample of Islamic Treasury Auction Results Announcement***

Tuesday, February 9, 2016

Notes	SPN-S 10082016	PBS006	PBS009	PBS011	PBS012
Incoming Bids	IDR5.291 trillion	IDR1.970 trillion	IDR5.213 trillion	IDR1.442 trillion	IDR0.910 trillion
The highest incoming yield	7.75000%	8.59375%	8.84375%	8.43750%	8.68750
The lowest incoming yield	6.25000%	8.06250%	8.00000%	8.75000%	9.03125
Weighted average winning yield	6.26562%	8.21733%	8.11879%	8.49921%	8.73996%
Coupon date	End Period	15 Mar & 15 Sept	15 Jan & 15 Jul	15 Feb & 15 Agt	15 May & 15 Nov
Coupon rate	Discount	8.25000%	7.75000%	8.75000%	8.87500%
Maturity Date	10 Agt 2016	15 Sep 2020	25 Jan 2018	15 Agts 2023	15 Nov 2031
Winning Bids	IDR0.5 trillion	IDR1.33 trillion	IDR2.92 trillion	IDR0.34 trillion	IDR0.16 trillion
Bid to cover ratio	10.58	1.48	1.79	4.24	5.69
Settlement Date	February 11, 2016				

2.6 Chapter Summary

This chapter provides a brief overview of the Indonesian treasuries market. It includes a discussion on the characteristics of the types of treasuries issued and the auction mechanism. Islamic bonds exist to facilitate MoF to diversify the source of financing of the state budget to include that concerned with Islamic principles.

Treasuries are essential to financing the state budget and an important asset in the Indonesian financial market. This instrument comprises a number of securities with different maturities, payment types, currencies, and tradability types, to cater for a range of investor demands. This means that the function of treasuries reaches many parties and interests. The presence of Islamic treasuries shows that the MoF is fully committed to developing the Islamic finance market in Indonesia.

Table 2.7: ***Primary Dealers in Conventional Treasury Auctions.***

Notes: SOE refers to state-owned enterprise. Data as of 20 January 2020.

No	Name	Type	Category
1	*Citibank*	Bank	Foreign
2	*Deutsche Bank A.G.*	Bank	Foreign
3	*PT. Bank Central Asia, Tbk*	Bank	Local (private)
4	*PT. Bank CIMB Niaga, Tbk*	Bank	Foreign
5	*PT. Bank Mandiri (Persero), Tbk*	Bank	Local (SOE)
6	*PT. Bank Maybank Indonesia, Tbk*	Bank	Foreign
7	*PT. Bank Negara Indonesia (Persero), Tbk*	Bank	Local (SOE)
8	*PT. Bank OCBC NISP, Tbk*	Bank	Local (private)
9	*PT. Bank Panin, Tbk*	Bank	Local (private)
10	*PT. Bank Permata, Tbk*	Bank	Local (private)
11	*PT. Bank Rakyat Indonesia, Tbk*	Bank	Local (SOE)
12	*Standard Chartered Bank*	Bank	Foreign
13	*The Hongkong and Shanghai Banking Corporation Limited*	Bank	Foreign
14	*PT. Bank Danamon Indonesia*	Bank	Local (SOE)
15	*PT. Bank ANZ Indonesia*	Bank	Foreign
16	*JP Morgan Chase Bank, N. A*	Bank	Foreign
17	*PT. Bahana Securities*	Securities Company	Local (SOE)
18	*PT. Danareksa Sekuritas*	Securities Company	Local (SOE)
19	*PT. Mandiri Sekuritas*	Securities Company	Local (SOE)
20	*PT. Trimegah Securities, Tbk*	Securities Company	Local (private)

Table 2.8: ***Auction Participants in Islamic Treasury Auctions.***

Notes: SOE refers to state-owned enterprise. Data as of 20 January 2020.

No	Name	Sharia/Conventional	Type	Category
1	*Citibank*	Conventional	Bank	Foreign
2	*Deutsche Bank A.G.*	Conventional	Bank	Foreign
3	*PT. Bank BNP Paribas Indonesia*	Conventional	Bank	Foreign
4	*PT. Bank Central Asia, Tbk*	Conventional	Bank	Local (private)
5	*PT. Bank CIMB Niaga, Tbk*	Conventional	Bank	Foreign
6	*PT. Bank Mandiri (Persero), Tbk*	Conventional	Bank	Local (SOE)
7	*PT. Bank Maybank Indonesia, Tbk*	Conventional	Bank	Foreign
8	*PT. Bank Negara Indonesia (Persero), Tbk*	Conventional	Bank	Local (SOE)
9	*PT. Bank OCBC NISP, Tbk*	Conventional	Bank	Local (private)
10	*PT. Bank Panin, Tbk*	Conventional	Bank	Local (private)
11	*PT. Bank Permata, Tbk*	Conventional	Bank	Local (private)
12	*PT. Bank Rakyat Indonesia, Tbk*	Conventional	Bank	Local (SOE)
13	*Standard Chartered Bank*	Conventional	Bank	Foreign
14	*The Hongkong and Shanghai Banking Corporation Limited*	Conventional	Bank	Foreign
15	*PT. Bank BRISyariah*	Sharia	Bank	Local (SOE)
16	*PT. Bank Negara Indonesia Syariah*	Sharia	Bank	Local (SOE)
17	*PT. Bank Syariah Mandiri*	Sharia	Bank	Local (SOE)
18	*PT. Bahana Securities*	Conventional	Securities Company	Local (SOE)
19	*PT. Danareksa Sekuritas*	Conventional	Securities Company	Local (SOE)
20	*PT. Mandiri Sekuritas*	Conventional	Securities Company	Local (SOE)
21	*PT. Trimegah Securities, Tbk*	Conventional	Securities Company	Local (private)

Chapter 3

Literature Review

3.1 Introduction

This chapter reviews previous research related to treasury auction design and bidding behaviour. It is organised into three sections as follows. Section 3.2 covers theoretical studies in multi-unit auctions. Section 3.3 reviews the empirical literature that compares and contrasts discriminatory and uniform-price auctions, based on either a laboratory experiment, a natural experiment or a structural approach. Section 3.4 discusses prior research on the factors driving the performance of treasury auctions. Section 3.5 summarises and concludes this chapter.

3.2 Theoretical Literature on Multi-unit Auctions

In 1961, Vickrey proposed the prominent *Revenue Equivalence Theorem* for a single indivisible auction as the essential part of the auction literature. The theorem proposes that any auction format that generates the same results (allocates items to the same bidders), given certain conditions, also has the same expected revenue, irrespective of whether these are Dutch auctions, English auctions, discriminatory-price (or multiple-price) auctions, or uniform-price (or single-price) auctions. The theory assumes that bidders are risk-neutral and have independent private value in the non-cooperative equilibrium approach. The other assumption is that sellers sell a single item and every bidder has a private value.

The theorem has a number of limitations[11] when applied to divisible or multi-unit auctions, such as treasury auctions. Theoretical comparisons become more complex and harder when assumptions about bidder's risk preferences are relaxed. For example, the single-unit auction literature often relies on the assumption that the set of available prices is tight. There is another opportunity in multi-unit auctions that bids can be submitted at a tight set of quantities (Geoffrion, 1976; Hall, 1986).

Several theoretical models have been proposed for multi-unit auctions. Smith (1966) proposes a bidding behaviour model for treasury bill auctions. He assumes that bidders will maximise their utility of expectation which will be higher than the lowest accepted bid with a subjective probability density function. He models the condition that price and quantity bids are optional variables and each bidder has a subjective probability position in the minimum winning bid. Using numerical examples, he shows that discriminatory price auctions produce lower cash proceeds than uniform price auctions. Although the paper is the first in recognising the specification of multi-unit auctions, it has a similar drawback to the single-unit auction model in that it presumes that bidders can submit only a single-bid price to maximise a single-period expected utility function.

Scoot and Wolf (1979) criticise the single-bid assumption in multi-unit auctions. They show that bidders can and do submit multiple bids (different amounts at different prices). They propose that the price in a single-price bid is optimal if placed at the lowest stop-out price and that in general, single-price bids are suboptimal. They show that discriminatory price auctions are optimal for treasury bills, confirming the actual practice of participants in these auctions. However, their proposal that single-price bids are

[11] The revenue equivalence principle exists if the number of units for sale is one and bidders have independent private values (Gretschko, 2012)

suboptimal does not represent the probability of strategic interactions among bidders, which may influence the market clearing price. They assume that bidders in multi-unit auctions are price takers and the expectation of profit function reflects the bidder's strategic interaction.

Harris and Raviv (1981) extend Smith's (1966) model by changing the assumption of the neutrality of bidder's risk preference to risk-averse bidders. However, they maintain the assumption that bidders merely submit one unit bid. They show that any auction method is equal to some type of sealed-bid auction in a simple setting. They characterise a *quasi-competitive* auction as the best format for the seller in this simple setting. Their theoretical model suggests that multiple-price auctions create more revenue than single-bid auctions when bidders are risk-averse. When bidders are risk-neutral, both the single and multiple price auctions generate equal revenue.

The studies cited above (Harris & Raviv, 1981; Scoot & Wolf, 1979; Smith, 1966) do not fully reflect the features of multi-unit auctions since bidders are assumed to submit only one unit bid. Bidders are also assumed to be price takers in the auctions. These assumptions are not appropriate in multi-unit auctions because bidders can submit multiple bids at different prices. Further, bidders tend to be risk-averse, implying that bidders may shade their bids in multiple-price auctions to take account of the winner's curse phenomenon.[12] Arguably, in the discriminatory price auctions, all bidders would shade the biddings by the same proportion regardless of whether they possess different levels of market power. In addition, bidders are more likely to submit their true value as their bids in the uniform-price auction.

[12] Winner's curse occurs when a winner's bid exceeds the weighted average winning yield/price of a good.

Wilson (1979) introduces explicit consideration as an essential property in multi-unit auctions. By presenting the notion of residual supply, he considers the downward sloping bid features and, more importantly, the strategic behaviour of bidders. He assumes that each bidder *i* can acquire the information of value summarised in an estimate or statistic. An approach is defined by a feature $x(p;s_i)$ of both the cost *p* and its data s_i for a bidder involved in a share auction. An ideal approach can be defined as follows under symmetric equilibrium. If each bidder, other than bidder *i*, uses the strategy, then when bidder *i* uses a strategy *y* the selling price *p** is the value for which

$$\sum_j X(p^*; s_j) + y(p^*; s_i) = 1.$$

Note that bidders must consider the fact that only the remaining supply, defined by complete supply minus the demand of others, and the chance that they may impact *p** by selecting a different bid schedule.

Recognising the bidding strategy through the remaining supply function and the results of numerical calculation, the estimation model demonstrates an important finding. The issuer may face more significant income loss in multi-unit auctions than in single-unit auctions. Based on the performance comparison between single-unit and multi-unit auctions, Wilson (1979) points out intuitively that the shift in the auction method has an insignificant influence on the results. The reason is that bidders anticipate the change of auction setting, changing their policies with the same distribution and payment as before.

Several studies suggest that predictions in single-unit auctions are not applicable to multi-unit auctions. Back and Zender (1993) suggest that auctions for divisible products are significantly different from auctions for indivisible products, or auctions where each bidder requires only one piece of product. In multi-unit auctions, bidders are

more concerned with marginal cost rather than price. Since the demand function determines marginal cost, bidders can make it very high for competitors through submitting very steep demand curves in single-price auctions. A drawback is that Back and Zender (1993) only analyse one-shoot auctions, whereas many distinct results can be achieved if bidders compete in repeated game auctions, like in treasury auctions.

Using a theoretical auction model, Engelbrecht-Wiggans and Kahn (1998a, 1998b) demonstrate that auctions where participants can buy more than one unit of the goods sold differ significantly from multi-unit auctions where participants can buy only one unit. They find that in multiple-price auctions, bidders shade bids more in the highest bid and less in the lowest bid. On the other hand, bidders in uniform-price auctions shade bids more in the lowest bid and less in the highest bid. The authors find that in multiple-price auctions bidders merge bidding at the same price even when the goods have different valuations. They conclude that both auction formats are inefficient, with revenue equivalence among them not working.

Using laboratory experiments, Engelmann and Grimm (2009) examine revenue equivalence between uniform, Ausubel, Vickrey, and discriminatory sealed-bid auctions, and test the demand reduction for sealed and open bid in single-price auctions. They find that demand reduction exists in single-price auctions, achieving both the lowest revenue and the lowest efficiency. The Ausubel auction appears to be the best mechanism in terms of efficiency output if participants have time to gain experience. In terms of revenue, sealed-bid auctions generate the best result due to over-bidding in the uniform-price sealed-bid and Vickery auctions, and the fact that offers in the discriminatory-price auction usually exceed the equilibrium offer. The authors suggest that whether uniform-

price or discriminatory-price auction outperforms cannot be decided based on theoretical arguments alone.

Ausubel et al. (2014) examine efficiency and revenue rankings between uniform-price and discriminatory-price auctions. They test the impact at a multi-unit demand setting in single-unit auctions and constant marginal utility in multi-unit auctions, as well as decreasing marginal utility when bidders' value is assumed to be symmetric and to decrease linearly. They find that both revenue and efficiency rankings between uniform-price and discriminatory-price auctions are ambiguous. However, discriminatory-price auctions outperform uniform-price auctions when bidders are symmetric in diminishing marginal utility and linearity. The authors argue that a single-price auction generates inefficient outcomes because of the presence of bid shading and because each bidder submits bids with different bid shading. Differential bid-shade incentives exist when a winner is likely to influence the price of market clearing and has greater incentives to shade bids in uniform-price auctions than in discriminatory-price auctions. However, when competition and innovation are promoted by the seller to encourage small participants to participate, single-price auctions are not always less efficient than multiple-price auctions. In single-price auctions, the level-playing field is the same for all bidders since the winner's curse phenomena is eliminated.

3.3 Empirical Literature on Multi-unit Auctions

The debate about which auction type can better serve a government treasury market have started early. In the US, the debate started when the former treasury method of selling, i.e., discriminatory-price auction, was criticised by academics. One of the proponents of uniform-price auctions in the US treasury market is Milton Friedman from the University of Chicago. He presented the idea to shift to the uniform-price auction in

1959 (Brimmer, 1962), claiming that bidders in discriminatory-price auctions are induced to shade their bids down, a problem which uniform-price auctions can eliminate (Friedman, 1959). Brimmer (1962), Rieber (1964) and Goldstein (1962), amongst others, found flaws in Friedman's presentation. Nevertheless, the debate has remained inconclusive till today.

Countries are polarised in employing the auction method for selling their treasuries. Several countries, including the US, Finland, and Switzerland employ uniform-price auctions to sell their treasuries. In the US, the decision to engage the uniform-price method was made after several years of experimentation. Nevertheless, many countries still employ discriminatory-price auctions, including the UK, Canada, Italy, Indonesia, Germany, and Sweden. Brenner et al. (2009) note that uniform-price auctions are used more widely in developed and common law countries. Developing and civil law countries tend to adopt the discriminatory-price auction design.

Empirical studies comparing the two auction types for treasuries have been carried out mostly on the US. The US initially employed the discriminatory-price auction mechanism, experimenting with the uniform-price auction since 1992 (Nyborg & Sundaresan, 1996). The US completely shifted to the uniform-price auction mechanism in August 1998. Even when economically small, the uniform-price auction tends to increase the revenue for the issuer through lower underpricing (Goldreich, 2007). However, the benefit of the uniform-price auction may well have been oversold because the evidence is relatively weak and unsubstantiated (Binmore & Swierzbinski, 2000). In general, the extant literature suggests that replacing the multiple-price auction format with a single-price auction does not improve the average revenues for the seller (Binmore & Swierzbinski, 2000).

There are three key empirical approaches to comparing auction revenue between discriminatory-price auctions and uniform-price auctions: laboratory experiments, natural experiments, and structural approaches. I will discuss each of these approaches next.

3.3.1 Laboratory Experiments

Laboratory research is complementary to natural experiments. The studies of Goswami, Noe and Rebello (1996) and Smith (1966) are most often referred to. In the laboratory experiment, the subjects, generally students, participate in an auction that is designed by the researcher. The auction process is simplified, with several factors, such as the value of goods, details of the auction institution, and available information, controlled by the researcher.

Goswami et al. (1996) find that auctioneers' surpluses are higher for uniform-price auctions than for discriminatory-price auctions when there is no communication between bidders. The surplus is lower for uniform-price auctions when communication is allowed. A similar experimental study conducted by Smith (1966) suggests that in a laboratory experimental settings, bidders in uniform-price auctions are more aggressive than those in discriminatory-price auctions. The average revenue is higher for uniform-price auctions; the average revenue per unit in discriminatory-price auctions is 88 percent from USD 1.55 on average winning price. For the uniform-price auctions, revenue increases to 96 percent from average winning price.

Using laboratory experiments, Binmore and Swierzbinski (2000) also find that bidders submit bids more aggressively in uniform-price auctions than in discriminatory-price auctions. In several cases, the bidding price in a uniform-price auction is higher than

the average market price in the secondary market. They conclude that uniform-price auctions yield higher revenue for the auctioneer than discriminatory-price auctions.

3.3.2 Natural Experiments

Natural experiments examine the coefficient of a dummy variable indicating a change in the auction design. The dependent variable in the regression model is a proxy for profit or mark-up by bidders. A significant coefficient on the dummy suggests that the specified auction mechanism (which takes the value of one) influences issuer revenue. The sign of the coefficient indicates which auction design generates higher average revenue. The natural experiment relies on data from both auction designs. The proxy for profit or mark-up is usually taken as the disparity between the secondary market price and the weighted average winning price in the auction. The key assumption of this approach is that secondary market prices reflect the true value of a security. This approach also assumes that one can control factors which may influence the security's price between the end of the auction and the start of trade in the secondary market.

There are several studies that adopt the natural experiment, examining revenue changes after a switch in auction design. A summary of previous studies is presented in Table 3.1. The results appear mixed across countries and study periods. Umlauf (1993) investigates the bidding behaviour in the Mexican treasury bill auction from 1986 to 1991. Mexico substituted discriminatory-price auction for uniform-price auction design in 1990 to eliminate collusion and to boost auction revenue. He argues that information asymmetries between small and large bidders in discriminatory-price auctions encourages collusions among dominant players. Bidders in Mexican treasury auctions estimate the existence of the winner's curse and submit bids cautiously when uncertainty arises. He

finds that after the treasury switched auction design from a discriminatory-price auction to uniform-price auction format, profits to the bidders declined significantly.

Simon (1994) examines whether uniform-price auctions lowered revenue to US treasury from 1973 to 1976. He applies a simple regression to test the profit of auctions over when-issued rates shortly after the auctions were completed by holding constant the effect of other factors. He finds that the borrowing cost of treasuries in uniform-price auctions is higher than in discriminatory-price auctions. He also finds that there is a positive relationship between the volatility of the bond market and the profits at the auctions. He concludes that discriminatory-price auctions generate higher average revenue for the Treasury relative to uniform-price auctions.

Using when-issued market data from the US treasury from July 1992 to August 1993, Nyborg and Sundaresan (1996) compare uniform-price and discriminatory-price auctions on when-issued market trading volume. When-issued markets exist within the period between the auction announcement date and the settlement date of auction results. Dealers and investors actively trade in the when-issued market before the Treasury's planned auction date. They find that under uniform-price auction format, the when-issued trading volume is larger than that under the discriminatory-price auction format. This suggests that the existence of the winner's curse in discriminatory-price auctions affects when-issued market trading volume. Further, they find higher information releases decrease uncertainty before the auction; volatility in the when-issued market drops after the auction date (and the announcement of auction results) under uniform-price auction format. On the other hand, volatility rises under discriminatory-price auction format in the same period. These findings suggest that uniform-price auctions increase pre-auction information and reduce the short squeeze.

Malvey and Archibald (1998) compare the performance of uniform-price auctions with discriminatory-price auctions for the 2-year, 3-year, 5-year and 10-year US treasury notes from 1990 to 1998. They test the impact of auction design by examining differences between bidding yields in the auction and the yield in the when-issued market. They find that uniform-price auctions generate higher than expected revenue than discriminatory-price auctions. The average auction yield spread in discriminatory-price auctions is statically significantly different from zero, but not in uniform-price auctions.

Berg et al. (2000) compare the performance of the uniform-price auctions (bonds) and discriminatory-price auction (bills) for Norwegian treasures from 1991 to 1996. They compare the average yield in treasury bonds and treasury bill auctions, while disregarding differences of the type of security issued. They find that uniform-price auctions generate higher revenue than discriminatory-price auctions, with the average difference amounting to 24 basis points.

Heller and Lengwiler (2001) investigate the impact on revenue on the Swiss treasury, when replacing uniform-price auctions with discriminatory-price auctions. Using treasury bond data from 1993 to 2000, they find higher than expected revenue in uniform-price auctions and conclude that the Swiss treasury has made the right decision by switching to a uniform-price auction format. They argue that shifting to the discriminatory-price auction will bring a more expensive cost of financing for the government. They also ascertain that the uniform-price auction is a simpler method since bidders do not have to work out how the other bidders behave, thus not requiring any strategic bidding.

Keloharju, Nyborg and Rydqvist (2005) study uniform-price treasury auctions in Finland from 1992 to 1999. Using bidder-level data, they find that bidders' behaviour is

influenced by the volatility of bond returns. An increase in volatility reduces the quantity demanded, and increases the price discount, as well as the dispersion of bids. They compare uniform-price auctions in Finland with discriminatory-price auctions in Sweden. They find fairly strong evidence that underpricing is lower in the Finnish sample and explain that this is due to Finnish market volatility being lower than that in Sweden. The results also indicate that uniform-price auctions have lower underpricing than discriminatory-price auctions. They conclude that volatility has an important effect on bidders' behaviour and that uniform-price auctions outperform discriminatory-price auctions by decreasing underpricing.

Goldreich (2007) examines a sample of 283 US treasury notes and bond auctions between June 1991 and December 2000, of which 105 are discriminatory-price and 178 are uniform-price auctions. The author compares the winning price in the auctions with secondary market price for the same or identical securities. He finds that the treasury's average price is less than the price in the secondary market for the same securities. He also finds that in uniform-price auctions, underpricing is reduced by half, compared to discriminatory-price auctions.

Table 3.1: ***Summary of Empirical Studies Based on Natural Experiments.***

Note: * differences between 0.12 - 1.5 percent. Source: author's table based on Monostori (2014) and Marszalec (2017).

Study	**Market**	**Time Period**	**Superiority***	
			Discriminatory	**Uniform**
Tsao and Vignola (1977)	US treasury bills	1973-1976		x
Umlauf (1993)	Mexico treasury bills	1986-1991		x
Simon (1994)	US treasury bills	1973-1976	x	
Archibald et al (1995)	US treasury bonds	1991-1995		x
Nyborg and Sundaresan (1996)	US treasury bills and bonds	1992-1993		Difficult to decide
Malvey and Archibald (1998)	US treasury bonds	1990-1998		x
Berg et al. (2000)	Norwegian treasury bills and bonds	1991-1996		x
Heller and Lengwiler (2001)	Swiss treasury bonds	1993-2000		x
Goldriech (2007)	US treasury bonds	1991-2000		x

3.3.3 Structural Approach

The third method used to compare revenue performance between discriminatory-price auctions and uniform-price auctions is the structural approach. The advantage of this approach is that it can provide an analysis of the bidding behaviour at the bidder level. The structural approach derives the best condition to observe bidding data. This method estimates the distribution and value of several unobservable. Table 3.2 provides a summary of previous empirical studies using a structural approach.

Fevrier, Preget, and Visser (2004) employ a structural econometric estimation to show that a two-step estimation procedure can calculate the merge distribution utility of the bidder' signals and the value of the good. They apply this approach to French treasury auction data in 1995. Using two procedures. Firstly, they utilise the association between the distribution of values from the structural model and the distribution of observed values to calculate the distribution of observable shares, which are input estimates in their first-order condition. Secondly, they use the fact that the Euler condition can create restriction moments. They find that discriminatory-price auctions generate on average a 5 percent points increase in revenue compared to uniform-price auctions. The ex-post estimation results may justify the reality that discriminatory-price auctions are more popular to sell treasuries.

Kang and Puller (2008) compare the revenue from uniform-price and discriminatory-price auctions counterfactually using data from Korean treasury auctions from September 1999 to April 2002. They use a model of structural equilibrium bid to compare revenue performance using the Vickrey auction as the benchmark. They find that the yield of discriminatory-price auctions is higher in terms of expected revenue and has a more efficient allocation of securities. However, these effects are economically small. It appears that the Korean treasury changed auction design from a discriminatory-

price auction to a uniform-price auction to encourage bidders to submit bids more aggressive and to improve participation rates.

Hortaçsu and McAdams (2010) perform a counterfactual analysis of 13-week treasury bills in Turkish treasury auctions between 1991 and 1993. They examine whether switching the auction format from a discriminatory-price auction to a uniform-price auction provides higher revenue. They estimate the marginal value of bidders using individual bid-level data. The difference between the auction price and the resale market price of the security across various samples of discriminatory and uniform-price auctions are then calculated. They find that issuers' revenue is not significantly increased by switching to the auction format. They argue that discriminatory-price auctions are closer to being fully efficient when compared to uniform-price auctions since in the latter format, bidders bid less than the true value in equilibrium. They suggest that bidders expected surplus is increased by at most 0.02 percent over the 3-year period by switching to a more efficient mechanism.

Kastl (2011) uses Czech Treasury bill auction data from November 1999 to December 2000 to examine a model of uniform-price auctions of a perfectly divisible good with private information. He shows that auction participants influence the equilibrium price; there is a correlation between an auction participant who has private information in multi-unit auctions and the bidding behaviour of oligopolies who deal with the uncertainty of demand. He suggests that the performance evaluation of uniform-price auctions versus discriminatory-price auctions should involve a direct comparison because it is easier than using hypothetical data simulations. Theoretically, bidders submit bids equal to their marginal valuation but empirically, bidders may submit bids higher than their marginal valuation when bid functions have the finite number of steps. He proposes

a new approach to analyse the performance of auction designs by calculating the effectiveness of the outcomes and the efficiency of allocation. He finds that the uniform-price auction treasury allocation is nearly efficient and extracts all auction participants' value.

Marszalec (2017) uses Polish Treasury bill auctions data to compare the performance of uniform-price auctions and discriminatory-price auctions. He employs two structural econometric models, i.e., Hortaçsu and McAdams (2010) (HMA, henceforth) and Fevrier et al. (2004) (FPV, henceforth) to investigate revenue superiority between the two auctions designs. He then estimates reciprocal stability results from the two models which generate revenue. He finds that discriminatory-price auctions generate more revenue than uniform-price auctions by approximately 0.12 percent on 52-week treasury bills and 1.5 percent on 2-year treasury bonds under the FPV model. The result from the FPV model is supported by the expected revenue from the HMA model, although the superiority of discriminatory-price auctions to uniform-price auctions is economically small. Lastly, he finds that both models generate results which are over 80 percent consistent with predictions, suggesting that the data drive the conclusions of the study, rather than the model.

Table 3.2: ***Summary of Empirical Studies Based on Structural Approach.***

Note: * differences between 0.12 - 1.5 percent. Source: author's table based on Monostori (2014) and Marszalec (2017).

Study	**Market**	**Time Period**	**Superiority***	
			Discriminatory	**Uniform**
Hortaçsu (2002)	Turkish treasury bills	1991-1993	x	
Fevrier, Preget and Visser (2004)	French treasury bills and bonds	1995	x	
Armantier and Sbai (2006)	French treasury bills and bonds	1998-2000		x
Castellanos and Oviedo (2008)	Mexico treasury bills	2001-2002		x
Kang and Puller (2008)	Korean treasury bonds	1999-2002	x	
Hortaçsu and McAdams (2010)	Turkish treasury bills	1991-1993	x	
Kastl (2011)	Czech treasury bills	1999-2000		x
Marszalec (2017)	Polish treasury bills	2004-2007	x	

3.4 Determinants of Auction Performance

3.4.1 Competition

With the exception of Burns (1985), few empirical studies have been published on the relationship between the number of bidders and the expected price at the auction. However, there are several theoretical studies dealing with this topic. Vickrey (1961) shows that under a collection of strong competition, as the number of bidder increases, each bid in a multi-unit progressive auction bidding model becomes closer to the value or reserve price of each bidder. Mcafee and Mcmillan (1987) discuss auctions with a stochastic number of bidders, where a certain number of actual bidders is likely to be found depending on the number of possible bidders. In an experimental environment, Burns (1985) measures buyers' reaction to shifts in the number of market participants. He finds that contrary to traditional standards, fewer bidders tend to lead to higher prices.

Multi-unit auctions are commonly used in practice and have rapidly enriched theoretical models, empirical studies, and experimental auctions (Klemperer, 2002). However, there is no research specifically analysing the effect of the competition structure on multi-unit auction results. For instance, in treasury auctions, the auctioneer chooses a target group to which the items will be sold or purchased and select the target group with the highest competition to provide maximum benefit. The level of competition can vary, depending on target group size, number of participants in the auction, and the level of demand to the goods being auctioned (McAfee & McMillan, 1987).

Similarly, if the auctioneer defines a unit's size as the minimum indivisible quantity that can be bid or purchased, this may affect the structure of competition (Pycia & Woodward, 2016). Thus, if a unit's size is small enough, more potential buyers will be

able to participate in the auction; in this case, larger buyers will want more units. Conversely, if a unit's size is large, only the big bidders can participate in the auction, unless the smaller buyers join to bid. Bidding strategies are responsive to the nature of the market: for a constant degree of competition, with a small number of large bidders, there is more strategic bidding than with a large number of small bidders (List & Jason, 1999). A higher variance of bids for the seller yield lowers expected revenue.

A pure common-value second-price auction model may help us better understand the market design auction of those involved in an environment where some bidders are known ex-ante to have better access to common-value information. Perhaps unexpectedly, some questions have been left open in the existing literature. For example, while it is well understood that the existence of an informationally advantageous bidder can reduce seller revenue moderately for a common-value item in a sealed-bid first-price (FPA) auction (Engelbrecht-wiggans et al., 1983; Hendricks & Porter, 1989; Milgrom & Weber, 1982; Weverbergh, 1979; Robert Wilson, 1977), in the background of second-price auctions, much less is known about the same problem. The well-known multiplicity of Bayesian Nash Equilibria in second-price common-value auctions has been one of the main obstacles to further our understanding (Milgrom, 1981). There are a number of explanations, as discussed below.

A larger dispersion in opinion among bidders indicates less severe competition since the bidder's valuations are just below the true value (Cammack, 1991). Bidders can lower their bids without severely diminishing their probability to win in the auction, thereby capturing a higher expected gain from average bidders' winning. The marginal probability of losing an auction from lowering the bid by a given amount increases with

increasing levels of competition. Therefore, increased competition on average results in higher bids and lower profits when there are a finite number of bidders.

Demand reduction reduces revenue and creates inefficiencies as bidders with lower-value units receive goods of higher value bidders (Ausubel et al., 2014). Krishna (2002) proves asymmetrical equilibrium to bid one's true value for the first unit and zero for the second if two units are available for sale and two bidders have ordered values drawn independently of a uniform distribution. Therefore, every bidder wins one unit in every realisation and the price is zero. In such cases, allocative efficiency is not guaranteed as the private value of one of the two bidders' first unit may be lower than the other bidder's second private value, even though he won.

Although structural properties of equilibrium bidding strategies can be deduced in simple examples, the existing literature offers little insight into their analytical characterisation. Multiple equilibria exist when there are more than two units and more than two bidders in the auction. Since the theoretical literature cannot provide a straightforward answer, a flurry of experimental studies has followed.

Most experimental auctions compare demand reduction in various multi-unit auction formats (Alsemgeest et al., 1998; Engelmann & Grimm, 2009; Kagel & Levin, 2001; Levin et al., 2005; Porter & Vragov, 2006). As the theory predicts, there is more demand reduction in the uniform-price open auction than in the uniform-price sealed-bid auction. However, Engelbrecht-Wiggans, List, and Reiley (2006) note that when the number of bidders increases, demand decreases but does not vanish. Nonetheless, these papers offer little insight into the essence of demand and suggest that theoretical explanation cannot decide between uniform-price and pay-as-bid auctions as the preferable format.

Coent, Préget, and Thoyer (2014) run an experiment on multi-unit auctions organised under two different competition structures. In the first structure, the auction includes six bidders with small individual demand for two units. In the second structure, the auction is characterised by two bidders with large individual demand for six units each. They find that aggregate demand and supply are the same in both structures. They demonstrate that Krishna's result (2002) cannot be extended to the cases when there are more than two bidders or when their individual demand is greater than two units. They deduce that in both structures, the auction revenue could potentially fall to zero if bidders were to coordinate on such equilibria. However, other revenue and efficiency outcomes can be expected, given the multiplicity of potential equilibria. They run controlled laboratory experiments to compare the efficiency outcomes of the two auctions. The main conclusion is that with a constant degree of competition, there is less strategic bidding (less demand reduction) when there are many bidders with a small demand than when there are fewer large bidders. This increases the expected revenue of the auctioneer and reduces the variance of bids. They also show that allocative efficiency is not significantly different under the two demand structures. Thus, it is important to consider the competition structure when designing a multi-unit auction (Klemperer, 2003). While the seller is typically unable to raise the level of competition, he can decide who is eligible to bid and the size of the unit, i.e., the minimum sum that can be bid on. The performance of the multi-unit auction system is all the more significant because the total demand is not concentrated in the hands of a few bidders, but is distributed evenly across more bidders (Klemperer, 2002).

The critical analysis between competition and auction performance is about whether the degree of competition affects the revenue and efficiency of the auction

(McAfee & McMillan, 1987). In multi-unit auctions, the competition structure depends on the number of bidders and bids, and the structure of each individual bidding. Furthermore, for the same degree of competition, i.e., identical aggregate demand and supply, the competition structure may be different. In other words, a Treasury auction with many small bidders and bids may generate more revenue and be more efficient than a multi-unit with few large bidders and bids.

In the treasury auction literature, the answer to the above question is rather vague. Previous explanations focus mainly on the impact of the auction format on auction demand. The demand reduction phenomenon exists for a given structure of competition, i.e., many indivisible units are put on sale and bidders submit a bid for each unit they want in the auction. There is an incentive for bidders who have a non-increasing multi-unit demand to bid under their true valuation (i.e., underbidding or bid shading) on all units but the first (the highest value unit). Bidders have market power in the uniform-price auction because their bid potentially determines the price they will have to pay. There is an increasing amount of bid shading occurring on subsequent units, but not for the first unit demanded (Engelbrecht-Wiggans & Kahn, 1998b; Noussair, 1995).

3.4.2 Uncertainty

Common-value auction theory states that if bidders are not sufficiently experienced, volatility in the bond market price can have a serious deleterious effect on the auction results (Keloharju et al., 2005). If a bidder considers his forecast of the market price to be greater than that of any of his rival bidders, he is likely to believe that his original prediction is too ambitious. A winning bidder may be cursed with unfavourable profits for bidding high in the auction, unless the bidder carefully incorporates information found in rival offers into his own bidding strategy. On average, bidders tend

to respond to fluctuations in the bond market to avoid constant negative gains synonymous with the winner’s curse (Kagel & Levin, 2002). Because of the winner's curse, there is also the possibility of winning ‘sorry’. It is therefore important for policymakers to determine if bidders are sufficiently clever and foresee the winner's curse in their deals (Charness & Levin, 2009).

The literature suggests that at least highly sophisticated and seasoned bidders involved in treasury bond auctions are anticipating the winner's curse. Implicit buyer commitments to retain a low price in the unfair transaction is likely to be more nuanced, which can contribute to higher expected sales (Bjonnes, 2001). Risk factors are likely to be significant determinants of bidding behaviour and return on auction activity. For each bidder, the most relevant source of risk is uncertainty about the aggregate demand: the cut-off price that the government will choose to impose, the covariance between returns, and other assets in the bidder's portfolio. Although risk-aversion decreases the degree of bid filtering, the higher the level of risk-aversion, the higher the expected return (Nautz & Wolfstetter, 1997).

Some of the above explanations are compatible with auction behaviour theories. In some ways, a differential pricing auction (with unit bids) is similar to a single-unit's first-price common-value auction, which has been thoroughly dealt with in the statistical literature as a winning bidder pays the pick offered in each situation. Such a scenario will give rise to the winner's disadvantage since winning means that all other bidders have a lower value of the item being offered. Therefore, on average the winner has an over-optimistic estimate of the worth of the product.

Bids are shaded downwards in order to compensate for the possibility of the winner’s curse (Hortaçsu et al., 2018). A greater dispersion in opinions on the true value

of the sale item and the cut-off point, as shown by the dispersion of bids for each bidder or across bidders, would lead to greater bid shading (Elsinger et al., 2007). The presence of more bidders intensifies the winner's curse, but at the same time increase competition. Hence, an exogenous increase in the number of bidders may increase or decrease profits, although the expected profits should eventually go to zero.

For a multi-unit auction, one may speculate on the winner's curse. Winning price does not reflect the true value of the information if more units are sold and each bid is accepted. Therefore, compensation for the winner's curse should be minimised with the share of bids reached. Bikhchandani and Huang (1989) set out a multi-unit auction model with a resale market. Although the bids are restricted to unit amounts, the authors demonstrate that bidders would still shadow their offers. On the basis of this model, Spindt and Stolz (1992) suggest that the estimated cut-off price will grow with the number of bidders and decrease with the amount sold. Nevertheless, as stated by Back and Zender (1993), the degree to which the effects of this literature on unit-bid transactions bear on the situation for multiple bids is unclear. Wilson (1979) indicates that the expected revenue for the seller and the expected profitability rate for the purchaser can be independent of the number of bidders in an auction.

3.5 Islamic and Conventional Financing Instruments

Raei & Cakir (2007) using a sample of sovereign Sukuk and Eurobond from the same issuer, to estimate and compare value-at-risk (VaR) for a portfolio that includes both instruments to a pure Eurobond portfolio. They examine the risk-reduction advantages of issuing sovereign Sukuk as alternative financing instruments compared to conventional sovereign bonds. They find that the VaR is reduced when Sukuk are added to the portfolio of fixed-income securities. They suggest that Sukuk are truly different

from conventional bonds and that these investment certificates create diversification benefits for investors. However, they do not conduct micro-level empirical analyses on domestic market with regularly issuance such as auction.

Miller et al (2007) claim that Sukuk and conventional bond are structured similarly so they will provide an equivalent return, however the difference is that the return on the Sukuk is not generated from the obligation to pay interest but an underlying asset.

Similarly, Wilson (2008) argues that the similar structure between Sukuk and conventional securities was made to help unfamiliar investors assess the risk of Sukuk which is still a new investment. AAOIFI Shariah Council, Mohammad Taqi Usmani, agreed that current practices of issuing Sukuk replicate the structure of conventional bonds only in terms of right to a fixed return, lack of ownership, and the guarantee of repayment of principal. Other than these, the Shariah scholars oppose that the structuring of Islamic financial instruments was made to please international investors because it can make them conventional interest-based products.

Godlewski et al (2013) find that investor behaviour is different in relation to Sukuk and bond issuance. Sukuk is considered as different instrument type to bonds due to price behaviour differences. The authors advise that incorporating Sukuk along with bonds in the investment portfolio can increase diversification benefits, especially for international fixed-income investors.

Kabir et al. (2015) examine credit risk in Islamic and conventional banks by employing market-based and accounting-based credit risk measurements. Using data of banks in 13 countries during 2000-2012, they find that Islamic banks have lower credit

risk than conventional banks. Hence, they cannot find any significant difference in credit risk between the two banking systems during the most recent financial crisis. Any policy and regulations regarding credit risk management are necessarily country specific. These findings suggest that investors and depositors should not have any predetermined position regarding the credit risk level in each banking system.

Godlewski et al. (2016) analyse the stock market reaction to Sukuk type and characteristics of the Shariah scholar certifying the issue, by employing event study methodology to measure abnormal returns for a sample of 131 Sukuk from eight countries over the period 2006–2013. They find that the announcements of conventional bond issues have neutral reaction for the stock market while for the announcements of Sukuk issues the market reacts negatively. This conclusion is taken through the Islamic investment certificates excess demand and to an adverse selection mechanism that shows Sukuk issuance is favour by lower-quality debtor companies. Investors expect that an adverse selection mechanism may encourage less-healthy companies to prefer Sukuk in the form of profit-and-loss sharing schemes over conventional bond financing. Companies with high profit expectations will opt for conventional bonds with a fixed repayment schedule that maximizes the upside potential, however companies with low profit expectations have incentives to finance their project through Sukuk arrangements.

Klein et al. (2017) examine whether trade on a stock market is affected by religious bias that is the month of Ramadan. They use a sample of 2140 issuances by 133 issuers in Malaysia over the period 2000–2013, and analyse the data using event study methodology. Their results show that religiosity affects stock returns, with a positive effect of the month of Ramadan on stock returns following a Sukuk issue, and negatively affects bond issues. Nevertheless, comparing firms which issue only Sukuk and issue both

Sukuk and conventional bonds, the authors find that investors react positively during Ramadan only to firms that issue Sukuk exclusively. However, there is no religious bias if the firms issue both Sukuk and conventional bonds.

Using the data of the Malaysian government securities return from January 31, 2007 to December 31, 2013, Haque et al. (2018) analyse the behavior of the securities return for conventional and Islamic. They examine the Granger-causality and lead–lag relationship between Sukuk and bond and find that Sukuk securities and conventional bonds have a causal relationship for a given period of time and that in the long-term, bond is led by Sukuk investment rather than in the short term. From the investor's perspective, Sukuks are more favourable than bonds in the long term (with maturities 1 year, 3 years and 5 years) for a given period, because the mean yield of Sukuk tends to be higher than of conventional bonds in the long run and on an average Sukuk are less volatile compared to bonds.

Hassan et al. (2018) examine the long-run relationship and dynamic correlations between major fixed-income markets and the Sukuk market in the sample period of 2010 to 2014. The data sample of major fixed income markets is collected from U.S., Europe, and emerging markets, for both investment-grade and high- yield fixed income. Meanwhile, the Sukuk market data gathered by building a cross-country index of highly liquid listed issues. The authors employ the Johansen cointegration test and the DCC-GARCH model under Student's t-distribution. They suggest that Sukuk and conventional bonds have valuable similarities due to their similar reaction to market shocks between Sukuk and investment grade bonds, that are lower reaction of conditional volatility to market shocks and a higher persistence. Nevertheless, Sukuk also provides impressive growth and diversification benefits due to their lower volatility and dynamic correlation

paths. Overall, the authors conclude that Sukuk has a challenge because its lower liquidity even though long-term institutional investors are suited more to this investment.

Comparing the financial performance of Islamic and conventional banking in Pakistan during 2006 to 2015, Khan et al. (2018) find that conventional banks are more profitable than Islamic banks. They suggest that conventional banks have a wider network compared to Islamic banks, which results to a better return on average equity ratio and return on average asset due to wide shares in financial markets. This also resulted in a better profit expense margin and net profit margin because conventional banks have superiority in managing assets for income generation.

Bhuiyan et al. (2020) compare Sukuk and conventional bond markets based on the value-at-risk (VaR) in developed and emerging market from January 2010 to December 2015. A portfolio's VaR of Sukuk and bond indices is taken to examine if the inclusion of the Sukuk index in the portfolio may reduce the VaR amount. The finding suggests that compared to the VaR of bond indices, the VaR amount of Sukuk indices is much lower by around 30 to 50 per cent for all the emerging and developed market bond indices. They conclude that the reduction occurs as a result of the different behaviour of Sukuk prices compared with conventional bonds in the secondary market. The authors claim that the VaR amount of Sukuk indices is much lower than the VaR of bond because the Islamic investments are better positioned due to its specific features, such as the limitation of interest-based leverage, ethical and ratio screenings, including the exclusion of highly leveraged firms, financial sector, and complicated structured of financial products or derivatives.

Khartabiel et al. (2020) examine issuance announcement effect on shareholder wealth. The study compares multiple economic settings, that are pre-, during, and post-

crisis, with a benchmark of recent financial crisis (2007–2009). By employing 237 Islamic bonds (Sukuk) versus 231 conventional bonds issued in twelve financial markets from 2005 to 2017, they use multiple regression model to identify the potential explanatory variables for the wealth effects. The finding suggests that an insignificant market reaction occurs, in the pre-global financial crisis period of 2008, for the announcements of Sukuk and conventional bonds. Market reaction is negative and significant for both groups during the crisis, while in the post-crisis period the reaction of the market for Sukuk is significantly positive. The positive reaction may be a result of market participants' awareness and increased demand for Islamic financial products, whereas in conventional bonds the market reaction is insignificant.

According to the literature, Sukuk are more competitive compared to conventional bonds. This is due to: (1) the possibility of obtaining higher revenue sharing than conventional bonds, (2) secure as Islamic bonds to fund prospective projects, (3) in the event of a loss (out of control) investors retain their assets, and (4) a breakthrough paradigm, not more debt but investment letters. While Sukuk has become one of the central features of Islamic finance so far, it has also been challenged by Islamic scholars in terms of its application in modern setting (Chowdhury et al., 2017). Sukuk is sometimes criticized as conventional bond due to its excessive replication from its counterpart (Wilson, 2004). Usmani (2007) criticized the existing Sukuk issues in terms of their violation of the Sharia standards and declared majority of today's Sukuk were very similar to the conventional bonds. The different Sukuk structures can be grouped into two main categories, namely, asset-backed and asset-based Sukuk (Amrani et al., 2017).

In fact, for the first category, the realization of the financing operation inevitably involves the actual transfer of ownership of the balance sheet, issuer to the subscriber

without there being a guarantee of either the principal or any losses that will result from the poor performance of the underlying asset. In this, it appears that this financial arrangement presents two financial risks, in this case, the market risk and the credit risk. In contrast, the second category allows the issuer of Sukuk to drain the funds it needs without the constraint of actual transfer of ownership but with a guarantee of the principal by the redemption of the property at maturity and a periodic payment of the coupons. Sukuk cases do not differ from conventional bonds in the sense that the credit risk that accompanies this transaction is the same as in the case of conventional bonds. This is why lawyers and specialists in Islamic finance consider Sukuk Asset-Backed to be closer to the spirit of Islamic finance than the Sukuk asset-based that do not give full ownership over the underlying assets.

3.6 Chapter Summary

This chapter discussed both extant theoretical and empirical literature on treasury auctions. Treasury auctions are categorised as multi-unit auctions and the development of theoretical models for these auctions is reviewed. Prior empirical studies primarily examine differences between uniform-price and discriminatory-price auctions for treasuries. These studies are discussed based on the approach used, namely laboratory experiment, natural experiment, and the structural approach. Comparison of the finding leads to inconclusive results: uniform-price auctions are generally found to be superior in laboratory and natural experiments, while discriminatory-price auctions are generally found to be superior in the structural approach.

The auction theory literature begins with Vickrey's *Revenue Equivalence Theorem*, proposing that the expected selling revenue is identical for all four known auction mechanisms when bidders are risk-neutral and have private value, and sellers sell a single item. This theorem, however, better suits single-unit auctions and is less applicable to multi-unit auctions. Several theoretical models have been proposed, but none are able to be applied to multi-unit auctions since they focus more on improving the effectiveness of single bids submitted by bidders. Further, multi-unit auction bidders tend to shade their bids under risk-averse conditions.

The problem of bid shading is partially solved by Wilson (1979) by introducing a prediction method where sellers may experience a significant reduction in multi-unit auctions, compared to single-unit auctions. However, this method has been heavily criticised for its limited application to multi-unit auctions. Bidders in multi-unit auctions are more commonly concerned about marginal cost rather than bidding price. They compare the highest accepted bid price to the lowest accepted bid price. Therefore, an

examination of the literature reveals mixed guidance in terms of which auction type is the preferred one for treasuries in terms of both revenue and efficiency.

In term of Islamic financing instruments, Sukuk does not differ from conventional bonds in the sense that the credit risk that accompanies this transaction is the same as in the case of conventional bonds. Sukuk asset-backed to be closer to the spirit of Islamic finance than the Sukuk asset-based that do not give full ownership over the underlying assets. Also, there is a lack of Sukuk study on micro-level empirical analyses such in a regular auction.

Chapter 4

Theoretical Framework and Hypotheses

4.1 Introduction

This chapter presents the theoretical framework and develops testable hypotheses linking treasury auction performance with the auction method. In Section 4.2, I predict that uniform-price auctions generate higher revenue than discriminatory-price auctions (H1) due to the existence of winner's curse in discriminatory-price auctions. In Section 4.3, I predict that uniform-price auctions are more efficient than discriminatory-price auctions (H2), because the former provides better allocation of bids. In Section 4.4, I hypothesise that the level of competition is negatively related to both revenue and efficiency gain when switching to uniform-price auctions (H3). In Section 4.5, I hypothesise that uncertainty is positively related to both revenue and efficiency gain when switching to uniform-price auctions (H4). This is followed by a summary in Section 4.6.

Analysing which auction type can better serve a government treasury market at a global level is useful as it can provide new insights about this phenomenon in a general context. However, availability of the data is the main driver of the majority of treasury auction studies. Exploring this phenomenon in an emerging market could possibly yield new insights. I focus on a single country, i.e., Indonesia. The reason is not simply because Indonesia is a developing country, but also because it has a dual treasury market, i.e. both conventional and Islamic treasury bonds are issued by the government. This is the first study in a dual treasury auction market. Besides, Indonesia is also my home and place of work. I have considerable knowledge and understanding of the country's treasury market,

and hence an in-depth study of the Indonesian treasury market is both appropriate and well-suited.

4.2 Revenue Superiority

The primary difference between the two most common auction designs is the price paid by the successful bidders (Becker, 2001). In uniform-price auctions, the winning bidders pay the same price, while in discriminatory-price auctions they pay different prices depending on their bids. As such, in discriminatory-price auctions, bidders are more strategic and will use their best effort to win the auction with the lowest price (Nautz & Wolfstetter, 1997). The reason is to maximise their profit and avoid excessively high bidding (Gordy, 1999). Bidders do not have such incentives in uniform-price auctions, where bidders using a true market value submit higher bids than in discriminatory-price auction (Damianov & Becker, 2010).

In discriminatory-price auctions, bidders with less analytical capacities shade their bidding to avoid the winner's curse problem (Wang, 2012). Adjusting for the presence of the winner's curse in discriminatory treasury auctions, bidders disperse their bids and reduce the price level at which they bid, thereby decreasing the demand for the issue (Bjonnes, 2001). Because of the absence of the winner's curse in uniform-price auctions, bidders bid more aggressively. Consequently, the market clearing price, on average, is expected to be higher than the weighted average winning price in discriminatory-price auctions.

Several papers argue that participants in uniform-price auctions can support a collusive Nash equilibrium, while such equilibria do not arise in discriminatory-price auctions (Chen & Tauman, 2006; Goswami et al., 1996). However, in Indonesia, a

collusive Nash equilibrium is hard to sustain for several reasons. Firstly, the government is not a price taker since the MoF sets a benchmark price as a reference for accepting or rejecting a bid. Although the government publicly announces its indicative target issuance, it is not mandatory to accept incoming bids. Based on the benchmark price, the MoF can accept or reject some or all incoming bids at the discretion of the MoF. For instance, in the Islamic treasury bond auction, on 26 July 2016 the MoF rejected bids even though the bid prices were all above the benchmark price and the total incoming amount was high (USD 331 million).

Secondly, the MoF prohibits adverse (unethical) action behaviour, including collusive bidding among auction participants, which can potentially result in a drop in revenue for the Indonesian government. The MoF has the option to sanction bidders, including revocation of membership and termination of cooperation association with the government. Specifically, PDs and APs are expected to support the MoF issuance program and not make contradictive activity. For example, on 17 November 2016, the MoF terminated cooperative association with JP Morgan as PDs because the bank released a report which drove investors to sell Indonesian treasuries, potentially raising the yield on Indonesian Treasuries. Based on the above discussion, I predict the following:

H1: In the Indonesian treasuries market, uniform-price auctions generate higher revenue than discriminatory-price auctions.

4.3 Efficiency Superiority

The valuation of bidders' bids distribution affects the efficiency performance of uniform and discriminatory-price auctions (Ausubel et al., 2014). Efficiency is accomplished when the goods end up with the bidders who place the highest bid.

Efficiency loss exists when the marginal value of winning bidders is less than the marginal value of unsuccessful bidders. Efficiency occurs if the marginal value of the winning bid is at least the same as the marginal value of the unsuccessful bids from one bidder. In the auction for private value component goods, issuers should take care whether the highest valuation can be achieved by the buyer for a particular auction design.

In uniform-price auctions, bidders have a lower incentive to do bid shading for more extensive demand quantities when each bidder shades in the same way proportionally. In contrast, in discriminatory-price auctions, the incentive to do bid shading does not necessarily increase with the demand quantity. Still, there are strong incentives to mark down the bids significantly for higher value units than the expected market clearing price. Bidders are sufficiently sophisticated to understand and adjust the winner’s curse. Bidders are shading and dispersing their bids more, and reducing the amount demanded when the number of competing bidders rises. The winner’s curse mainly drives the marginal effect of the additional bidders. Bid shading increases with the volatility of the bidder's signals, and the dispersion of information is used to hedge against the possibility of the winner's curse (Wu, 2016).

The demand function and the rate of asymmetry across bidders influence the efficiency of auction type. The differential incentives to do bid shading for different units of the goods indicate that inefficient equilibria exist both in uniform-price auctions (Engelbrecht-Wiggans & Kahn, 1998a) and discriminatory-price auctions (Engelbrecht-Wiggans & Kahn, 1998b and Swinkles, 1999). However, under tight competition, bidders in uniform-price auctions have more incentives to submit bids at their true valuation, or the maximum price, since all winning bidders will pay the same price. If bidders submit bids below the true value or the market clearing price, the opportunity to acquire the

security will be lower or lost. In addition, when competition is tight, and demand is higher than supply, highly sophisticated and experienced bidders will submit bids consistent with their valuation. In summary, uniform-price auctions allocate the security to higher bids, more so than in discriminatory-price auctions.

Based on the theoretical link between auction format and efficiency discussed above, I thus predict the following:

H2: In the Indonesian treasuries market, uniform-price auctions are more efficient than discriminatory-price auctions.

4.4 Competition

In markets with a small number of large bidders, there may be incentives for bidders to act strategically, within the limits set by the auction protocol. By reducing the importance of specialised knowledge regarding market demand and the information costs associated with its collection, uniform-price auctions are thought to encourage more aggressive bidding than discriminatory-price auctions. Malvey et al. (1998) find that in a discriminatory-price auction there is a relatively tight distribution of large bids around the auction average price because successful bidders pay the bid price. The distribution of bids is more extensive in uniform price auctions because there is no consequence for submitting bids over the market price unless there is a restriction rule.

With increasing competition, bidders in a uniform-price auction have less incentive for bid shading with increasing demand quantity, with each bidder shading in the same way proportionally. Bidders have more incentive to submit bids with their true valuation, or the maximum price, since all winning bidders will pay the same price. If bidders submit bids below the true value or market clearing price, the opportunity to

acquire the security will be lower or be lost. In contrast, in discriminatory-price auctions, there are strong incentives to mark down the bids for higher value units. Bidders are sufficiently sophisticated to appreciate the winner's curse. When the number of competing bidders rises, they shade and disperse their bids more, and reduce the amount demanded. This is because the winners' curse mainly drives the marginal effect of the additional bidders. There is a positive relationship between the marginal probability of losing an auction by reducing the price of bids and increasing the degree of competition. In general, the degree of competition positively influences bid price and seller revenue.

Previous explanations focus mainly on the impact of the auction format on auction demand. The demand reduction phenomenon exists for a given structure of competition, i.e., many indivisible units are put on sale and bidders submit a bid for each unit they want in the auction. There is an incentive for bidders who have a non-increasing multi-unit demand to bid under her true valuation (bid shading) on all units but the first (the highest value unit). Bidders have market power in the uniform-price auction because their bids potentially determine the price they will have to pay. An increasing amount of bid shading occurs on subsequent units, but not for the first unit demanded (Engelbrecht-Wiggans & Kahn, 1998b; Noussair, 1995).

An increasing number of bidders and bids may reflect increasing competition, thus potentially raising revenue and efficiency for the seller in both auction formats. Each bid in a multi-unit progressive auction bidding model becomes equal to the value or reserve price of each bidder under strong competition as the number of bidders increases (Vickrey, 1961). In contrast to standard expectation, an experimental study shows that fewer bidders results in higher prices (Burns, 1985). The increasing number of bids and bidders raise an interesting question about the impact of competition on revenue and

efficiency gains when switching to a uniform-price auction. Specifically, does the level of competition reduce the revenue and efficiency gaps between discriminatory and uniform-price auctions?

Auction theory explains the relationship between the number of bidders and the expected price at the auction. A certain number of actual bidders are likely to be found depending on the number of possible bidders (McAfee & McMillan, 1987). Ordinarily, the average market price tends to be higher in markets where there are relatively more buyers. In Indonesia, there are a lot of potential bidders, with 20 primary dealers in conventional treasury auctions and 21 auction participants in Islamic treasury auctions. Both PDs and APs submit bids for themselves as well as on behalf of their clients.

The performance of the multi-unit auction system is more significant when total demand is not concentrated in the hands of a few bidders, but is distributed across many bidders (Jackson & Kremer, 2006). The competition structure depends on the number of bidders and bids or the structure of each individual bidding. Furthermore, given the same degree of competition, i.e., identical aggregate demand and supply, there are multiple competition structures. In other words, a treasury auction with many small bidders and bids may generate more revenue and be more efficient than a multi-unit auction with few large bidders and bids.

The degree of competition can vary, depending on the size of the group (number of bidders) and the level of demand by each potential bidder in the group. The seller is typically unable to raise the degree of competition but can decide who is eligible to bid and the size of the unit, i.e., the minimum sum that can be bid on, boosting the expected income. The auctioneer therefore should choose the target group which provides the

highest degree of competition. This condition exists in Indonesian treasury auctions where primary dealers and auction participants are appointed by the MoF.

The level of competition influences the magnitude of bid shading where bidders place a bid below what they believe is the true value of the security. In discriminatory-price auctions, participants do bid shading to maximise their expected profit. This condition is not applicable in uniform-price auctions since all winners pays the same price. However, when the level of competition is high, bidder who submit lower bids have a smaller opportunity to win the bid. Thus, bidders use a higher price to raise the probability of success in the auction. Consequently, a high level of competition reduces profits for bidders but increases revenue for the issuer (Goldreich, 2007).

Undersubscription indicates that the issuer is not able to get enough bids in the treasuries on offer. Undersubscribed auctions potentially decrease governments' credibility, pushing up future borrowing costs (Beetsma & Giuliodori, 2017). If the demand for treasuries is declining, this more than likely reflects investors moving their portfolios into riskier financial assets and consequently they desire less treasuries. Since an optimal portfolio is one that seeks to maximise the expected risk-adjusted return (Sharpe, 1966), auction demand is related to bidders' profit and volatility. This association gives the treasury both a way to understand, predict, and influence auction competitiveness, and a way to reduce bidders' profit.

On the other hand, higher demand is associated with a lower secondary market yield after the auction relative to before the auction. This implies that decreasing demand in the auction is followed by increasing borrowing cost for the government. When the bid-to-cover ratio is high, the uniform-price system is more advantageous (Monostori, 2014). High demand eliminates the opportunity to set up a collusion equilibrium. The

increased volume demand sends the expected revenue from uniform-price auctions above that from discriminatory-price auctions (Bolten, 1973; Friedman, 1959; Milgrom & Weber, 1982).

Demand reduction reduces revenue and creates inefficiencies as bidders with lower-value units receive goods instead of higher value bidders (Ausubel et al., 2014). For instance, Krishna (2002) proves asymmetrical equilibrium to bid one's true value for the first unit and zero for the second if two units are available for sale and two bidders have ordered values drawn independently of a uniform distribution. Therefore, every bidder wins one unit in every realisation, and the price is zero. In such a case, allocative efficiency is not guaranteed, as the private value of one of the two bidders' first unit may be lower than the other bidder's second private value, but it is won. Increasing the number of bidders weakly diminishes incentives for demand reduction in second unit bids (Engelbrecht-Wiggans, List, & Reiley, 2006). Thus, there is less demand reduction when there are many bidders with a small demand than when there are fewer large bidders. This increases the expected revenue of the auctioneer and reduces the variance of bids.

Another motivation is grounded on revenue equivalence theorem, arguing that under certain conditions, various auction formats generate the same expected result to the issuer (Vickrey, 1961, 1962). In other words, discriminatory and uniform-price auctions generate the same revenue and efficiency in an environment setting with high competition. Thus, predicting a negative association between competition and revenue and efficiency gain when switching to a uniform-price auction. Further, weak demand causes an increasing possibility of collusion among bidders in uniform-price auctions. However, this condition is unlikely to exist in Indonesia because MoF is not a price taker

and collusion is not a tolerated activity. If demand is high, the bidders raise their bids and minimise bid shading because the opportunity to win also drops.

Based on the multiple arguments provided above, I predict the following:

H3a: In the Indonesian treasuries market, competition reduces revenue gains, when switching to the uniform-price auction format.

H3b: In the Indonesian treasuries market, competition reduces efficiency gains, when switching to the uniform-price auction format.

4.5 Uncertainty

When determining their bid schedules, bidders face a number of uncertainties, including the cut-off price decided upon by the issuer,[13] the aggregate demand, the dispersal of bids from each bidder and all bids in a specific auction, the value of the treasury, and the expected secondary market price following the auction (Cammack, 1991; Haile, 2003; Holmberg & Wolak, 2018). Price uncertainty influences the performance of a discriminatory-price auction differently from a uniform-price auction. Lu and Yang (2003) find that bidders in a discriminatory-price auction bid more cautiously when market uncertainty is high in order to avoid the winner's curse: bidders lower their bids and disperse their bids more to avoid over-bidding.

Price uncertainty influences the revenue and efficiency level between the single price and multiple prices auctions, with some issuers intentionally introducing quantity uncertainty into their auctions. Randomness eliminates informed bidders' willingness to thwart uninformed bidder entry; thus so they must raise the price paid compared to a non-

[13] In Indonesia, MoF always announces its indicative target issuance.

random distribution or quantity uncertainty (Daripa, 2001). It boosts the Treasury's income from a standardised market auction with random distribution above that of any arbitrary auction. Uniform-price auctions entail more uncertainty than discriminatory-price auctions because more equilibria are possible, with new entrants potentially causing significant price swings. The supply-quantity can often be changed, thus participants employ mixed strategies and assuming private valuation; there is always considerably more uncertainty as to the outcome of the game, which affects revenue and efficiency (Binmore & Swierzbinski, 2000).

The auction format that is more advantageous in a given situation depends on circumstances, in particular market uncertainty (Monostori, 2014). Bidders lower and disperse their bids to avoid over-bidding as a response to increasing market volatility. The difference between bid yield and market yield increases with market volatility because bidders drop their bidding size with increased market volatility (Bikhchandani & Huang, 1993; Cammack, 1991; Keloharju et al., 2005; Nyborg et al., 2002). Bidders change their bidding strategy to generate more profit when market volatility rises (Lu & Yang, 2003). Bidders bid more cautiously in the discriminatory-price auction when market uncertainty is high to avoid the winner's curse. For example, in Mexico, the issuer's revenue increased and bidder's profit dropped when the government replaced discriminatory-price auction with uniform-price auctions (Umlauf, 1993).

Bidders in uniform-price auctions decrease bid shading and enhance the bidding price when they are unsure of the demand from other bidders. In single-price auctions, profit is the gap between the value of the security and the price of the stop-out multiplied by the number of winning units. Because the stop-out price is decided by aggregate demand, it is impossible for a bidder to reduce his bids (Spindt & Stolz, 1989). The

submitted bids would drop the stop-out price unless the volume is relatively large, or many bidders submit similar bids. However, if a bidder decreases his price, it reduces his chance to win—thus bidders would instead submit higher bid prices.

In discriminatory-price auctions, profit is the spread between the value of the security and the submitted bid multiplied by the number of accepted units. Reducing the bid by one unit raises one unit's benefit per good, but whether it reduces the number of winning bids depends on the demand from other bidders. If there is uncertainty about the demand from other bidders, the willingness of a bidder to reduce his offer price would exceed his incentive to increase the bid price. Bidders understand that while lowering their bid would boost their profit per unit, it may also reduce their total winning bids.

Auction theory predicts that in a discriminatory-price auction, a higher dispersion of information results in lower bids and lower revenue for the issuer. When information asymmetry is high, bidders (especially small bidders with less analytical abilities), are inclined to shade their bids to avoid the winner's curse; they disperse their bids and reduce the price level at which they bid, thereby decreasing the revenue for the issuer (Cammack, 1991; Bjonnes, 2001; Wang, 2012). Bidders lower their bids without severely reducing their probability of winning at an auction, thereby capturing a higher than expected gain from average bidders' winnings. Hence, in a discriminatory-price auctions, higher dispersion of information results in lower bids and lower revenue for the issuer. Because of the absence of the winner's curse, with all winning bidders paying the same price, bidders in a uniform-price auction have more incentive to submit bids with an accurate valuation or the maximum amount (Cole et al., 2018). Consequently, when uncertainty about the value of the treasury is high, the market clearing price in uniform-price auctions

is expected to be higher than the weighted average winning price in discriminatory-price auctions.

The literature has documented that there is a link between uncertainty and bid prices for each auction format (Haile, 2003; Holmberg & Wolak, 2018). Bidders will submit a high bid to guarantee their allocation without increasing the stop-out price in uniform-price auctions. However, bidders in discriminatory-price auctions decrease their bids in response to uncertainty since bidders pay their bids. The information dispersion seems to be a more important source of uncertainty about the value of goods than asymmetric information between bidders and issuers (Yin, 2006). In other words, bidders estimate the dispersion of information when submitting bids in the auction. The revenue and efficiency gaps between discriminatory and uniform-price auctions are likely to increase with the uncertainty due to the presence of the winner's curse phenomenon in discriminatory-price auctions. Therefore, based on the above discussion, I predict the following:

H4a: In the Indonesian treasures market, uncertainty increases revenue gains, when switching to the uniform-price auction format.

H4b: In the Indonesian treasures market, uncertainty increases efficiency gain, when switching to the uniform-price auction format.

4.6 Chapter Summary

This chapter presents the theoretical framework and testable hypotheses. In discriminatory-price auctions, bidders use their best effort to win the auction at the lowest price because they pay a different price depending on their bids. In uniform-price auctions, bidders bid more aggressively due to the absence of the winner's curse. Bidders

in uniform-price auctions are therefore more likely to submit higher bids. I thus predict that uniform-price auctions generate higher revenue than discriminatory-price auctions (H1).

In term of efficiency, bidders have lower incentives to do bid shading in uniform-price auctions than in discriminatory-price auctions. In discriminatory-price auctions, bidders have less incentives to submit bids based on their true value because successful bidders pay their submitted bids. Since uniform-price auctions allocate the security to bidders with higher bids, I predict that uniform-price auctions are more efficient than discriminatory-price auctions (H2).

In an environment setting with high competition, discriminatory and uniform-price auctions generate the same revenue and efficiency. Therefore, I argue that competition reduces both the revenue and efficiency gain when switching to uniform-price auction format (H3a and H3b). Bidders will submit a high bid to guarantee allocation without increasing the stop-out price in uniform-price auctions. However, bidders in discriminatory-price auctions decrease their bids in response to uncertainty since bidders pay their bids. Therefore, my final hypotheses predict that uncertainty is positively related to both revenue and efficiency gains when switching to uniform-price auctions (H4a and H4b).

Chapter 5

Data and Research Methodology

5.1 Introduction

This chapter provides a discussion on the data and research methods used to test my hypotheses on the performance of discriminatory-price auctions relative to uniform-price auctions developed in Chapter 4. It begins with a discussion on the data sources and sample construction procedure in Section 5.2. Section 5.3 explains the counterfactual revenue approach. Estimations of auction revenue and efficiency are discussed in Section 5.4 and Section 5.5, respectively. This is followed by a discussion of the methodology used to test the determinants of both revenue and efficiency in Section 5.6. Measurement of the test variables is discussed in Section 5.7. Finally, Section 5.8 provides a summary of this chapter.

5.2 Data

My primary data source is the MoF, which provided detailed individual bidding data on Indonesian treasuries. For conventional bonds, the sample period covers 2003 to 2017, while for Islamic bonds, the sample period covers 2009 to 2017.[14] The raw data comprise: (i) 117 Islamic treasury bill auctions; (ii) 319 conventional treasury bill auctions; (iii) 534 Islamic treasury bond auctions; and (iv) 689 conventional treasury bond auctions.

Each record contains the following bidding data: the auction date; settlement date; series; submitted bids; awarded bids; bidding yield; winning price; investor code, and

[14] The MoF started to issue Islamic treasuries through an auction in 2009 to regularly maintain issuance, diversify treasury instruments, and widen the investor base.

dealer code. I use the weighted average awarded price (yield) as the price (yield) in non-competitive bids, since bidders only submit an amount without the price (yield). Data on exchange rates and price quotations are collected from Bank Indonesia and Bloomberg Professional Service database.

Since I need the residual supply function and invert it to conduct counterfactual calculations for the performance of uniform-price auctions, I exclude unsuccessful auctions and winning auctions with a 100 percent success rate. There are several reasons for unsuccessful auctions. The main consideration is that there are no bids with prices (yields) over (below) the market clearing price (yield). Another reason is that the number of accepted bids is too few, thus not meeting the minimum amount set by the MoF. The government needs to show certainty in future outlooks, consistency in issuance size or supply side, and credible asset-liability (cash) management. If the MoF absorbs too few bids, this will deliver a negative signal to the market that the government faces a shortage in cash and an uncertain future prospect.

Table 5.1 displays the sample selection process. Islamic treasury bills auctions commenced in 2011. In total 117 auctions were held from 2011 to 2017. Based on the benchmark price, the MoF accepted bids from 100 auctions. I exclude one auction with 100 percent winning rate, thus giving a final sample of 99 auctions.

Starting in 2009, there has been a total of 534 Islamic treasury bonds auctions. The MoF accepted 341 auctions and rejected 193 auctions, with 10 auctions having a success rate of 100 percent. This gives a final sample of Islamic treasury bonds of 331 auctions.

Commencing in 2007, there is a total of 319 conventional treasury bills auctions. The MoF absorbed bids from 298 auctions and disapproved bids from 21 auctions. Since

all winning auctions have a success rate below 100 percent, the final sample is 298 auctions.

Finally, conventional treasury bond auctions started in 2003, giving a total of 689 auctions over the sample period. The MoF accepted 634 auctions and declined 55 auctions, with 2 winning auctions with a success rate of 100 percent. This gives a final sample of conventional treasury bond auctions of 632 auctions.

Table 5.1: ***Sample Selection Process.***

Note: The discount rate issues have maturities less than 1 year, while the coupon rate issues have maturities exceeding 1 year.

Securities	Description	Issue Period	# Auctions	#Winning Auctions	Final Sample
Islamic	Treasury bills	2011-2017	117	100	99
	Treasury bonds	2009-2017	534	341	331
Conventional	Treasury bills	2007-2017	319	298	297
	Treasury bonds	2003-2017	689	634	632

5.3 Counterfactual Revenue Approach

The method of quantitative data analysis used is the counterfactual approach or income estimation model. I use this method because it is not possible to carry out a natural experimental method given that Indonesia has not yet applied a uniform-price auction for her treasuries. Neither is the laboratory experiment method chosen because changing the complex treasury auction practices into a simple process requires many unrealistic assumptions. The laboratory experiment method also has the weakness of not adequately describing the actual conditions due to test and perception bias of experimental participants about reality.

Referring to the models developed by Hortaçsu and McAdams (2010), Kang and Puller (2008), and Kim, Oh, and Ryu (2003), the following tasks are required in order to compare the performance of discriminatory-price auctions with uniform-price auctions: (i) develop a theoretical model of both uniform-price and discriminatory-price auctions; (ii) develop a theoretical association between uniform-price and discriminatory-price auctions; (iii) produce counterfactual revenue (estimated income) method using the theoretical association that has been formulated above; and (iv) perform a large number of simulations and comparisons between the two auction methods.

I apply the structural model of strategic bidding developed by Hortaçsu and McAdams (2010), where counterfactual estimations are conducted by converting mark-down orders to diagnose unobserved marginal valuation. The structural model results in a first-order crucial circumstance in which the bidding price for a specific quantity is equivalent to the marginal value of the bidder. That is, the marginal value of the bidder is the inverted elasticity of the residual supply function of every bidder. This model has several advantages compared to other models: (i) it provides a full structural model for

carrying out a counterfactual approach; and (ii) it provides several methodological advances in structural econometric modelling of auctions, i.e., it provides an empirical algorithm that has low computational demand which minimises the impact of distribution assumptions on the estimation results. The key assumption using this approach is that the ex-ante information of bidders is stable.

In every auction, the upper and lower bounds are estimated by computing the marginal value schedule of each bidder. Estimates of a bidder's probability of winning a specific quantity at a certain price may induce consistency in bounds calculation. For instance, the lower bound of the marginal value of bidders is estimated consistently with an observed bidding function by estimating the probability of winning for all quantities. A resampling procedure is then implemented to set the consistency of estimation as the number of treasury auctions approaches infinity, holding the number of participants in each auction constant. Bidders influence the residual distribution supply of bids since each bidder has his own estimation and is concerned with the bidding strategies from other bidders. Estimation of this distribution then depends on resampling from bidding observations.

I make the following assumptions in order to perform the counterfactual revenue simulation: (i) when the price rises, the bid volume decreases; (ii) the auction has private values; (iii) the market clearing price is based on the total volume of treasuries sold; and (iv) a comparison of the results of the estimated income is possible and can be achieved either ex-post or ex-ante. I use ex-post data as a basis for comparison. In the ex-post analysis, the simulation process is based on the observed results of the discriminatory-price auction and the estimated revenue from the uniform-price auction where the bid value is derived from the marginal function of each bidder. In the ex-ante model, the bid

value and marginal function are random variables which result in random values revenue. The simulation compares the expected revenue value of the two auction mechanisms. This can also be achieved using a statistical test of income similarity from the two auction methods utilising the bootstrap approach.

Hortaçsu and McAdams (2010) suggest that the following resampling procedure simplifies the estimation problem: (i) set bidder x has a bid z; (ii) substitute N_t sample of bids in the data set will gain a subsample of random N_{t-1} bid vectors; (iii) determine the actual market clearing price to build the actual residual supply of the bidder x. Then estimate whether the bidder x would have won quantity y at a specific price (p) for all (y,p); (iv) repeat the process multiple times to compute the winning probability for each bidder. The probability of winning is taken as the portion of bidders winning a unit at price p in all subsamples.

The demand function can be expressed as y (., v(.), x,N), with private marginal value plan as v(.), covariates x, and N bidders in each auction. Using the assumption that the demand function is symmetric with non-random supply, the winning probabilities can be stated as follows:

$$H(y;p|x,N) = E[1\{\sum_{j=1}^{N-1} Yj \ \leq Q - y\}|x,N] \qquad (1)$$

where the probability that the bidder submits bids at the highest quantity y with specific price (p) can be stated as H(y; p|x) = Prob(y(p, v(.), x) ≤ y|x), and dH(y; p|x) = Prob(y(p, v(.), x) = y|x).

The equations to compute the lower bounds ($\underline{\hat{\boldsymbol{v}}}^{\boldsymbol{T}}$) and upper bounds ($\overline{\hat{\boldsymbol{v}}}^{\boldsymbol{T}}$) of the marginal values of the bidders are as follows:

$$\underline{\hat{v}}^T|(y,p(.)|x) = p(y) + \frac{\Delta\sum_{q=y}^{\bar{y}}\hat{H}^T(q;p(y)-\Delta|x)}{\sum_{q=y}^{\bar{y}}[\hat{H}^T(q;p(y)|x)-\hat{H}^T(q;p(y)-\Delta|x)]} \quad (2)$$

$$\hat{\bar{v}}^T(y,p(.)|x) = p(y) + \Delta + \frac{\Delta\sum_{q=\underline{y}}^{\bar{y}}\hat{H}^T(q;p(y)|x)}{\sum_{q=\underline{y}}^{\bar{y}}[\hat{H}^T(q;p(y)+\Delta|x)-\hat{H}^T(q;p(y)|x)]} \quad (3)$$

The consistency of the bounds (as T → ∞) follows the consistent estimation of $\hat{H}^T$ (q; p(y)|x). The smooth functions of $\left\{H(q;p(y)): q = \left[\underline{y}, \bar{y}\right], p \in \{p - \Delta, p, p + \Delta\}\right\}$ are $\bar{v}$ (y; p(.)) and $\underline{v}$ (y; p(.)). In each component, there is a limited number of bounds. I invoke the delta method for bootstrapping and estimate a bootstrap range of confidence for $\underline{\hat{v}}^T(y,p(.))$ and $\hat{\bar{v}}^T(y,p(.))$.

Based on the Hortaçsu and McAdams (2010) model, the ex-post method is as follows. Firstly, with a resampling algorithm, compute $H(p_k, \bar{y}_t)$, the probability that the value of the market clearing price will decrease from p_k conditional towards the vector bid value from *i*- bidder of $\bar{y}_t$. The simulation algorithm is a value of market clearing price shown to *i*- bidder. The value of $H(p_k, \bar{y}_t)$ is calculated as a frequency of values of market clearing price produced by the value of over p_k value. The value of $H(p_k, \bar{y}_t)$ can also be determined from the density estimates.

Next, compute the estimated value of the marginal utility of the *i*- bidder when the bidder wins *q* unit, also called the *marginal valuation function* $v_t(q, s_t, s_{-t})$. The value of marginal valuation will maximise the value of expected profit from *i*- bidder if he wins the demand of $\bar{y}_{tk}$ unit (in the discrete model), as shown in the following formula:

$$v_t(y_{tk}, s_t) = p_k + H(p_{k-t}, \bar{y}_t)\frac{(p_k - p_{k-t})}{[H(p_k, \bar{y}_t) - H(p_{k-t}, \bar{y}_t)]}$$

This yields $K+1$ estimated values of the marginal valuation $v_t(y_{tk}, s_t)$, with the assumption that $K+1$ of y_{tk} value on vector $\bar{y}_t$, that $\bar{y}_t = \{y_{i0} \geq y_{i1} \geq \cdots \geq y_{iK+1}\}$. Perform repeatedly the two steps outlined above for all bidders in an auction.

Thirdly, compute the value of the best-case uniform-price auction using the structural model, the observed data from the discriminatory-price auction, and the value of marginal valuation as a proxy for the true marginal valuation of all bidders. This step is achieved by combining all estimated values of the marginal valuation of all bidders and estimating the intersection between marginal value and total supply, so that we obtain the value of the market clearing price. The auction revenue is obtained by multiplying the value of the best-case market clearing price with the total number of units sold.

Fourthly, estimate the market clearing price. Each auction t has data for the bidding function $\{Y_{it}(p, S_i), i = 1,\cdots, N_t; t = 1,\cdots, \mathrm{T}\}$. Based on the data obtained, estimate the distribution of the market clearing price that is expected by each bidder prior to the auction. The resampling method (non-parametric approach) can be used for data retrieval. For the non-parametric method or resampling method (Hortaçsu & McAdams, 2010), I assume that the price-quantity pair form the bidding function. With this notation, the lattice of the price is $\boldsymbol{\rho_0} < \boldsymbol{\rho_1} < \cdots < \boldsymbol{\rho_{K+1}}$, and the value of the bid quantity forms a vector value specified for each price value, as shown by: $\overrightarrow{\boldsymbol{y_i}}: \{\boldsymbol{y_{i0}} \geq \boldsymbol{y_{i1}} \geq \cdots \geq \boldsymbol{y_{iK+1}}\}$. The resampling method is performed as follows. Determine a random bidder i among N_t bidders in auction t. From N_t vector of bidding data in the auction t, taken a random sample with a return of N_{t-1} of bidding vector, where each data vector in the initial data has an equal chance to be taken which is $1/ N_t$. Use the bidding vector data of the bidder i and N_{t-1} of resample vector data where the total demand is equal to the total number of offers. Perform the steps above M times (about 1000 times) for all bidders in auction t. This

procedure will produce M values of the conditional market clearing price towards the value $\overrightarrow{\bar{y}_{\iota}}$. The estimated value for H (pk, $\bar{y}$t) is then obtained by computing the frequency (percentage) of the emergence of values of the market clearing price produced with the method above using a smaller value for $\boldsymbol{\rho_k}$.

5.4 Revenue Comparison

Each bid in an auction can be rationalised using the computed lower and upper bounds on marginal valuations. A counterfactual revenue comparison can be conducted for each bound of the marginal value. The comparison is based on the actual marginal value taken as ex-post revenue in the auction. The upper bound of the counterfactual revenue from the uniform-price auction is generated using the seller's revenue in the equilibrium of bidding, since bidders submit bids below true values in the uniform-price auction. In these auctions, the upper bound is equivalent to the marginal value of bidders because bidders submit fair bids based on increasing valuation. Since the bidding observations and the computed bounds on bidders' marginal value are random variables, another method for estimating counterfactuals is to use the empirical bids distribution by computing the bidder's marginal value bounds. This allows a comparison of the expected income under both auction designs. I resample 1,000 i.i.d. random variables based on the initial bid vector to create 1,000 bidding data in the discriminatory-price auction. Each of the bidding data generated are used to estimate the seller's revenue. For uniform-price auctions, I compute the counterfactual results using fair bidding data from each bidder. Ex-ante revenue disparity is computed by comparison of the mean percentage disparity between the realised revenue and the counterfactual revenue from 1,000 bidding data.

5.5 Efficiency Comparison

Upper bounds of efficiency in the discriminatory-price auction can also be estimated from the upper and lower bounds of the marginal value. Efficiency loss occurs when the marginal value of the winning bidders is less than the marginal value of the unsuccessful bidders. A comparison between the upper bounds at the marginal value of unsuccessful bidders and the lower bounds at the marginal value of successful bidders generates an upper bound inefficiency in discriminatory-price auctions. The upper bound, as a measure of efficiency loss in the discriminatory-price auction, is a result of inefficiency when bidders have marginal values at the upper and bottom lower bounds of the band estimation. The efficiency can be computed both ex-ante and ex-post, like the estimation of the counterfactuals of revenue bounds.

5.6 Regression Model

To examine the determinants of revenue and efficiency gains, I employ a linear regression specification that takes the following general form:

$$Gain = \alpha + \beta 1\ Competition + \beta 2\ Uncertainty + \beta 3\ Controls + e$$

where *Gain* is the estimated gain in performance when switching from a discriminatory-price to a uniform-price auction. The two main test variables are *Competition* and *Uncertainty*. The regression model includes *Maturity* as control variable. Definition and measurement of the regression variables are summarised in Table 5.2.

5.7 Measurement of Variables

5.7.1 Dependent Variable

<u>Ex-post Upper Revenue Gain</u>

$$= Hypothetical\ Upper\ Revenue\ Uniform\ Price_i - Actual\ Revenue\ Discriminatory\ Price_i$$

<u>Ex-ante Upper Revenue Gain</u>

$$= Hypothetical\ Upper\ Revenue\ Uniform\ Price_i - Hypothetical\ \ Revenue\ Discriminatory\ Price_i$$

<u>Ex-post Lower Revenue Gain</u>

$$= Hypothetical\ Lower\ Revenue\ Uniform\ Price_i - Actual\ Revenue\ Discriminatory\ Price_i$$

<u>Ex-ante Lower Revenue Gain</u>

$$= Hypothetical\ Lower\ Revenue\ Uniform\ Price_i - Hypothetical\ \ Revenue\ Discriminatory\ Price_i$$

<u>Ex-post Efficiency Gain</u>

$$= Hypothetical\ Efficiency\ Uniform\ Price_i - Actual\ Efficiency\ Discriminatory\ Price_i$$

<u>Ex-ante Efficiency Gain</u>

$$= Hypothetical\ Efficiency\ Uniform\ Price_i - Hypothetical\ \ Efficiency\ Discriminatory\ Price_i$$

5.7.2 Independent Variables

Competition:

I employ three proxies for the level of competition in the auction. My first proxy is *Bidding*, the natural logarithm of the total number of bids received in each auction. The larger *Bidding*, the more competitive the auction. My second proxy for competition is *Bidders*, the natural logarithm of the number of unique bidders in each auction. The more bidders at the auction, the more competitive. My final proxy is bid-to-cover ratio (*BTC*), the dollar amount of the bids received in an auction over the dollar amount sold. The larger *BTC*, the more competitive the auction.

$$BTC_i = \frac{\Sigma\ Incoming\ Bids_i}{\Sigma\ Winning\ Bids_i}$$

Uncertainty:

I employ three proxies for uncertainty. My first proxy is *Spread*, the difference between the highest and the lowest bidding prices at each auction. The higher the *Spread*, the more dispersed the information in the auction. My second proxy for uncertainty is *Tail*. *Tail* indicate the dispersion of information, the difference between the market clearing yield and the weighted average awarded yield in each auction. The higher *Tail*, the more dispersed the information in the auction. My final proxy for uncertainty is *Volatility*, the standard deviation of the bidding yields in each auction. The higher the *Volatility*, the higher the uncertainty.

5.7.3 Control Variables

I include *Maturity* as the control variable. *Maturity* is computed as the natural logarithm of the time between the auction date and maturity date of the treasury.

Table 5.2: ***Variable Definitions.***

Variable	Definition
Dependent Variable	
Ex-post Upper Revenue Gain	Difference between the hypothetical income of the uniform-price auction obtained from upper bound estimation and the actual income of the discriminatory-price auction.
Ex-ante Upper Revenue Gain	Difference between the hypothetical income of the uniform-price auction obtained from upper bound estimation and the hypothetical income of the discriminatory-price auction.
Ex-post Lower Revenue Gain	Difference between the hypothetical income of the uniform-price auction obtained lower bound estimation and the actual income of the discriminatory-price auction.
Ex-ante Lower Revenue Gain	Difference between the hypothetical income of the uniform-price auction obtained from lower bound estimation and the hypothetical income of the discriminatory-price auction.
Ex-post Efficiency	Difference between the hypothetical income allocation of the uniform-price auction and the actual income allocation of the discriminatory-price auction.
Ex-ante Efficiency	Difference between the hypothetical income allocation of the uniform-price auction and the hypothetical income allocation of the discriminatory-price auction.
Independent Variables	
Bidding	Number of bids received in an auction.
Bidders	Number of unique bidders in an auction.
Bid-to-Cover (BTC)	Dollar amount of the bids received in an auction over the dollar amount of treasuries sold.
Spread	Difference between the price of the highest incoming bid and the price of lowest incoming bid in an auction (in percent).
Tail	Difference between the yield of the weighted average accepted bids and the yield of the lowest accepted bid in an auction (in basis points).
Volatility	Standard deviation of bidding yields in an auction (in percent).
Control Variable	
Maturity	Number of years between issue date and maturity date.

5.8 Chapter Summary

This chapter discusses the sample selection and research methodology applied to test the predictions. For the Indonesian treasury auction market, which has been operating since 2003, I obtain a final sample of 99 Islamic treasury bill auctions (short-term); 297 conventional treasury bill auctions (short-term); 331 Islamic treasury bond auctions (long-term); and 632 conventional treasury bond auctions (long-term) between 2003 and 2017.

I employ a counterfactual approach or income estimation model to compare the revenue and efficiency of discriminatory-price auctions with uniform-price auctions. I use a counterfactual analysis given that it is not possible to carry out a natural experimental in Indonesia. Based on the structural model of strategic bidding of Hortaçsu & McAdams (2010), I apply ex-post and ex-ante revenue estimations. To investigate the determinants of the gains in revenue and efficiency results when switching to a uniform-price auction, I apply a multiple regression model.

Chapter 6

Empirical Results: Auction Performance

6.1 Introduction

In this chapter I compare the performance of discriminatory-price auctions relative to uniform-price auctions for Indonesian treasuries. Even though both auction types generate bid shading and demand reduction, there is no explicit functional form which links optimal bidding in discriminatory-price auctions to uniform-price auctions. Hence, to establish revenue ranking and efficiency performance, I adopt the approach of Hortaçsu and McAdams (2010) using a structural model of equilibrium bidding in discriminatory-price auctions. I perform a non-parametric estimation of the bidding valuation and use the results to establish a counterfactual performance comparison. For completeness, I report the results separately for Islamic and conventional treasuries, even though my hypotheses do not distinguish between the two type of treasuries.[15]

6.2 Descriptive Statistics

Tables 6.2 and 6.3 shows the descriptive statistics of the sample. Conventional treasury bonds receive the highest number of bids per auction, with mean (median) *Bidding* of USD 465 million (USD 410 million). This is followed by conventional treasury bills, with mean (median) *Bidding* of USD 422 million (USD 360 million), Islamic treasury bills, with mean (median) *Bidding* of USD 286 million (USD 217 million), and Islamic treasury bonds with mean (median) *Bidding* of USD 133 million (USD 89 million). Conventional treasury bonds have by far the largest mean (median)

[15] I have no good reasons to suspect that the results are likely to be different across the two treasuries, given that both treasuries have the same underlying issuers, risks and credit rating.

number of institutional bidders (*Bidders*) per auction at 21 (20). This is about twice the average number of institutional bidders observed per auction for the other treasuries.

Based on the dollar amount of bids received in an auction over the dollar amount sold (*BTC*), there is stronger average demand for treasury bills than for treasury bonds. Islamic treasury bills have a mean (median) *BTC* ratio of 4.28 (2.98), followed by conventional treasury bills at 3.37 (2.49), and Islamic treasury bonds at 2.87 (1.77). Conventional treasury bonds have the lowest average (median) *BTC* ratio of 2.62 (1.93).

Islamic treasury bond auctions have the highest spread, with a mean *Spread* of 3.81 percent, followed by conventional treasury bonds, with a mean *Spread* of 3.69 percent. Islamic and conventional treasury bills have the lowest average *Spread* at 0.55 percent and 0.50 percent, respectively. That treasury bills have much smaller spread than treasury bonds is not surprising, given their short maturity.

Islamic treasury bills have the highest average bid dispersion (*Tail)* per auction, with a mean (median) *Tail* of 13 basis points (9.21 basis points), followed by conventional treasury bills, with a mean (median) *Tail* of 7.56 basis points (4.37 basis points), and Islamic treasury bonds, with a mean (median) *Tail* of 3.52 basis points (0.44 basis points). Conventional treasury bonds have the lowest mean (median) bid dispersion, at 2.16 basis points (0.44 basis points).

Islamic treasury auctions have the highest average volatility in bid spreads, with a mean (median) *Volatility* of 1.5 percent and 1.41 percent (1.62 percent and 1.34 percent), respectively. Conventional treasury bond auctions have substantially less dispersion in bid spreads, with a mean (median) *Volatility* of 0.53 percent (0.09 percent).

Conventional treasury bill auctions have the lowest mean (median) *Volatility*, at 0.46 percent (0.22 percent).

The conventional treasury bonds issued have a mean (median) *Maturity* of 13.85 years (14.54 years), followed by Islamic treasury bonds at 10.34 years (6.05 years), and conventional treasury bills at 0.71 years (0.96 years). Islamic treasury bills have the shortest maturity, with a mean (median) *Maturity* of 0.48 years (0.5 years).

Table 6.1: ***Summary Statistics of Islamic Treasury Auctions.***

Variable definitions are provided in Table 5.2.

	Islamic Treasury Bills (N = 99 auctions)							Islamic Treasury Bonds (N = 331 auctions)						
	Mean	Median	Min	Max	Std Dev	25th	75th	Mean	Median	Min	Max	Std Dev	25th	75th
Bidding	286.78	217.02	23.54	1347.04	219.48	151.25	374.25	133.31	89.13	1.48	907.48	138.83	44.06	161.28
Bidders	11	11	4	24	3	8	13	12	11	2	30	5	8	15
BTC	4.28	2.98	1.04	14.49	2.95	2.43	5.12	2.87	1.77	1	34.71	3.42	1.24	2.91
Spread	0.55	0.49	0.2	2.99	0.36	0.37	0.61	3.81	2.49	0.21	19.58	3.56	1.16	5.28
Tail	13	9.21	0	48.37	12.72	0.77	24.31	3.52	0.44	0	41.12	5.87	0.04	5.19
Volatility	1.41	1.34	0.33	2.44	0.44	1.12	1.65	1.5	1.62	0.04	5.62	1.11	0.26	2.26
Maturity	0.48	0.5	0.38	0.54	0.03	0.46	0.5	10.34	6.05	0.72	29.95	8.94	3.95	14.88

Table 6.2: ***Summary Statistics of Conventional Treasury Auctions.***

Variable definitions are provided in Table 5.2.

	Conventional Treasury Bills (N = 297 auctions)							Conventional Treasury Bonds (N = 632 auctions)						
	Mean	Median	Min	Max	Std Dev	25th	75th	Mean	Median	Min	Max	Std Dev	25th	75th
Bidding	422.08	360.77	13.72	1523.62	292.01	292.01	360.77	465.66	410.31	22.71	2193.78	291.84	291.84	410.31
Bidders	10	10	4	24	3	3.31	10	21	20	6	96	9	8.56	20
BTC	3.37	2.49	1.02	16.88	2.5	2.5	2.49	2.62	1.93	1.01	56.63	3.15	3.15	1.93
Spread	0.5	0.41	0.02	2.49	0.37	0.37	0.41	3.69	2.72	0.43	35.22	3.34	3.34	2.72
Tail	7.56	4.37	0	35.79	8.64	8.64	4.37	2.16	0.44	0	28.56	3.66	3.66	0.44
Volatility	0.46	0.22	0.03	2.58	0.54	0.54	0.22	0.53	0.09	0.03	3.62	0.89	0.89	0.09
Maturity	0.71	0.96	0.17	1.19	0.34	0.34	0.96	13.85	14.54	1.91	30.64	6.8	6.8	14.54

6.3 Islamic Treasury Bill Auctions

6.3.1 Counterfactual Approach

To illustrate the counterfactual approach, I next discuss the results for Islamic treasury bill auction #465, which was held on the 22 March 2016. There were 28 bids submitted by 10 bidders, constituting a total bid value of USD 193 million. About 40 percent of the submitted bids were accepted by the MoF, generating USD 75 million in revenue.

Figure 6.1 shows the resampling distribution of the market clearing price which accommodates all bids. In the construction of the weights, I use the kernel-based resampling method in which the Islamic treasury bill supply and number of bids are used as covariates in the auction. I observe that the high bid at 97.4 percent looks like an outlier among the next 14 bids. The probability of the market clearing price being above this price is small. In this auction, I pay attention to bidder #214 who submitted 10 bids, equivalent to 20 percent of the demand for the issue. This bidder has a very high value for the initial bid. A high valuation makes sense since auction #465 happened in the first quarter of the year when market liquidity is typically high and the probability of obtaining Islamic treasury bills in the secondary market is relatively small. Banks tend to use Islamic treasury bills as liquid asset reverse requirements.[16]

[16] Banks recognise Islamic treasury bills as a primary liquid asset if put in a trading account and as a secondary liquid asset if it is in the hold-to-maturity account. Majority bank purchases are Islamic treasury bills in hold until maturity.

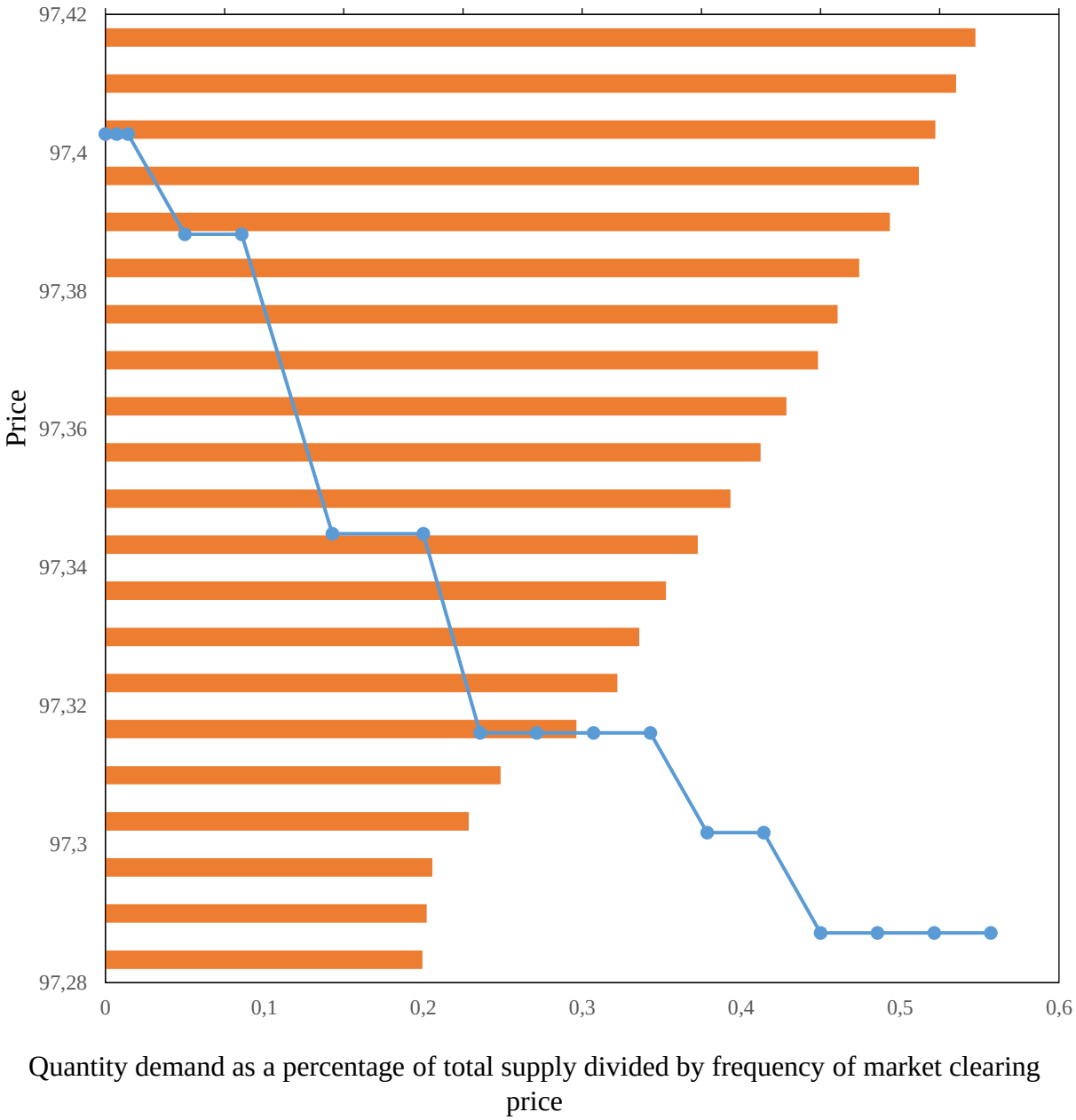

Figure 6.1: Bidder #214 in Auction #465 and Market Clearing Price Distribution Generated by the Resampling Procedure.

To compute the bounds on bidder #214's marginal assessment v(y) for each quantity y, I assume that $\Delta = 0.03125$, corresponding to the increase in the smallest bid price as shown in the data set. The computation of $G(q;\cdot)$ is performed using a resampling algorithm for each intermediate quantity q in the bid step at the same price as y. Computation of the standard errors of bid data is done using 1,000 resamples (bootstraps).

The results for bidder #214 are displayed in Figure 6.2 for 1,000 realisations of the resampled market clearing price. The horizontal axis corresponds to the quantity requested by bidder #214, with the prices displayed on the vertical axis. The staircase represents the bid vector. I plot the upper and lower bound point estimates for different amounts on the marginal valuation of bidder #214. The 5 and 95 percent confidence level of these points are estimated around the bands.

The estimate of the marginal values of the upper and lower bounds, and the quantities of bids are relatively tight for bids corresponding to 4 percent of the supply and above. For the marginal value, the initial step is higher and somewhat less accurately estimated. This lower accuracy indicates that for the initial steps, the residual value calculations, will depend highly on being able to pin down the market clearing demand distributions' upper tail.

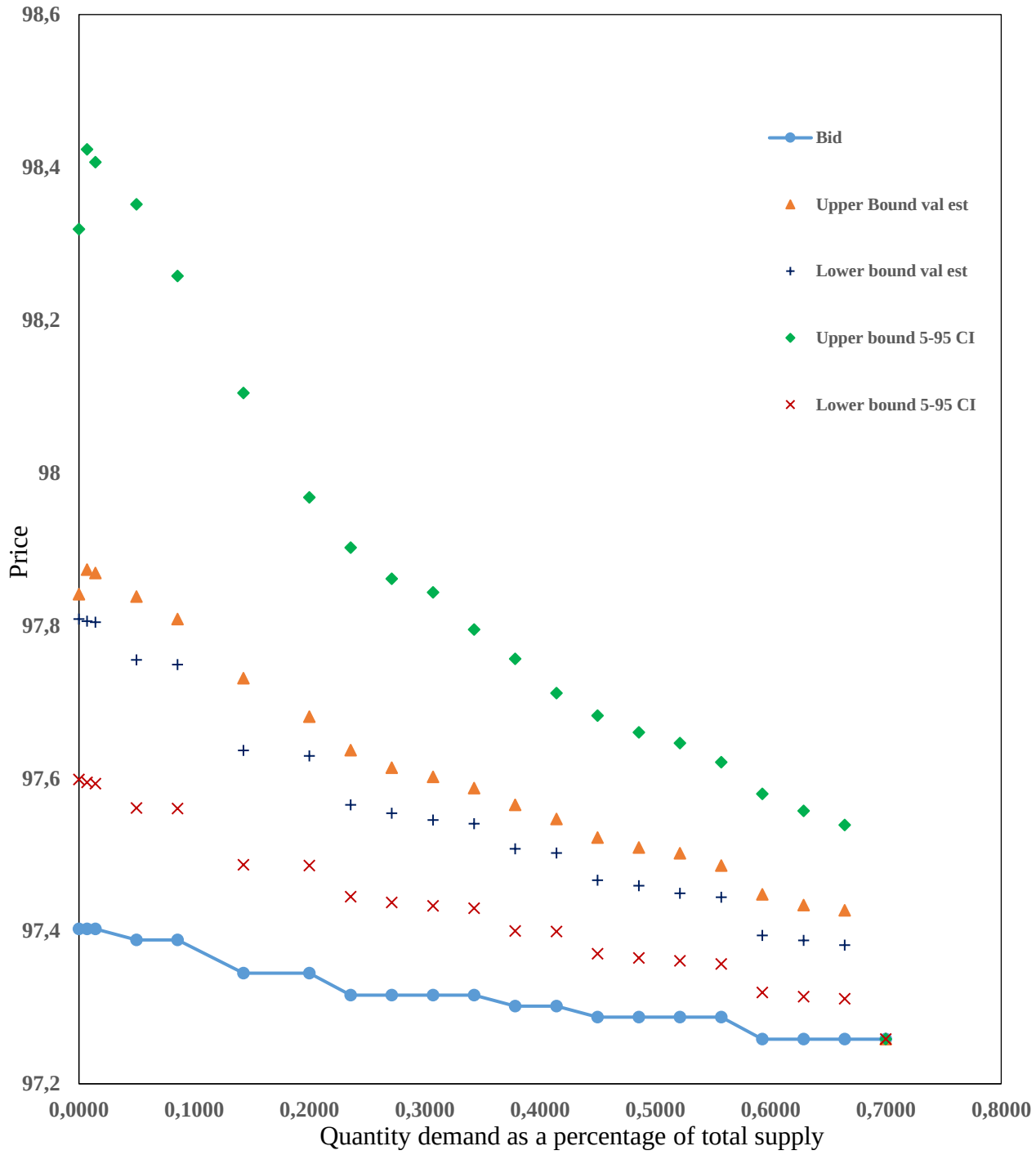

Figure 6.2: Estimation Results for Bidder #214 in Auction #465.

Repetition of the analysis for every bidder in auction #465 generates upper and lower bounds on marginal valuations, which allow for a counterfactual comparison of the revenue that would have been generated in the uniform-price auction vs. the discriminatory-price auction. I refer to the difference between the realised marginal values and hypothetical marginal values in auction #465 as the ex-post revenue in that auction. In uniform-price auctions, bidders submit bids that are less than the true value in equilibrium. Hence, it is more appropriate to use the hypothetical uniform-price auction with truthful bidding (HUPATB). Since truthful bids increase bidder value, the marginal values are equal to the upper bound. The sellers' revenue in HUPATB provides an upper bound on the counterfactual estimation.

Figure 6.3 demonstrates the results of the HUPATB method for auction #465, where the aggregate bids and the aggregated marginal values of the upper and lower bounds are overlaid with supply, normalised to one. I note that the aggregate bid revenue is below the marginal assessment intersection on the upper and lower bound with supply. As the marginal value estimation of the upper bound is above the bids, it appears from the top of the figure that the counterfactual revenue is higher than the revenue generated by the discriminatory-price auction. Therefore, the estimation of the HUPATB revenue is based on the variation in resampling (bootstrap).

For auction #465 the realised revenue of the discriminatory-price auction is 0.14 percent less than the 95 percent (the bootstrap distribution) of the HUPATB revenue at the time when the values are the same as the estimated values of the upper bound. Using the counterfactual approach, I find that moving from a discriminatory-price auction to HUPATB increases the ex-post revenue of the Islamic treasury bill auction.

However, not all auctions in my sample generate similar results. As an illustration, I plot the aggregate bid and bounds valuation for auction #423 in Figure 6.4. The HUPATB revenue is close to the marginal assessment inter section on the upper and lower bound with supply. As the marginal value estimate of the upper bound is nearly equal to the bids, it appears from the bottom of Figure 6.4 that the revenue generated by the discriminatory-price auction is higher than the counterfactual. In fact, computing the point estimate of the upper bound marginal valuation, the counterfactual HUPATB generates 0.09 percent lower revenue than the discriminatory-price auction. Considering sampling variability, I therefore reject the hypothesis that the counterfactual HUPATB generates lower revenue than the discriminatory-price auction.

Another method for establishing counterfactual revenue is based on the concept that the estimated marginal value bounds and the observed bids are random variables. The comparison of expected revenue under both methods uses the empirical bids distribution and the estimated bounds of marginal evaluation. For the discriminatory-price auction, 1,000 resamples are drawn from the original bid vector set, generating 1,000 data sets of bids. The revenue of the auctioneer is computed for each data set generated. The revenue of the counterfactual HUPATB is computed as the difference of the average percentage between the actual and counterfactual revenue. The estimation point of the difference in the ex-ante revenue is obtained using 1,000 simulations.

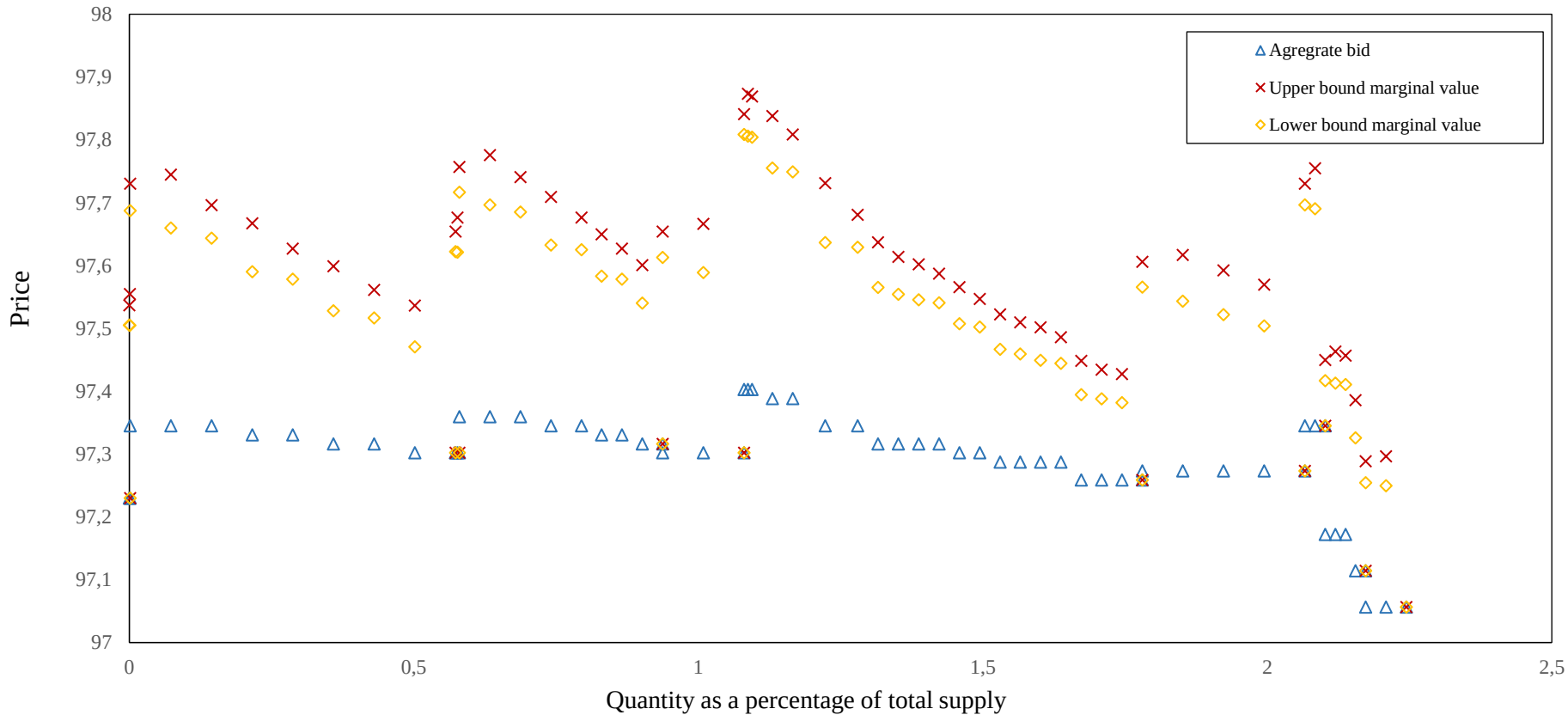

Figure 6.3: Counterfactual Revenue Comparison for Auction #465.

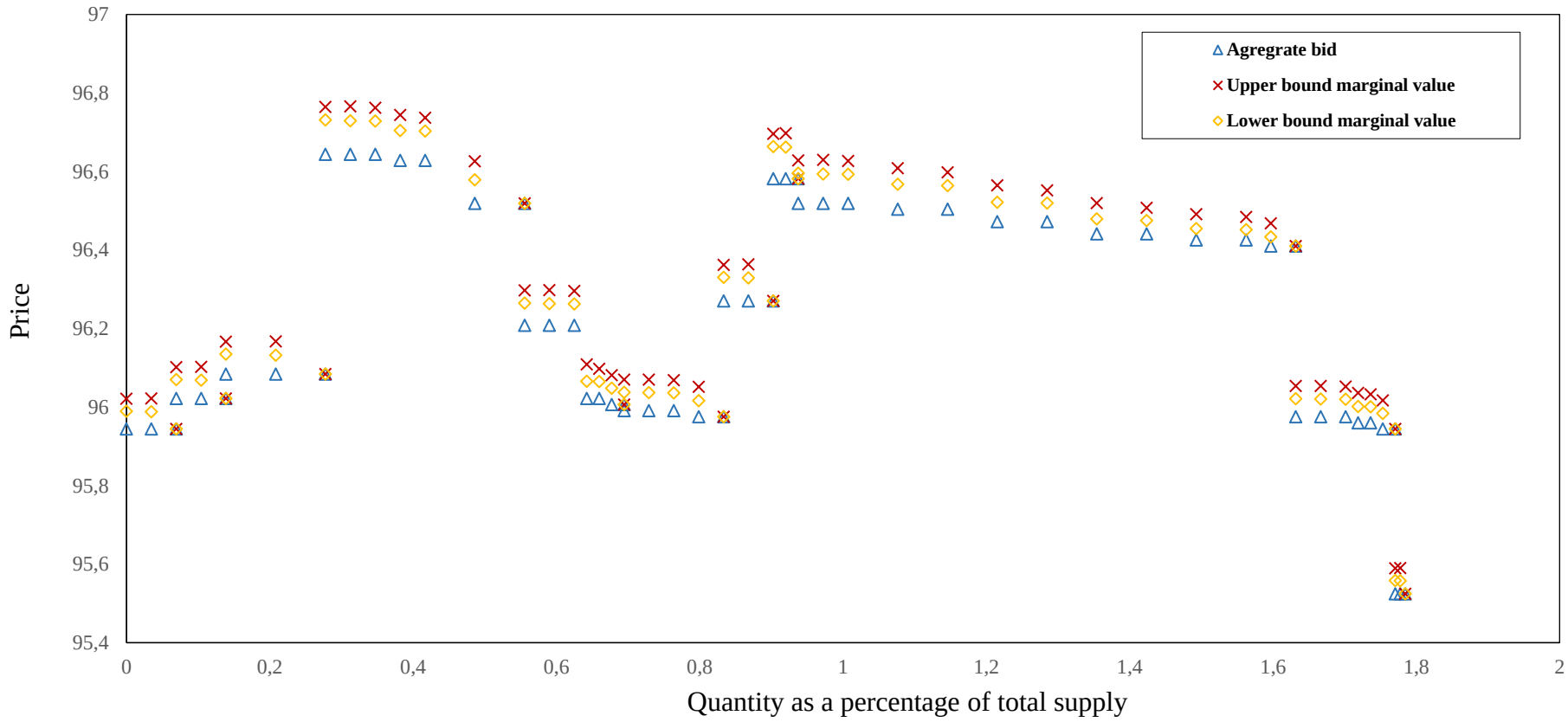

Figure *6.4*: Counterfactual Revenue Comparison for Auction #423.

6.3.2 Revenue Comparison

Table 6.3 presents the counterfactual revenue estimation results for Islamic treasury bill auctions. For the upper value, the mean (median) value of the ex-post revenue is 0.44 percent (0.27 percent), with a t-statistic of 9.03. This figure is larger than the ex-ante estimate, which has a mean (median) value of 0.30 percent (0.20 percent), with a t-statistic of 8.97. Since the average revenue gain values is positive and significant, I infer from these results that uniform-price auctions generate higher revenue than discriminatory-price auctions, on average.

I also group the results based on whether they show a loss or gain in revenue. There are 93 auctions with an ex-post revenue gain, with a mean (median) value of 0.47 percent (0.31 percent). For the six auctions with ex-post revenue loss, the mean (median) value is 0.04 (0.03). I similarly group the ex-ante revenue estimation results based on the gain-loss category. I find 95 auctions with an ex-ante revenue gain, with a (median) median value of 0.31 percent (0.21 percent). There are only three auctions with ex-ante revenue loss, with a mean (median) value of 0.03 percent (0.03 percent). In addition, I find one auction with revenue equivalence. In sum, the results for the upper bound estimations strongly suggest that switching to a single-price auction system increases revenue of Islamic treasury bill auctions.

Table 6.4 displays the counterfactual revenue estimation results for Islamic treasury bill auctions using the lower bound. The results are consistent with the upper bound estimations in that uniform-price auctions generate higher average revenue than discriminatory-price auctions, on average. The ex-post estimations show slightly higher revenue gains than the ex-ante estimations. The mean (median) ex-post revenue gain is 0.26 percent (0.13 percent), with a t-statistic of 6.31, compared to a mean (median) ex-

ante revenue gain of 0.18 percent (0.11 percent), with a t-statistic of 6.47. Focusing on the results based on whether they show a loss or gain in revenue, I find 86 auctions with an ex-post revenue gain, with a mean (median) value of 0.31 percent (0.16 percent). For the 11 auctions with ex-post revenue loss, the mean (median) value is 0.04 percent (0.03 percent). There are two auctions with revenue equivalence. There are 84 auctions with ex-ante revenue gain, with a mean (median) value of 0.22 percent (0.12 percent). There are 10 auctions with ex-ante revenue loss, with the mean (median) value of 0.03 percent (0.02 percent). Moreover, five auctions show revenue equivalence. In sum, the estimation results for the lower bounds suggests that for Islamic treasury bills, single-price auctions generate higher revenue than multiple-price auctions.

Table 6.3: ***Counterfactual Revenue Estimation Results for Islamic Treasury Bill Auctions (Upper Bound).***

This table presents the counterfactual revenue estimation results for 99 Islamic treasury bill auctions using upper bound revenue. Ex-post is comparing the actual income value of a discriminatory-price auction and the hypothetical income generated by a uniform-price auction. Ex-ante is comparing the hypothetical income value of a discriminatory-price auction and the hypothetical income generated by uniform-price auction. Confidence interval level is 95 percent.

			Ex-Post	**CI Level**			**Ex-Ante**	**CI Level**	
All	*Mean*	N = 99	0.436	0.34	0.532	N = 99	0.298	0.232	0.364
	Median		0.27				0.2		
	Std Dev		0.481				0.33		
	Max		2.1				1.87		
	Min		-0.09				-0.06		
	t-value		9.029				8.969		
Gain	*Mean*	N = 93	0.467	0.368	0.566	N = 95	0.312	0.244	0.379
	Median		0.31				0.21		
	Std Dev		0.48				0.331		
	Max		2.1				1.87		
	Min		0.01				0.01		
	t-value		9.388				9.186		
Loss	*Mean*	N = 6	-0.042	-0.077	-0.006	N = 3	-0.033	-0.096	0.029
	Median		-0.035				-0.03		
	Std Dev		0.034				0.025		
	Max		-0.01				-0.01		
	Min		-0.09				-0.06		
	t-value		-3.027				-2.294		
Equivalence		N = 0				N = 1			

Table 6.4: ***Counterfactual Revenue Estimation Results for Islamic Treasury Bill Auctions (Lower Bound).***

This table presents the counterfactual revenue estimation results for 99 Islamic treasury bill auctions using lower bound revenue. Ex-post is comparing the actual income value of a discriminatory-price auction and the hypothetical income generated by a uniform-price auction. Ex-ante is comparing the hypothetical income value of a discriminatory-price auction and the hypothetical income generated by uniform-price auction. Confidence interval level is 95 percent.

			Ex-Post	**CI Level**			**Ex-Ante**	**CI Level**	
All	*Mean*	N = 99	0.262	0.18	0.345	N = 99	0.182	0.126	0.238
	Median		0.13				0.11		
	Std Dev		0.414				0.28		
	Max		2.1				1.66		
	Min		-0.13				-0.09		
	t-value		6.306				6.468		
Gain	*Mean*	N = 86	0.307	0.216	0.398	N = 84	0.218	0.155	0.281
	Median		0.16				0.12		
	Std Dev		0.426				0.289		
	Max		2.1				1.66		
	Min		0.01				0.01		
	t-value		6.683				6.913		
Loss	*Mean*	N = 11	-0.041	-0.067	-0.015	N = 10	-0.031	-0.049	-0.013
	Median		-0.03				-0.02		
	Std Dev		0.039				0.025		
	Max		-0.01				-0.01		
	Min		-0.13				-0.09		
	t-value		-3.516				-3.899		
Equivalence		N = 2				N = 5			

For illustrative purposes, Figure 6.5 provides a scatter plot of the counterfactual revenue estimation results for the Islamic treasury bill auctions over time. The month of the auction and gain/loss value in each auction are represented on the x-axis and y-axis, respectively. The scatterplot shows that the potential revenue gains for the first 11 auctions is extraordinarily high (2.1 percent versus 0.4 percent), plausibly driven by low competition amongst bidders. There is a weak negative correlation between revenue gain and time, which seems to indicate some seasonal patterns. Revenue gains appear higher in quarters 1 and 3 than in quarters 2 and 4 of the year. Although the gains in revenue seems to decline somewhat over the years, importantly for Islamic treasury bills the results are consistent with the proposition that replacing the discriminatory-price auction with a uniform-price auction format increases revenue for the MoF.

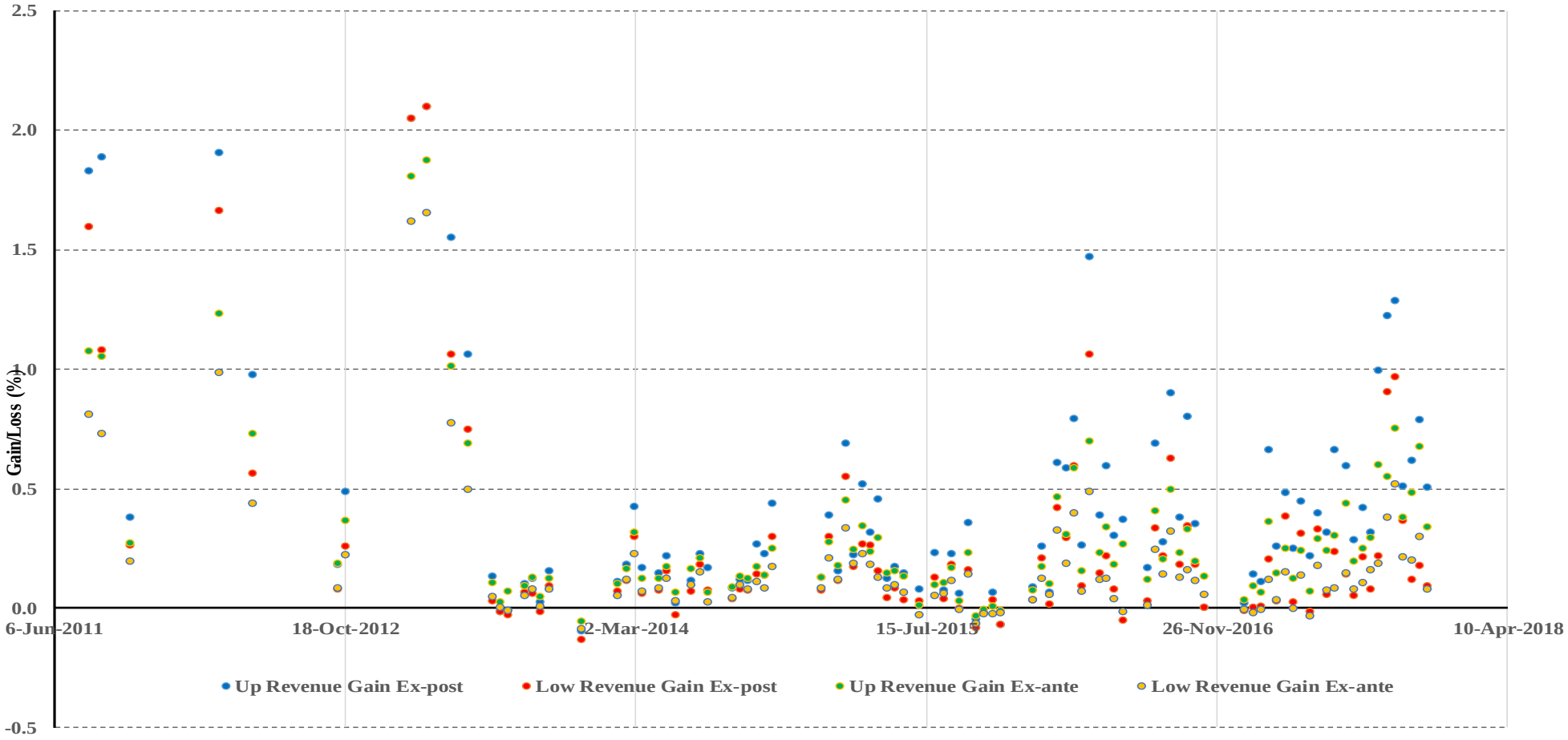

Figure 6.5: Counterfactual Revenue Estimates for Islamic Treasury Bill Auctions.

6.3.3 Efficiency Comparison

Maximising the efficiency of the bid allocation process is another important objective of the Indonesian treasury. The efficiency of the upper bound of the discriminatory-price auction can be computed through a comparison of the upper bound on the marginal value for rejected quantities with the lower bound on the marginal value for accepted quantities. Specifically, imagine that minimal marginal values are assigned to all bidders on the quantities they win and maximum marginal values on quantities that are lost. Upper bound inefficiency occurs when the bidders have marginal values at the top and bottom of the estimation band. The efficiency of the upper bound is estimated ex-post and ex-ante.

The counterfactual efficiency gains for Islamic treasury bill auctions are provided in Table 6.5. The mean (median) value of the ex-post efficiency gain indicates that uniform-price auctions have a significantly higher average (median) efficiency of 0.061 percent (0.030 percent) compared to discriminatory-price auctions. The ex-ante efficiency estimates are slightly smaller, with a mean (median) value of 0.033 (0.020) percent, and a t-statistic of 9.05. I find 79 auctions with positive ex-post efficiency gains, with a mean (median) value of 0.076 (0.050) percent. There are no auctions with a loss in efficiency, while 20 (14) auctions show ex-post (ex-ante) revenue equivalence. Overall, the results suggest that for Islamic treasury bills, switching to a uniform-price auction improves auction efficiency marginally.

Figure 6.6 depicts the ex-post and ex-ante efficiency gains over time. The graph's x-axis depicts the month the auction is held, while the y-axis depicts the percentage of efficiency gains in each auction. I observe little change, except that the efficiency gains appear more dispersed in the early and later years.

Table 6.5: ***Counterfactual Efficiency Estimation Results for Islamic Treasury Bill Auctions.***

This table presents the counterfactual efficiency estimation results for 99 Islamic treasury bill auctions. Ex-post is comparing the actual income value of a discriminatory-price auction and the hypothetical income generated by a uniform-price auction. Ex-ante is comparing the hypothetical income value of a discriminatory-price auction and the hypothetical income generated by uniform-price auction. Confidence interval level is 95 percent.

			Ex-Post	**CI Level**			**Ex-Ante**	**CI Level**	
All	*Mean*	N = 99	0.061	0.046	0.075	N = 99	0.033	0.025	0.040
	Median		0.030				0.020		
	Std Dev		0.074				0.036		
	Max		0.370				0.200		
	Min		0.000				0.000		
	t-value		8.177				9.053		
Gain	*Mean*	N = 79	0.076	0.059	0.093	N = 85	0.038	0.031	0.046
	Median		0.050				0.030		
	Std Dev		0.075				0.036		
	Max		0.370				0.200		
	Min		0.010				0.010		
	t-value		8.978				9.741		
Equivalence		N = 20				N = 14			

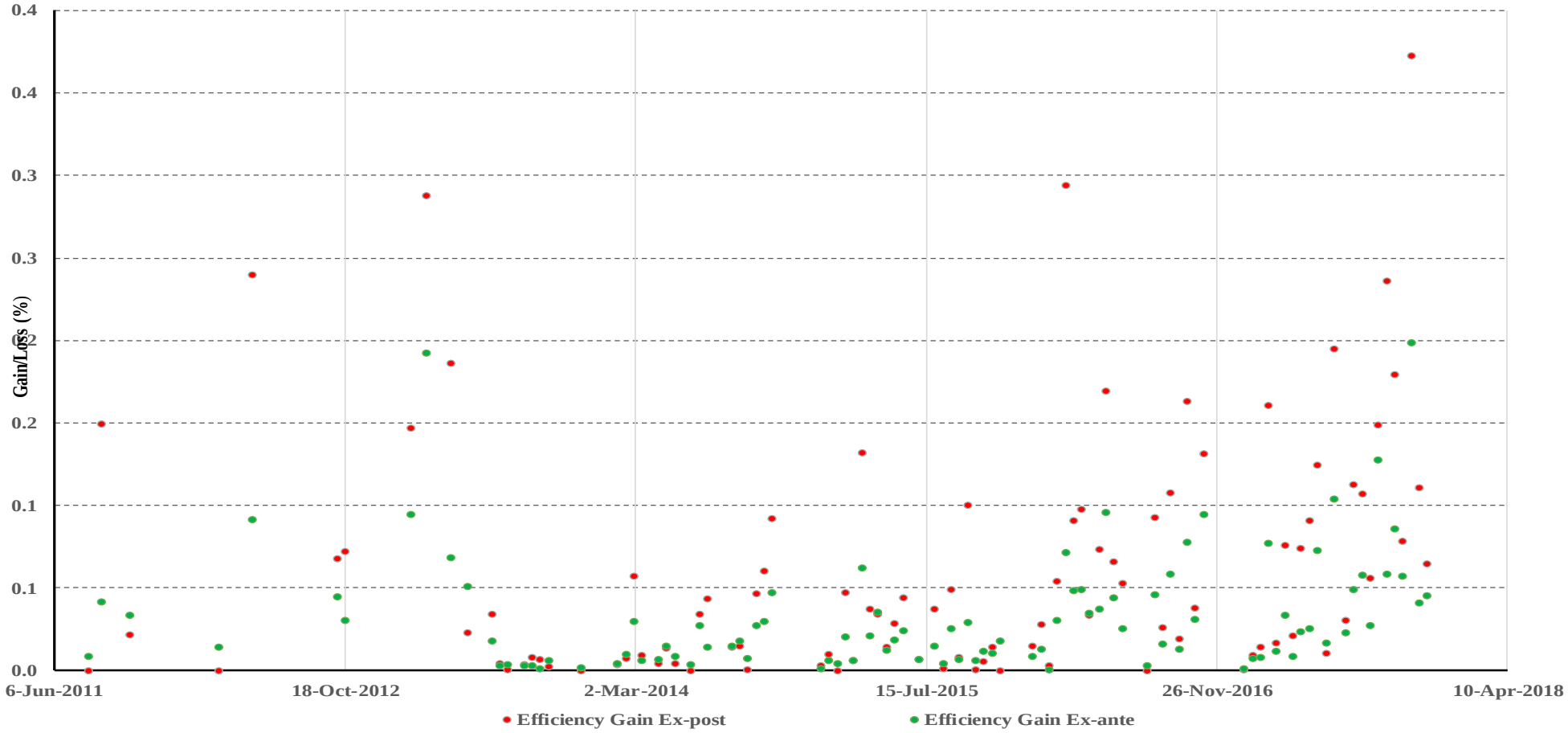

Figure 6.6: Counterfactual Efficiency Estimates for Islamic Treasury Bill Auctions.

6.4 Conventional Treasury Bill Auctions

6.4.1 Revenue Comparison

I rerun the counterfactual approach for conventional treasury bill auctions. Table 6.6 shows that for the upper bound estimation, the mean (median) ex-post revenue gain is 0.98 (1.02) percent, with a t-statistic of 23.78. The ex-ante revenue gain is slightly lower, with a mean (median) value of 0.84 (0.83) percent and a t-statistic of 30.03. Grouping the results based on the loss-gain categories, I find 267 auctions with ex-post revenue gain, with a mean (median) of 1.10 (1.17) percent. There are 293 auctions with ex-ante revenue gain, with a mean (median) value of 0.85 percent (0.83 percent). There are 15 auctions with ex-post revenue loss, with a mean (median) of 0.056 (0.030) percent. There are just three auctions with ex-ante revenue loss, with a (mean) median of 0.067 (0.050) percent. Overall, this result indicates that for conventional treasury bills, uniform-price auctions generate higher revenue than discriminatory-price auctions, on average.

Table 6.7 displays the results for the lower bound estimation. The results are consistent with those for the upper bound in that uniform-price auctions tend to outperform discriminatory-price auctions for conventional treasury bills in terms of revenue generation. The ex-post estimation results are slightly more dramatic than the ex-ante results: the mean (median) ex-post revenue is 0.91 (0.95) percent, compared to a mean (median) ex-ante revenue of 0.77 (0.76) percent. There are 264 auctions with ex-post revenue gain, with a mean (median) of 1.025 (1.070) percent. There are 291 auctions with ex-ante revenue gain, with a mean (median) of 0.79 (0.76) percent. There are 16 auctions with ex-post revenue loss and four auctions with ex-ante revenue loss. The mean (median) ex-post revenue loss is 0.068 (0.035) percent, compared to a mean (median) ex-ante revenue loss of 0.090 (0.080) percent.

For completeness, Figure 6.7 shows the scatterplot of the counterfactual revenue estimation results for conventional treasury bill auctions over time. Revenue gain on 19 August 2016 is surprising high (4.2 percent versus a long-term average of 0.9 percent), possibly due to the low level of competition amongst bidders the day before Indonesia's Independence Day.

Table 6.6: ***Counterfactual Revenue Estimation Results for Conventional Treasury Bill Auctions (Upper Bound).***

This table presents the counterfactual revenue estimation results for 297 treasury bill auctions using upper bound revenue. Ex-post is comparing the actual income value of a discriminatory-price auction and the hypothetical income generated by a uniform-price auction. Ex-ante is comparing the hypothetical income value of a discriminatory-price auction and the hypothetical income generated by uniform-price auction. Confidence interval level is 95 percent.

			Ex-Post	**CI Level**			**Ex-Ante**	**CI Level**	
All	*Mean*	N = 297	0.983	0.902	1.064	N = 297	0.838	0.783	0.893
	Median		1.020				0.830		
	Std Dev		0.712				0.481		
	Max		4.220				3.480		
	Min		-0.240				-0.110		
	t-value		23.779				30.028		
Gain	*Mean*	N = 267	1.097	1.017	1.176	N = 293	0.850	0. 796	0.905
	Median		1.170				0.830		
	Std Dev		0.660				0.473		
	Max		4.220				3.480		
	Min		0.020				0.030		
	t-value		27.133				30.780		
Loss	*Mean*	N = 15	-0.056	-0.094	-0.018	N = 3	-0.067	-0.160	0.027
	Median		-0.030				-0.050		
	Std Dev		0.068				0.038		
	Max		-0.010				-0.040		
	Min		-0.240				-0.110		
	t-value		-3.212				-3.049		
Equivalence		N = 15				N = 1			

Table 6.7: ***Counterfactual Revenue Estimation Results for Conventional Treasury Bill Auctions (Lower Bound).***

This table presents the counterfactual revenue estimation results for 297 treasury bill auctions using lower bound revenue. Ex-post is comparing the actual income value of a discriminatory-price auction and the hypothetical income generated by a uniform-price auction. Ex-ante is comparing the hypothetical income value of a discriminatory-price auction and the hypothetical income generated by uniform-price auction. Confidence interval level is 95 percent.

			Ex-Post	**CI Level**			**Ex-Ante**	**CI Level**	
All	*Mean*	N = 297	0.908	0.108	0.235	N = 297	0.773	0.126	0.238
	Median		0.950				0.760		
	Std Dev		0.695				0.475		
	Max		4.220				3.250		
	Min		-0.290				-0.160		
	t-value		22.488				28.009		
Gain	*Mean*	N = 264	1.025	0.172	0.329	N = 291	0.789	0.154	0.278
	Median		1.070				0.760		
	Std Dev		0.647				0.464		
	Max		4.220				3.250		
	Min		0.010				0.020		
	t-value		25.737				6.485		
Loss	*Mean*	N = 16	-0.068	-0.069	-0.031	N = 4	-0.090	-0.037	-0.009
	Median		-0.035				-0.080		
	Std Dev		0.085				0.051		
	Max		-0.010				-0.040		
	Min		-0.290				-0.160		
	t-value		-3.214				-3.530		
Equivalence		N = 17				N = 2			

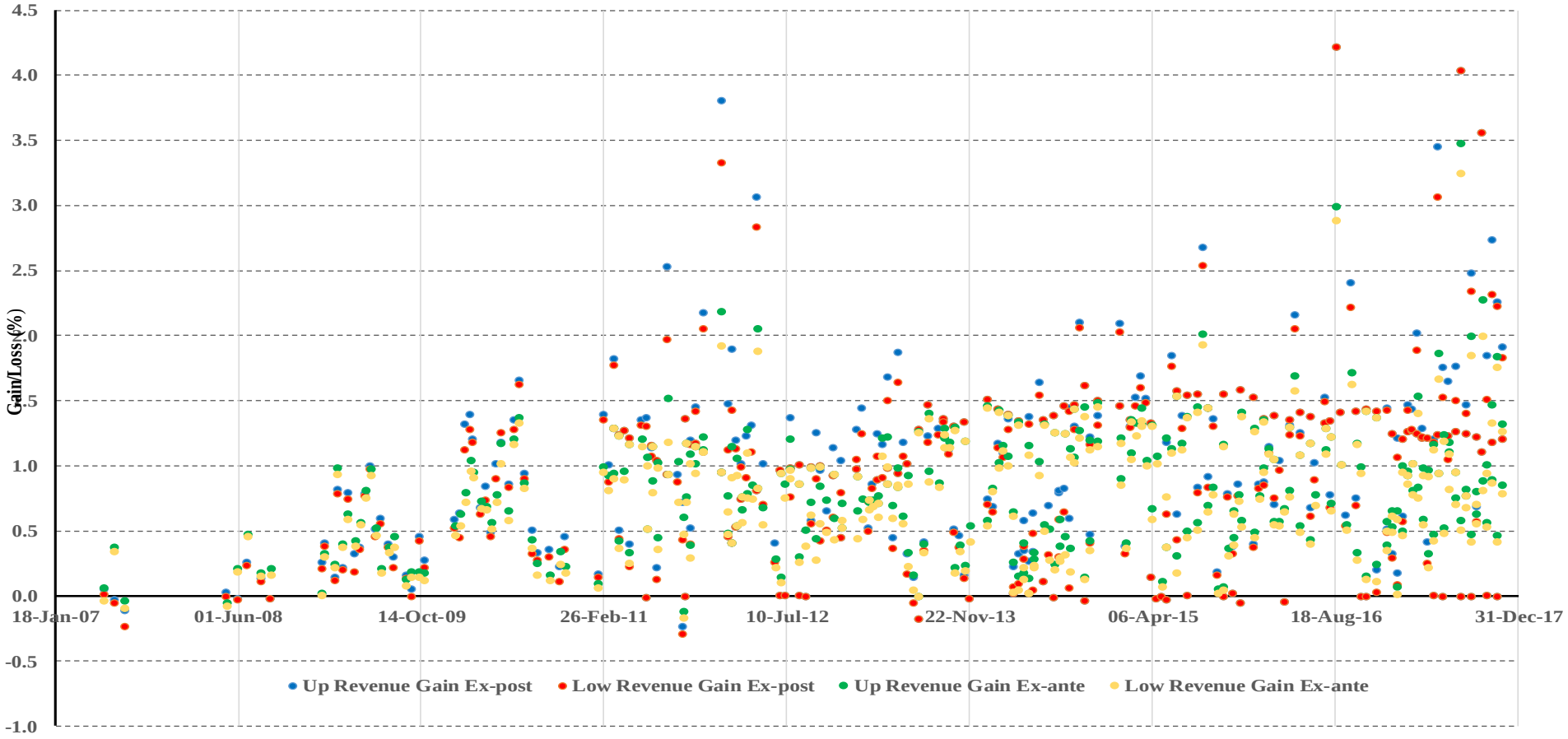

Figure 6.7: Counterfactual Revenue Estimates for Conventional Treasury Bill Auctions.

6.4.2 Efficiency Comparison

Table 6.8 displays the counterfactual efficiency estimation results for the conventional treasury bill auctions. The mean (median) ex-post efficiency gain is 0.23 (0.18) percent, with a t-statistic of 19.20. The ex-post estimation results are slightly more dramatic than the ex-ante results; the mean (median) ex-ante efficiency gain is 0.15 (0.13) percent, with a t-statistic of 24.64. There are 275 auctions with ex-post efficiency gain, with a mean (median) of 0.25 (0.19) percent. The are 296 auctions with ex-ante efficiency gain, with a mean (median) of 0.15 (0.13) percent. None of the auctions shows a loss in efficiency, while 22 (1) auctions show ex-post (ex-ante) revenue equivalence. Overall, the results suggest that switching to a uniform-price auction leads to gains in efficiency for conventional treasury bill auctions.

Figure 6.8 shows the scatterplot of the ex-post and ex-ante efficiency gains for conventional treasury bill auctions over time. The gains in efficiency seem to be relatively consistent over time. The highest potential efficiency gain is 1.4 percent, which occurred on 21 April 2011.

Table 6.8: ***Counterfactual Efficiency Estimation Results for Conventional Treasury Bill Auctions.***

This table presents the counterfactual efficiency estimation results for 297 treasury bill auctions. Ex-post is comparing the actual income value of a discriminatory-price auction and the hypothetical income generated by a uniform-price auction. Ex-ante is comparing the hypothetical income value of a discriminatory-price auction and the hypothetical income generated by uniform-price auction. Confidence interval level is 95 percent.

			Ex-Post	**CI Level**			**Ex-Ante**	**CI Level**	
All	*Mean*	N = 297	0.235	0.211	0.259	N = 297	0.152	0.140	0.164
	Median		0.180				0.130		
	Std Dev		0.211				0.106		
	Max		1.400				0.670		
	Min		0.000				0.000		
	t-value		19.204				24.641		
Gain	*Mean*	N =275	0.254	0.229	0.279	N = 296	0.152	0.140	0.165
	Median		0.190				0.130		
	Std Dev		0.208				0.106		
	Max		1.400				0.670		
	Min		0.010				0.010		
	t-value		20.237				24.727		
Equivalence		N = 22				N = 1			

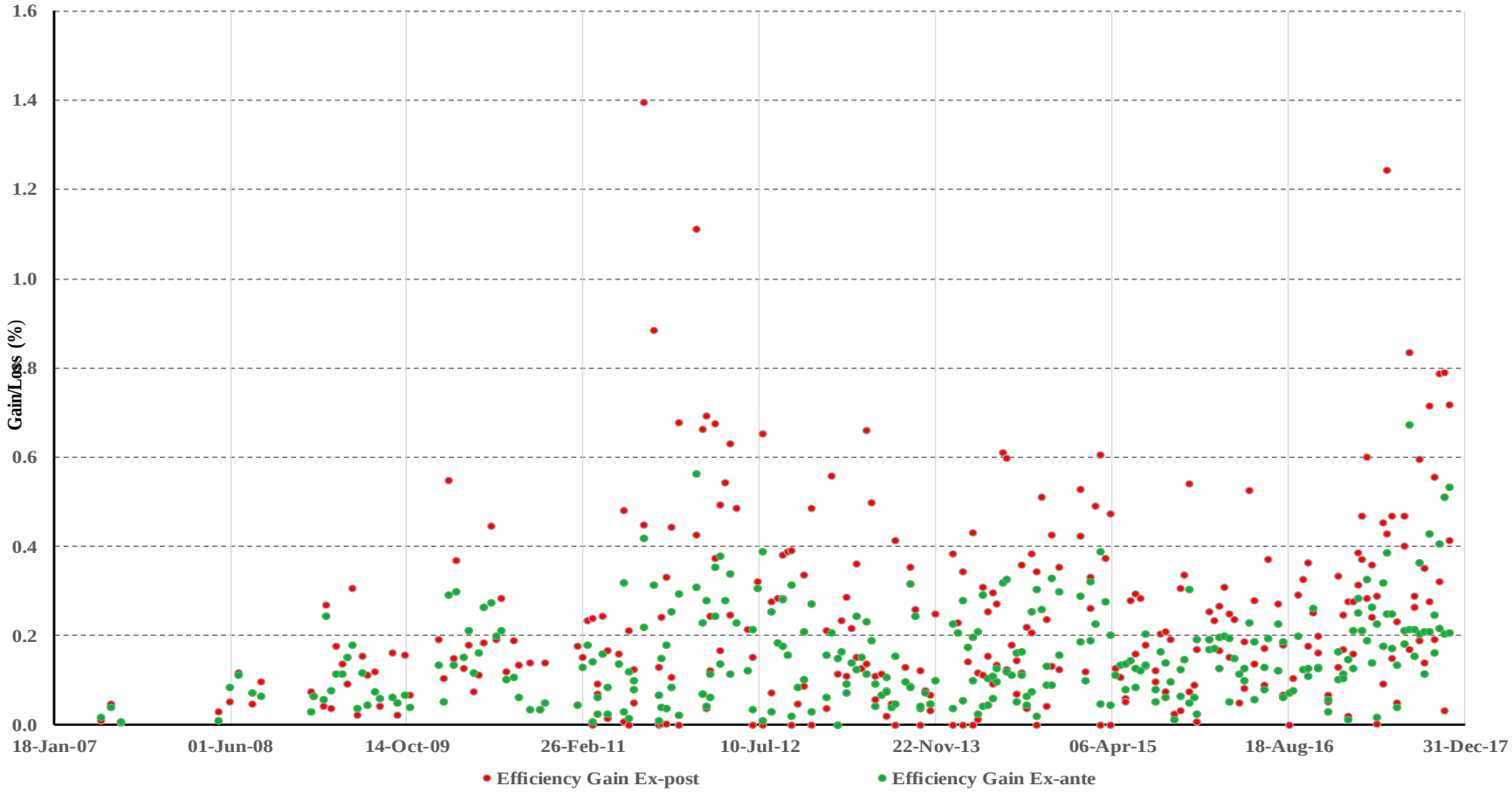

Figure 6.8:Counterfactual Efficiency Estimates for Conventional Treasury Bill Auctions.

6.5 Islamic Treasury Bond Auctions

6.5.1 Revenue Comparison

Table 6.9 presents the counterfactual revenue estimation results for Islamic treasury bond auctions. Unlike treasury bill auctions, the mean (median) ex-post upper revenue value for Islamic treasury bonds is negative at -1.64 (-0.29) percent, with a t-statistic of -6.72. This loss is slightly smaller for the ex-ante estimates, which has a mean (median) of -1.53 (-0.02) percent and a t-statistic of -6.35. There are 208 auctions with ex-post upper revenue loss, with a mean (median) of 3.80 (2.30) percent. There are 168 auctions with ex-ante upper revenue loss, with a mean (median) of 4.59 (4.15) percent. In contrast, there are 120 (163) auctions with ex-post (ex-ante) revenue gain, with a mean and median value of 2.08 (1.63) percent and 1.28 (1.63) percent, respectively. This result suggests that switching the auction format to a uniform-price auction is likely to decrease the auction efficiency for Islamic treasury bond auctions.

Table 6.10 shows that the lower revenue bound generates similar results. The mean (median) ex-post revenue value is -1.67 (-0.29) percent, with a t-statistic of -6.91. The mean (median) ex-ante revenue value is -1.57 (-0.02) percent, with a t-statistic -6.60. There are 208 (169) auctions with ex-post (ex-ante) revenue loss.

Finally, Figure 6.9 depicts the scatter plot of the counterfactual revenue estimation results for Islamic treasury bond auctions over time. Few auctions yield a revenue gain when switching to a uniform-price auction format.

Table 6.9: ***Counterfactual Revenue Estimation Results for Islamic Treasury Bond Auctions (Upper Bound).***

This table presents the counterfactual revenue estimation results for 331 Islamic treasury bonds auctions using upper bound revenue. Ex-post is comparing the actual income value of a discriminatory-price auction and the hypothetical income generated by a uniform-price auction. Ex-ante is comparing the hypothetical income value of a discriminatory-price auction and the hypothetical income generated by uniform-price auction. Confidence interval level is 95 percent.

			Ex-Post	**CI Level**			**Ex-Ante**	**CI Level**	
All	*Mean*	N = 331	-1.636	-2.115	-1.157	N = 331	-1.528	-2.002	-1.055
	Median		-0.290				-0.020		
	Std Dev		4.426				4.381		
	Max		7.500				6.220		
	Min		-26.950				-26.92		
	t-value		-6.725				-6.348		
Gain	*Mean*	N = 120	2.077	1.017	1.176	N = 163	1.629	1.430	1.828
	Median		1.635				1.280		
	Std Dev		1.699				1.289		
	Max		7.500				6.220		
	Min		0.030				0.010		
	t-value		13.384				16.137		
Loss	*Mean*	N = 208	-3.802	-0.094	-0.018	N = 168	-4.593	-5.223	-3.962
	Median		-2.305				-4.150		
	Std Dev		4.103				4.139		
	Max		-0.010				-0.020		
	Min		-26.950				-26.920		
	t-value		-13.365				-14.382		
Equivalence		N = 3				N = 0			

Table 6.10: ***Counterfactual Revenue Estimation Results for Islamic Treasury Bond Auctions (Lower Bound).***

This table presents the counterfactual revenue estimation results for 331 Islamic treasury bond auctions using lower bound revenue. Ex-post is comparing the actual income value of a discriminatory-price auction and the hypothetical income generated by a uniform-price auction. Ex-ante is comparing the hypothetical income value of a discriminatory-price auction and the hypothetical income generated by uniform-price auction. Confidence interval level is 95 percent.

			Ex-Post	**CI Level**			**Ex-Ante**	**CI Level**	
All	*Mean*	N = 331	-1.669	-2.144	-2.044	N = 331	-1.574	-2.044	-1.105
	Median		-0.290				-0.020		
	Std Dev		4.392				4.342		
	Max		7.500				6.160		
	Min		-26.950				-26.920		
	t-value		-6.914				-6.597		
Gain	*Mean*	N = 120	1.986	1.687	2.284	N = 162	1.552	1.362	1.743
	Median		1.580				1.270		
	Std Dev		1.651				1.225		
	Max		7.500				6.160		
	Min		0.030				0.010		
	t-value		13.172				16.131		
Loss	*Mean*	N = 208	-3.802	-4.363	-3.241	N = 169	-4.572	-5.201	-3.944
	Median		-2.305				-4.130		
	Std Dev		4.102				4.137		
	Max		-0.010				-0.020		
	Min		-26.950				-26.920		
	t-value		-13.366				-14.366		
Equivalence		N = 3				N = 0			

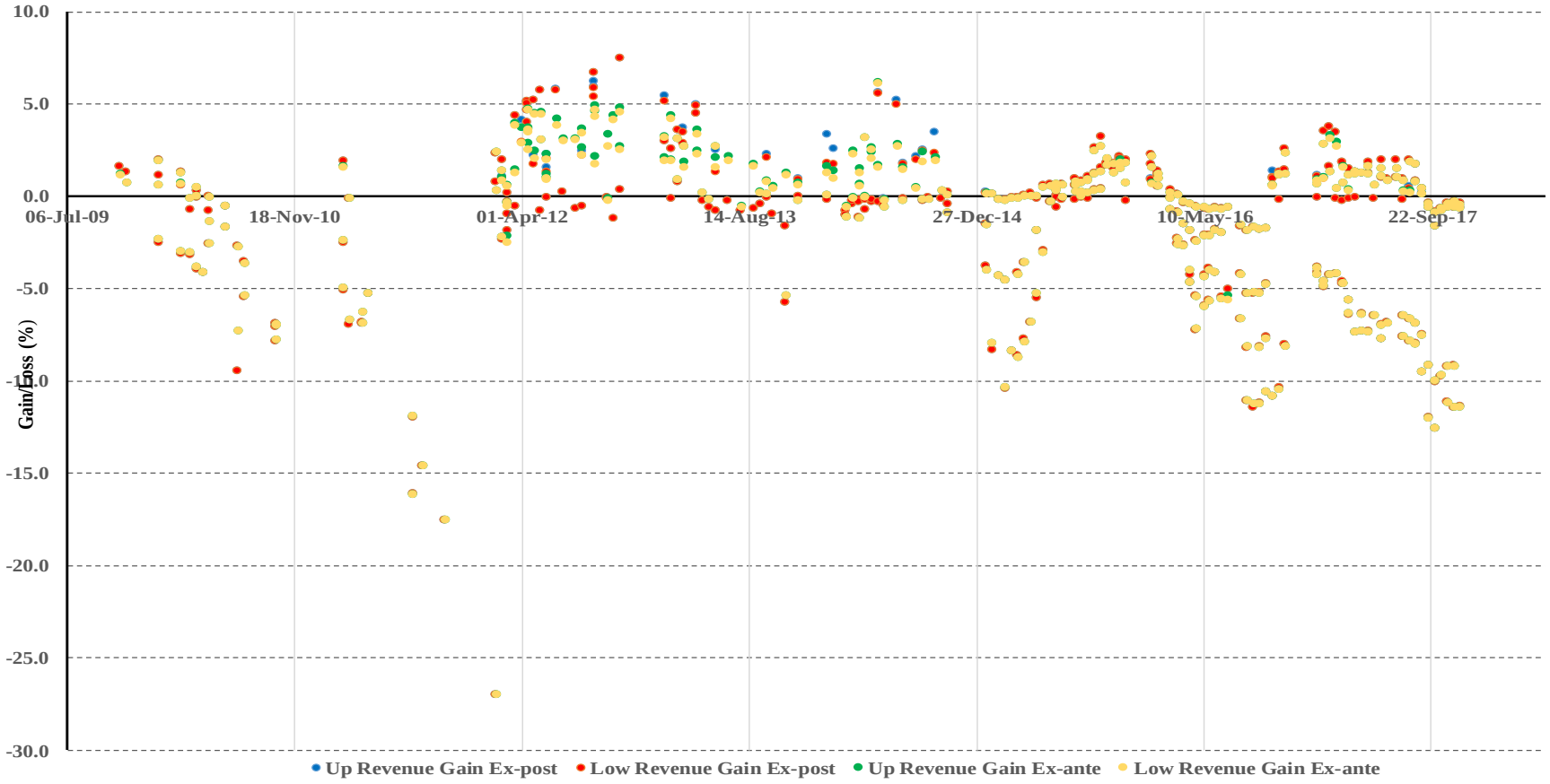

Figure 6.9: Counterfactual Revenue Estimates for Islamic Treasury Bond Auctions.

6.5.2 Efficiency Comparison

Table 6.11 displays the counterfactual efficiency estimation results for Islamic treasury bond auctions. In contrast to the findings for Islamic treasury bills, there is no significant economic gain in efficiency from switching to a uniform-price auction. For the upper revenue estimates, the mean (median) ex-post efficiency gain is a mere 0.17 percent (0.00 percent), with a t-statistic of 7.80. The mean (median) ex-ante efficiency is slightly lower at 0.11 percent (0.01 percent), with a t-statistic of 9.50. There are 173 (160) auctions with an ex-post (ex-ante) efficiency equivalence. There are 158 (171) auctions with ex-post (ex-ante) efficiency gain, with a mean and median of 0.36 (0.21) percent and 0.16 (0.12) percent, respectively. This result suggests that switching to a uniform-price auction format would not significantly increase auction efficiency for Islamic treasury bonds.

Figure 6.10 shows the scatter plot of the ex-post and ex-ante efficiency values for each of the auctions over time. Most efficiency values are well below zero, implying that uniform-price auctions generate lower efficiency than discriminatory-price auctions. The efficiency gains appear more dispersed in the period from 2012 to 2014. The highest (ex-post) efficiency gain occurs on 13 November 2012, which shows a net gain of 3.17 percent vs. 0.081 percent long-term average.

Table 6.11: ***Counterfactual Efficiency Estimation Results for Islamic Treasury Bond Auctions.***

This table presents the counterfactual efficiency estimation results for 331 Islamic treasury bonds auctions. Ex-post is comparing the actual income value of a discriminatory-price auction and the hypothetical income generated by a uniform-price auction. Ex-ante is comparing the hypothetical income value of a discriminatory-price auction and the hypothetical income generated by uniform-price auction. Confidence interval level is 95 percent.

			Ex-Post	**CI Level**			**Ex-Ante**	**CI Level**	
All	*Mean*	N = 331	0.170	0.128	0.212	N = 331	0.111	0.088	0.134
	Median		0.000				0.010		
	Std Dev		0.387				0.213		
	Max		3.170				1.850		
	Min		0.000				0.000		
	t-value		7.994				9.502		
Gain	*Mean*	N = 158	0.356	0.278	0.434	N = 171	0.215	0.177	0.254
	Median		0.160				0.120		
	Std Dev		0.498				0.256		
	Max		3.170				1.850		
	Min		0.010				0.010		
	t-value		8.990				11.001		
Equivalence		N = 173				N = 160			

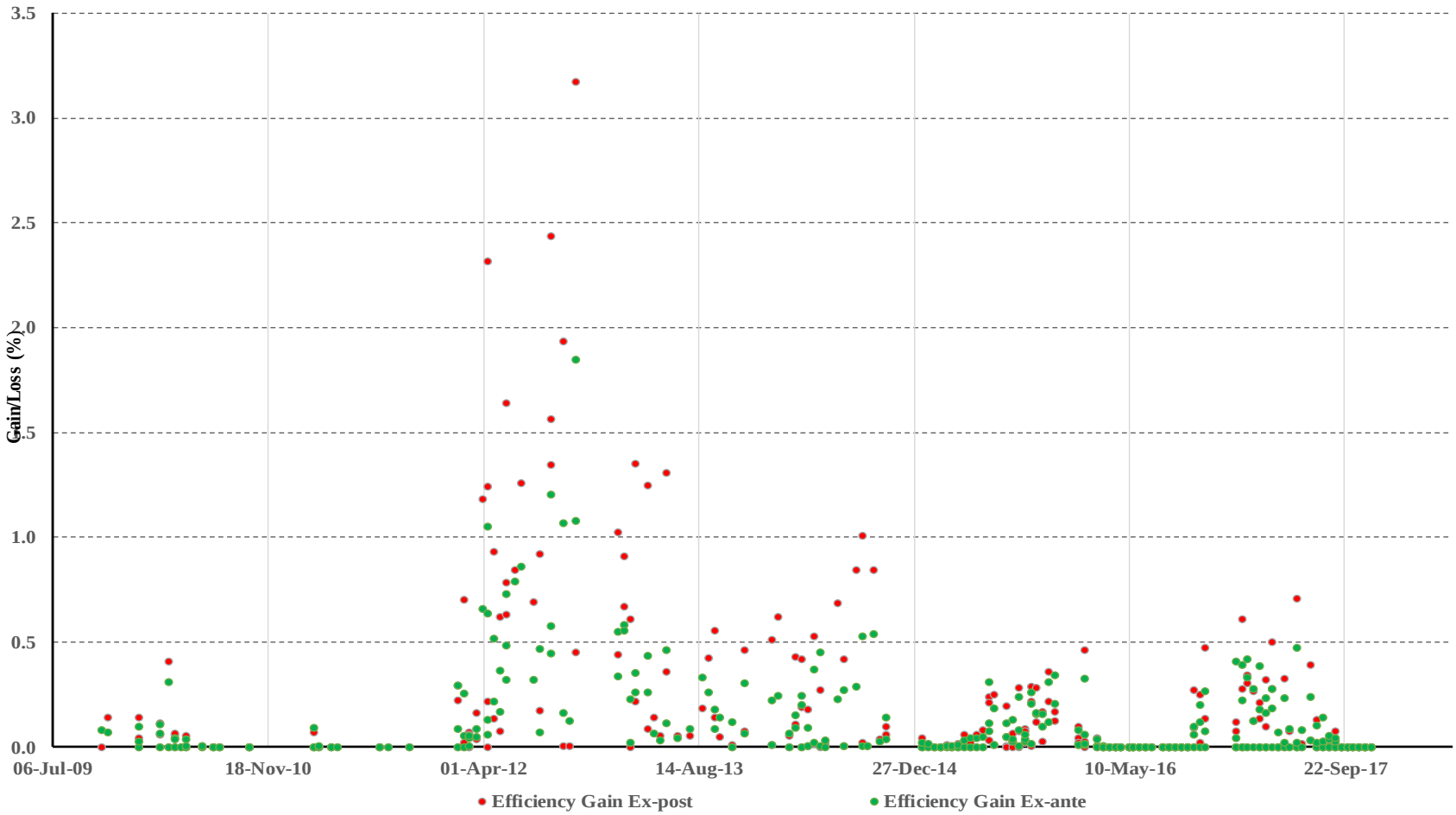

Figure 6.10: Counterfactual Efficiency Estimates for Islamic Treasury Bond Auctions.

6.6 Conventional Treasury Bond Auctions

6.6.1 Revenue Comparison

Table 6.12 displays the counterfactual revenue estimation results for conventional treasury bond auctions. The mean (median) ex-post upper revenue gain is -1.67 (-0.71) percent, with a t-statistic of -10.49. The ex-ante estimation results are similar, with a mean (median) revenue gain of -1.87 percent (-0.72 percent) and a t-statistic of -12.52.

More dramatically, more than half the auctions show a revenue loss. There are 366 auctions with ex-post revenue loss, with a mean (median) value of 4.01 (3.31 percent). For the ex-ante estimation, 369 auctions show a revenue loss, with a mean (median) of 3.97 (3.07) percent. In contrast, there are 260 auctions with ex-post upper revenue gain, with a mean (median) of 1.58 (1.15) percent). There are 163 auctions with ex-ante upper revenue gain, with a mean (median) of 1.08 (0.90) percent. These results are consistent with the results obtained for Islamic treasury bond auctions: switching to a uniform-price auction is likely to result in a loss of revenue for the MoF.

Table 6.13 provides counterfactual estimation results for conventional treasury bond auctions for the lower bound. In general, the lower bound estimation results are consistent with those for the upper bound, with uniform-price auctions generating lower revenue than discriminatory-price auctions on average. The mean (median) ex-post revenue gain is -1.71 (-0.71) percent, with a t-statistic of -10.84. The mean (median) ex-ante revenue gain is -1.90 (-0.72) percent, with a t-statistic of -12.78. Again, most of the auctions show a revenue loss. There are 368 auctions with ex-post revenue loss, with a mean (median) of 3.99 (3.12) percent. There are 372 auctions with ex-ante revenue loss, with a mean (median) of 3.94 (3.06) percent. In contrast, there are 258 auctions with ex-post revenue gain, with a mean (median) of 1.51 (1.13) percent. There are 258 auctions

with ex-ante revenue gain, with a mean (median) of 1.04 (0.86) percent. Overall, the results suggest that for conventional treasury bonds, discriminatory-price auctions favour uniform-price auctions in terms of revenue.

Figure 6.11 shows the scatter plot of revenue estimates for conventional treasury bond auctions, indicating that most revenue values are negative. In more recent periods, the revenue loss is more dispersed. The highest revenue loss is 19.9 percent, occurring on 21 February 2012.

Table 6.12: ***Counterfactual Revenue Estimation Results for Conventional Treasury Bond Auctions (Upper Bound).***

This table presents the counterfactual revenue estimation results for 632 conventional treasury bonds auctions using upper bound revenue. Ex-post is comparing the actual income value of a discriminatory-price auction and the hypothetical income generated by a uniform-price auction. Ex-ante is comparing the hypothetical income value of a discriminatory-price auction and the hypothetical income generated by uniform-price auction. Confidence interval level is 95 percent.

			Ex-Post	**CI Level**			**Ex-Ante**	**CI Level**	
All	*Mean*	N = 632	-1.671	-2.115	-1.157	N = 632	-1.872	-2.002	-1.055
	Median		-0.715				-0.720		
	Std Dev		4.004				3.759		
	Max		10.960				8.600		
	Min		-19.900				-19.870		
	t-value		-10.494				-12.522		
Gain	*Mean*	N = 260	1.581	1.017	1.176	N = 261	1.080	1.430	1.828
	Median		1.155				0.900		
	Std Dev		1.578				1.060		
	Max		10.960				8.600		
	Min		0.010				0.010		
	t-value		16.162				16. 455		
Loss	*Mean*	N = 366	-4.010	-0.094	-0.018	N = 369	-3.971	-5.223	-3.962
	Median		-3.135				-3.070		
	Std Dev		3.589				3.579		
	Max		-0.010				-0.010		
	Min		-19.900				-19.870		
	t-value		-21.373				-21.310		
Equivalence		N = 6				N = 2			

Table 6.13: ***Counterfactual Revenue Estimation Results for Conventional Treasury Bond Auctions (Lower Bound).***

This table presents the counterfactual revenue estimation results for 632 conventional Treasury bonds auctions using lower bound revenue. Ex-post is comparing the actual income value of a discriminatory-price auction and the hypothetical income generated by a uniform-price auction. Ex-ante is comparing the hypothetical income value of a discriminatory-price auction and the hypothetical income generated by uniform-price auction. Confidence interval level is 95 percent.

			Ex-Post	**CI Level**			**Ex-Ante**	**CI Level**	
All	*Mean*	N = 632	-1.708	-2.144	-2.044	N = 632	-1.898	-2.044	-1.105
	Median		-0.715				-0.720		
	Std Dev		3.960				3.734		
	Max		10.470				8.320		
	Min		-19.900				-19.870		
	t-value		-10.841				-12.776		
Gain	*Mean*	N = 258	1.509	1.687	2.284	N = 258	1.036	1.362	1.743
	Median		1.135				0.865		
	Std Dev		1.475				1.007		
	Max		10.470				8.320		
	Min		0.010				0.010		
	t-value		16.427				16.519		
Loss	*Mean*	N = 368	-3.991	-4.363	-3.241	N = 372	-3.942	-5.201	-3.944
	Median		-3.120				-3.065		
	Std Dev		3.590				3.579		
	Max		-0.010				-0.010		
	Min		-19.900				-19.870		
	t-value		-21.326				-21.246		
Equivalence		N = 6				N = 2			

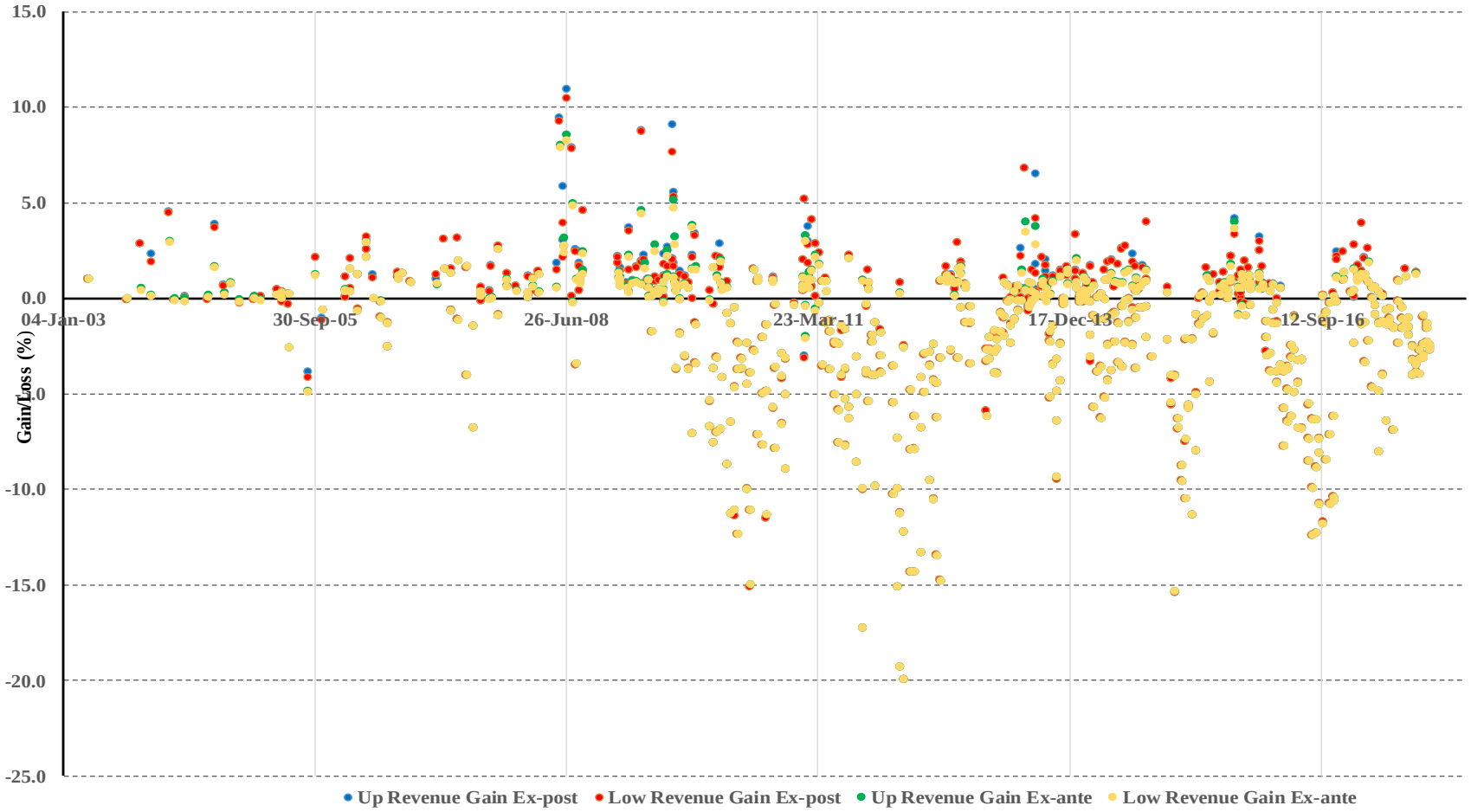

Figure 6.11: Counterfactual Revenue Estimates for Conventional Treasury Bond Auctions.

6.6.2 Efficiency Comparison

Table 6.14 shows the counterfactual efficiency estimation results for conventional treasury bond auctions. The mean (median) ex-post efficiency gain is positive but small at 0.08 (0.00) percent, with a t-statistic of 9.68. The mean (median) ex-ante efficiency gain is also positive but slightly smaller at 0.04 (0.00) percent, with a t-statistic of 11.94. While none of the auctions shows a loss in efficiency, more than half of the auctions show efficiency equivalence. In contrast, 258 (283) auctions show ex-post (ex-ante) efficiency gain, with a mean and median of 0.19 (0.10) percent and 0.10 (0.07) respectively. The results are in line with those of Islamic treasury bond auctions in that switching to a uniform-price auction provides no significant improvement in efficiency.

Figure 6.12 exhibits the scatter plot of the efficiency gains for conventional treasury bond auctions over time. Like the results for Islamic treasury bonds, most efficiency values are near zero. The highest efficiency gain is 2.3 percent for the auction on 30 July 2013. The efficiency gains appear to be relatively stable over time.

Table 6.14: ***Counterfactual Efficiency Estimation Results for Conventional Treasury Bond Auctions.***

This table presents the counterfactual efficiency estimation results for 632 conventional treasury bond auctions. Ex-post is comparing the actual income value of a discriminatory-price auction and the hypothetical income generated by a uniform-price auction. Ex-ante is comparing the hypothetical income value of a discriminatory-price auction and the hypothetical income generated by uniform-price auction. Confidence interval level is 95 percent.

			Ex-Post	**CI Level**			**Ex-Ante**	**CI Level**	
All	*Mean*	N = 632	0.081	0.064	0.097	N = 632	0.045	0.038	0.053
	Median		0.000				0.000		
	Std Dev		0.210				0.096		
	Max		2.340				0.960		
	Min		0.000				0.000		
	t-value		9.681				11.941		
Gain	*Mean*	N = 258	0.191	0.156	0.225	N = 283	0.102	0.087	0.116
	Median		0.100				0.070		
	Std Dev		0.288				0.122		
	Max		2.340				0.960		
	Min		0.010				0.010		
	t-value		10.823				14.045		
Equivalence		N = 364				N = 349			

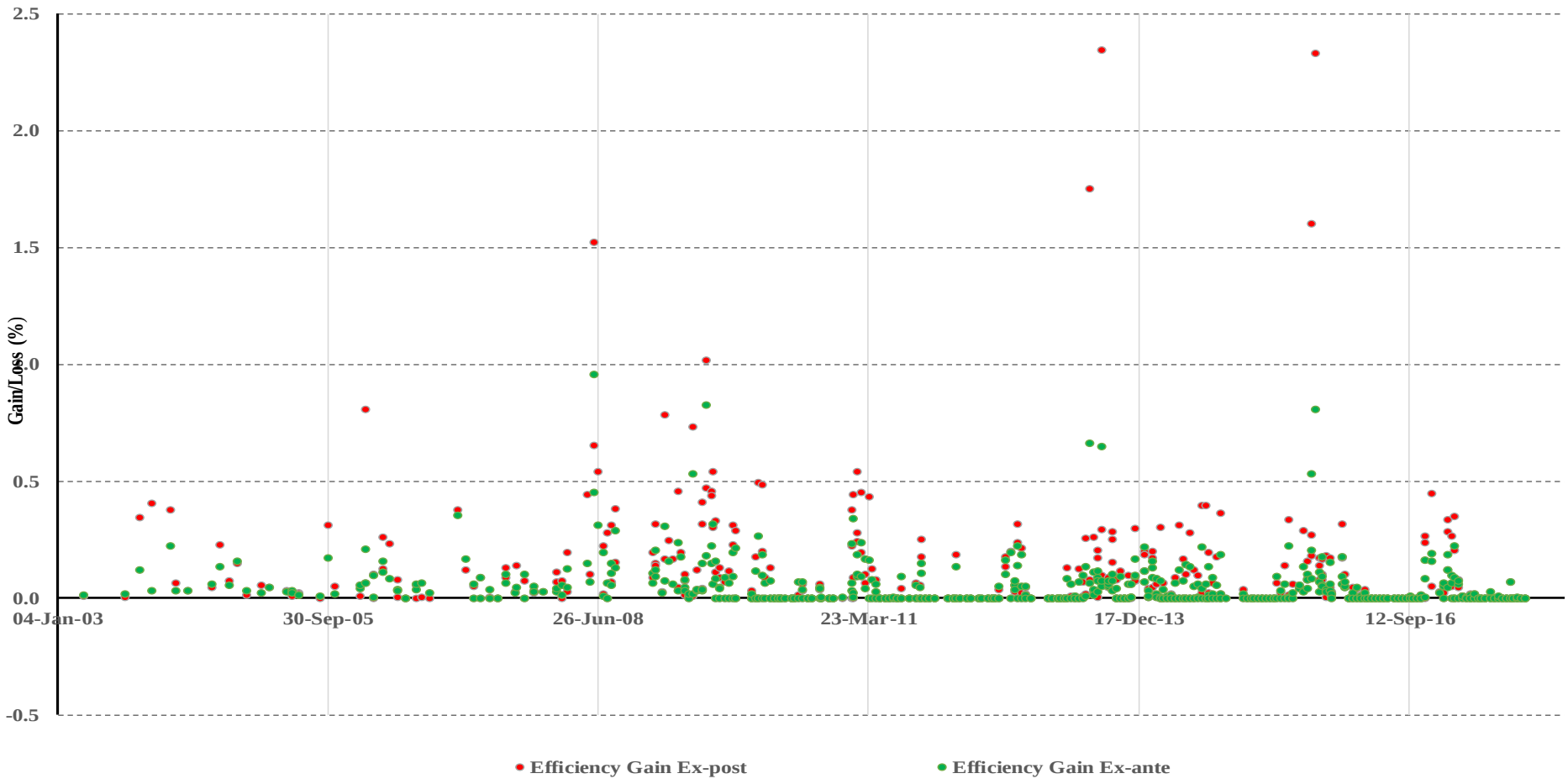

Figure 6.12: Counterfactual Efficiency Estimates for Conventional Treasury Bond Auctions.

6.7 Conclusion

In this chapter, I discuss the estimation results of performance gains achieved when switching from a discriminatory-price to a uniform-price auction format. I apply the structural econometric model of treasury auctions developed by Hortaçsu and McAdams (2010) on a data set of Islamic and conventional treasury auctions in Indonesia.

Table 6.15 provides a summary of the empirical results presented in this chapter, together with the findings of published studies on the topic that employ a structural model. Firstly, consistent with the proposition of multi-unit auction theory, I find that for both Islamic and conventional treasury bills, (ex-post and ex-ante) switching to a uniform-price auction yields improvements in revenue and efficiency (at the 95 percent confidence level). This finding is consistent with the argument that in discriminatory-price auctions, bidders will use their best effort to win the auction with the lowest price (Nautz & Wolfstetter, 1997). Doing so helps bidders maximise their profit and avoid the winner's curse by summitting excessively high bids (Gordy, 1999). Since bidders submit higher bids in uniform-price auctions than in discriminatory-price auctions (Damianov & Becker, 2010), uniform-price auctions generate higher revenue for the issuer.

Secondly, for Islamic and conventional treasury bonds, I find discriminatory-price auctions outperform (at the 95 percent confidence level) uniform-price auctions, yielding higher revenue. While inconsistent with the empirical results for the treasury bills, it is consistent with the view that participants in uniform-price auctions can support a collusive Nash equilibrium; thus bidders can steepen their schedules, enough to neutralise the elasticity supply of sellers (Goswami et al., 1996). Bidders prefer to coordinate on equilibrium with a highly inelastic announced demand so that the awarded price will be low and less than the common knowledge lower bound on all bidders' marginal value.

Winning bidders affect the market clearing price with a positive probability. Therefore, bidders are incentivised to shade their bids in uniform-price auctions (Ausubel et al., 2014). However, each bidder wins in zero bid equilibria at a maximum of one item, even if his value for another item exceeds that of other bidders in a discriminatory-price auction (Engelbrecht-Wiggans & Kahn, 1998a). Bidders symmetrically create an inelastic or flat demand function, thus discriminatory-price auctions generate a higher revenue than uniform-price auctions (Daripa, 2001). When the dispersion of information is high and the level of competition is low, bidders prefer to submit bids close to the true value to guarantee their position in the discriminatory-price auction. Since bidders prefer to secure their position, the winning price in discriminatory-price auctions is on average higher than the winning price in uniform-price auctions.

Lastly, for both Islamic and conventional treasury bonds, I find efficiency equivalence between discriminatory-price and uniform-price auctions (at the 95 percent confidence level). Thus, switching to a uniform-price auction has inconsequential effects on auction efficiency. This is consistent with the argument that in discriminatory-price auctions, bidders shade their bids according to the effectiveness of getting an allocation. This is because a bid for an additional unit does not affect the price paid for previous units (Engelbrecht-Wiggans & Kahn, 1998b). However, in uniform-price auctions, differential bid incentives occur when a winner has a favourable likelihood of influencing the market clearing price (Ausubel et al., 2014). Differential shading is present as the incentive to lower demand rises with the amount of uncertainty.

Table 6.15: ***Summary of Results for Structural Models.***

Source: Author's table based on Marszalec (2017). H indicates the model of Hortaçsu (2002). HMA indicates the model of Hortaçsu & McAdams (2010). FPV indicates the model of Fevrier, Preget and Visser (2004). Revenue superiority refers to the performance of uniform-price auctions vs. discriminatory-price auctions. The model used by Armantier and Sbai (2006) is parametric and includes risk-aversion.

Study	Method	Country	Time Period	Revenue Superiority	Efficiency Superiority
Panel A: Treasury bonds					
This thesis	HMA	Indonesia	Islamic treasury bonds	D by up to 1.64 percent, significant	Efficiency equivalence
This thesis	HMA	Indonesia	Conventional treasury bonds	D by up to 1.67 percent, significant	Efficiency equivalence
Kang & Puller (2008)	H	Korea	3-year treasury bonds	D 0.1 percent, significant	D 0.1 percent, significant
Armantier & Sbai (2006)	Other	France	Treasury bonds	UP by 4.8 percent, significant	-
Panel B: Treasury bills					
Marszalec (2017)	HMA	Poland	53-week treasury bills, 2-year treasury bonds	D by up to 0.07 percent, significant	-
Marszalec (2017)	FPV	Poland	53-week treasury bills, 2-year treasury bonds	D by up to 1.5 percent, significant	-
Hortaçsu (2002)	H	Turkey	13-week treasury bills	D by 14 percent, not significant	UP by 11 percent, significant
Fevrier, Preget, & Visser (2004)	FPV	France	Treasury bills	D by 2 percent, significant	-
Castellanos & Oviedo (2008)	FPV	Mexico	Treasury bills	UP by 2 percent, significant	-
Hortaçsu & McAdams (2010)	HMA	Turkey	13 weeks treasury bills	UP by 0.1 percent, not significant	UP by 0.035 percent, significant
Kastl (2011)	H-type	Czech	3-month treasury bills	UP price may exceed truthful bidding	UP by 0.04 percent, significant
This thesis	HMA	Indonesia	6-month Islamic treasury bills	UP by up to 0.43 percent, significant	UP by up to 0.06 percent, significant
This thesis	HMA	Indonesia	3- and 12-month conventional treasury bills	UP by up to 0.98 percent, significant	UP by up to 0.23 percent, significant

Chapter 7

Determinants of Revenue and Efficiency Gains

7.1 Introduction

This chapter discusses the empirical results of the cross-sectional determinants of revenue and efficiency gains when switching to a uniform-price auction. Section 7.2 presents the pairwise-correlation analysis. Section 7.3 and 7.4 present the results from the multiple variable regressions, where I estimate the effect of auction characteristic on revenue and efficiency gains using the ex-post estimation results. This is followed by a robustness test using ex-ante estimation results. Finally, I summarise and conclude the chapter in Section 7.5.

7.2 Pearson Correlation Analysis

Table 7.1 displays the Pearson (linear) pairwise correlations between the test variables. *Revenue* is significantly positively correlated with *Efficiency*. *Upper Revenue* has a very strong positive correlation with *Lower Revenue* as expected. *Bidding* and *Bidders* have a small negative correlation with *Efficiency* and *Revenue*, indicating that an increase in the number of *Bidders* reduce the gains in *Revenue* and *Efficiency*. *BTC* is negatively and significantly correlated with *Efficiency*, suggesting that increasing demand for the treasury reduces gains in *Efficiency*. *Spread* is significantly negative correlated with *Revenue*, but not with *Efficiency*, suggesting that higher bid dispersion in auctions reduces gains in *Revenu*e. *Tail* is positively correlated with *Revenue*, *Bidding* and *BTC*, but it is negatively correlated with *Bidders* and *Spread*. These results suggest that more disperse bids (*Tail)* yields higher *Revenue* gains. *Volatility* has a significant negative correlation with *Revenue*, but a significant positive correlation with *Efficiency*, suggesting

that higher uncertainty reduces gains in *Revenue* but increases gains in *Efficiency*. *Maturity* is significantly correlated with all variables, except *Bidding and Volatility*, suggesting that longer maturity reduces gains in *Revenue* and *Efficiency*.

7.3 Testing for Multicollinearity

I analyse the variance inflation factor tolerance statistic to confirm if there is a multicollinearity problem. The results from Table 7.2 show that multicollinearity is not a problem because all p-values of VIF are smaller than 10.

Table 7.1: ***Pearson Correlation Matrix of Regression Variables.***

This table presents Pearson pairwise correlation matrix of test variables. Variable definitions are provided in Table 5.2. ***, **, * denote significance at the 1, 5, and 10 percent level, respectively. ***, **, * denote significance at the 1, 5, and 10 percent level, respectively.

		1	*2*	*3*	*4*	*5*	*6*	*7*	*8*	*9*	*10*	*11*
1	*Ex-post Up Revenue*	1.0000										
2	*Ex-post Low Revenue*	0.9992*	1.0000									
3	*Ex-post Efficiency*	0.4521*	0.4361*	1.0000								
4	*Ex-ante Efficiency*	0.4878*	0.4771*	0.9019*	1.0000							
5	*Bidding*	-0.0119	-0.0103	-0.1229*	-0.1517*	1.0000						
6	*Bidders*	-0.2584*	-0.2533*	-0.2555*	-0.2954*	0.6390*	1.0000					
7	*BTC*	-0.0152	-0.0176	-0.0716*	-0.0930*	0.1889*	-0.0231	1.0000				
8	*Spread*	-0.1800*	-0.1805*	0.005	-0.0154	-0.1253*	0.2651*	-0.1169*	1.0000			
9	*Tail*	0.1578*	0.1564*	0.0467	0.0540*	0.0579*	-0.1814*	0.1390*	-0.1278*	1.0000		
10	*Volatility*	-0.1573*	-0.1597*	0.0059	0.0071	-0.2440*	-0.0755*	-0.0138	0.0834*	-0.0648*	1.0000	
11	*Maturity*	-0.3438*	-0.3432*	-0.0937*	-0.1487*	-0.0886*	0.4691*	-0.2244*	0.6214*	-0.3788*	0.0416	1.0000

Table 7.2: ***Variance inflation factor results of test variables***

	Full Sample		Bills		Bonds		Islamic		Conventional	
	VIF	1/VIF	VIF	1/VIF	VIF	1/VIF	VIF	1/VIF	VIF	1/VIF
Bidding	3.28	0.305337	2.15	0.46611	3.48	0.287077	3.20	0.312584	3.96	0.252776
Bidders	2.78	0.359568	2.02	0.494957	3.43	0.291221	2.73	0.366643	3.13	0.319391
BTC	2.75	0.363642	1.49	0.669918	1.37	0.727784	2.36	0.423522	2.14	0.467087
Spread	1.69	0.590307	1.35	0.742437	1.35	0.742975	1.91	0.52385	1.64	0.608017
Tail	1.20	0.830148	1.22	0.816448	1.09	0.920065	1.21	0.826182	1.23	0.813499
Volatility	1.10	0.910847	1.21	0.829143	1.09	0.92119	1.14	0.877687	1.14	0.876934
Maturity	1.09	0.917775	1.11	0.902123	1.04	0.964075	1.13	0.887597	1.07	0.938283
Mean VIF	1.98		1.51		1.84		1.95		2.04	

7.3 Determinants of Revenue Gain

7.3.1 Upper Bound Analysis

Table 7.3 reports the results for the determinants of ex-post upper revenue gain. I run the regression for the full sample and the subsamples of treasury bills versus treasury bonds, and Islamic versus conventional treasuries. I note that the explanatory power of the regression for the subsample of treasury bills is substantially higher than for the subsample of treasury bonds (is 0.184 percent vs. 0.076 percent). I attribute this to the finding that the gains in auction revenue are substantially smaller for treasury bonds than for treasury bills, as extensively documented in the previous chapter.

In specification (1) for the full sample, the coefficient on *Bidding* and *Bidders*, which proxy for the level of competition in the auction, is negative and significant (at the 1 and 5 percent level respectively). These results are replicated across the subsamples of treasury bills, treasury bonds and Islamic treasuries for *Bidders*. The coefficient on *BTC* is also negative, it is statistically significant for all the subsample indicating strong consistency. Overall, these findings are consistent with my proposition (H3a) that higher competition reduces revenue gains when switching to a uniform-price auction format. When the level of competition amongst bidders is high, the existence of the winner's curse in discriminatory-price auctions is reduced, while the opportunity for collusive equilibria decreases in uniform-price auctions (Ausubel, 2004; Bolten, 1973; Milgrom & Weber, 1982).

For treasury bills, the coefficient on *Bidding* and *Bidders*, which proxy for the level of competition in the auction, is negative and significant (at the 1 and 5 percent level respectively). These results are replicated across the subsamples of treasury bills, treasury bonds and Islamic treasuries for *Bidders*. The coefficient on *BTC* is also negative, it is

statistically significant for all the subsample indicating strong consistency. Overall, these findings are consistent with my proposition (H3a) that higher competition reduces revenue gains when switching to a uniform-price auction format.

The coefficients on *Spread* and *Tail*, the proxy for uncertainty, are mostly inconsistent across the regression specifications, and mostly statistically insignificant. On the other hand, the coefficient on *Volatility* is negative and significant at the 1 percent level in all specifications, except for the subsample of treasury bills. This finding does not lend any evidence for hypothesis H4a, that higher uncertainty favours the uniform-price auction in terms of revenue for the issuer. Uniform-price auctions entail more uncertainty than discriminatory-price auctions, and thus more equilibria are possible for the former. Consequently, in uniform-price auctions, a new entrant may cause major price swings. The quantity to be sold can often be changed, and participants may employ mixed strategies. Assuming private valuation, there is greater uncertainty as to the outcome of the uniform-price auctions. This may reduce demand and revenue (Binmore & Swierzbinski, 2000). The higher the uncertainty, the more cautious (lower) the bidders are in the price submitted. Market participants understand that absorbing uncertainty will provide economic benefits to them. Others have shown that higher uncertainty reduces bidding prices and thus auction revenue (Bjonnes, 2001; Goldreich, 2007; Nyborg et al., 2002; Nyborg & Strebulaev, 2004). When uncertainty increases, bidders reduce the price level at which they bid, reducing the quantity demanded thus decreasing the revenue for issuers (Cardozo, 2013).

The control variable is statistically significant in explaining revenue gain. Maturity has a negative association, suggesting that for longer *maturity* treasuries switching from a discriminatory-price to a uniform-price auctions are less beneficial to

the issuer in terms of revenue generating. Longer *maturity* suggests higher uncertainty, so bidders may prefer to reduce their bid prices. The diminishing bid price results in a decrease in issuer's revenue. Thus, when switching to a uniform-price auction format, revenue drops.

For robustness and to arrive at reliable conclusions, I rerun the tests on the determinants of revenue gain using the ex-ante estimates. The results are reported in Table 7.4. Overall, the results from ex-post and ex-ante analysis are remarkably similar. The coefficient on *Bidding* and *Bidders* remain negative and significant (at the 1 percent level) across the regressions, indicating that increasing level of competition reduces revenue gain, consistent with H3a. None of the proxies for uncertainty are in the predicted directions, lending no support for H4a.

Table 7.3: ***Regressions of Determinants of Ex-post Upper Revenue Gain.***

This table presents the results of the OLS regression model for auctions' characteristics on revenue gain. Variable definitions are provided in Table 5.2. ***, **, * denote significance at the 1, 5, and 10 percent level, respectively. T-statistics in parentheses.

	Full Sample	**Bills**	**Bonds**	**Islamic**	**Conventional**
	(1)	(2)	(3)	(4)	(5)
Bidding	-0.308*	-0.543***	0.269	0.51	-0.247
	(-1.83)	(-7.396)	(1.11)	(1.583)	(-1.035)
Bidders	-1.500***	-0.603***	-1.712***	-3.134***	-0.396
	(-3.912)	(-4.260)	(-2.994)	(-4.748)	(-0.818)
BTC	-0.164***	-0.054***	-0.242***	-0.180***	-0.113**
	(-4.122)	(-4.059)	(-3.914)	(-2.606)	(-2.559)
Spread	0.089*	-0.227*	0.089	0.191	0.053
	(1.682)	(-1.928)	(1.619)	(1.617)	(0.843)
Tail	0.011	-0.001	0.013	-0.001	0.017
	(0.921)	(-0.346)	(0.426)	(-0.024)	(1.005)
Volatility	-0.567***	-0.062	-0.646***	-0.455**	-0.445***
	(-5.024)	(-0.945)	(-4.938)	(-2.073)	(-2.805)
Maturity	-1.046***	-0.096	-0.976***	-1.531***	-1.186***
	(-6.644)	(-0.458)	(-4.771)	(-3.903)	(-6.184)
Constant	3.822***	-0.408	4.681***	6.987***	4.194***
	(5.966)	(-1.301)	(5.374)	(4.914)	(4.339)
Observations	1359	396	963	430	929
adj. R-sq	0.163	0.184	0.076	0.213	0.152

Table 7.4: ***Regressions of Determinants of Ex-ante Upper Revenue Gain.***

This table presents the results of the OLS regression model for auctions' characteristics on revenue gain. Variable definitions are provided in Table 5.2. ***, **, * denote significance at the 1, 5, and 10 percent level, respectively. T-statistics in parentheses.

	Full Sample	**Bills**	**Bonds**	**Islamic**	**Conventional**
	(1)	(2)	(3)	(4)	(5)
Bidding	0.236	0.383***	0.241	0.501	-0.319
	(1.463)	(8.256)	(1.041)	(1.579)	(-1.453)
Bidders	-1.589***	-0.489***	-1.895***	-3.333***	-0.242
	(-4.248)	(-5.451)	(-3.414)	(-5.070)	(-0.546)
BTC	-0.147***	-0.044***	-0.223***	-0.185***	-0.085**
	(-3.877)	(-5.206)	(-3.739)	(-2.880)	(-2.007)
Spread	0.102*	-0.179**	0.104*	0.251**	0.03
	(1.953)	(-2.324)	(1.929)	(2.117)	(0.493)
Tail	0.008	-0.002	0.007	-0.004	0.016
	(0.792)	(-0.876)	(0.234)	(-0.261)	-0.987
Volatility	-0.520***	-0.160***	-0.581***	-0.467**	-0.431***
	(-4.761)	(-3.336)	(-4.578)	(-2.194)	(-2.817)
Maturity	-1.034***	-0.374**	-0.981***	-1.546***	-1.211***
	(-6.872)	(-2.581)	(-5.009)	(-4.006)	(-6.828)
Constant	4.209***	0.233	5.117***	7.444***	4.053***
	(6.77)	(1.085)	(6.022)	(5.27)	(4.483)
Observations	1359	396	963	430	929
adj. R-sq	0.171	0.277	0.082	0.22	0.169

7.3.2 Lower Bound Analysis

The previous analysis for upper bound estimation shows support for H3a in that increasing competition amongst bidders reduces the gains in auction revenue when switching to a uniform-price auction. To establish robustness, I rerun the regressions using the lower bound ex-post and ex-ante estimates of revenue gain in Table 7.5 and 7.6, respectively. Overall, the results from upper bound and lower bound analysis are remarkably similar, providing support for H3a—competition amongst bidders in the auction decreases revenue gain for the issuer when switching from a discriminatory-price to a uniform-price auction format. I find no evidence consistent with H4a in that uncertainty increases revenue gain when switching to a uniform-price auction format.

Table 7.5: ***Regressions of Determinants of Ex-post Lower Revenue Gain.***

This table presents the results of the OLS regression model for auctions' characteristics on revenue gain. Variable definitions are provided in Table 5.2. ***, **, * denote significance at the 1, 5, and 10 percent level, respectively. T-statistics in parentheses.

	Full Sample	**Bills**	**Bonds**	**Islamic**	**Conventional**
	(1)	(2)	(3)	(4)	(5)
Bidding	0.278*	0.530***	0.246	0.467	-0.274
	(1.671)	(7.627)	(1.027)	(1.462)	(-1.170)
Bidders	-1.393***	-0.497***	-1.621***	-3.023***	-0.261
	(-3.687)	(-3.747)	(-2.873)	(-4.603)	(-0.556)
BTC	-0.166***	-0.062***	-0.242***	-0.180***	-0.114***
	(-4.218)	(-5.065)	(-3.958)	(-2.656)	(-2.624)
Spread	0.087*	-0.232**	0.087	0.188	0.05
	(1.65)	(-2.119)	(1.595)	(1.608)	(0.793)
Tail	0.01	-0.002	0.013	-0.001	0.017
	(0.868)	(-0.475)	(0.416)	(-0.087)	(1.021)
Volatility	-0.572***	-0.124*	-0.644***	-0.478**	-0.439***
	(-5.114)	(-1.896)	(-4.970)	(-2.206)	(-2.778)
Maturity	-1.054***	-0.318	-1.004***	-1.517***	-1.209***
	(-6.805)	(-1.576)	(-4.983)	(-3.929)	(-6.448)
Constant	3.682***	-0.489	4.588***	6.878***	3.989***
	(5.796)	(-1.600)	(5.315)	(4.871)	(4.173)
Observations	1359	396	963	430	929
adj. R-sq	0.162	0.215	0.076	0.212	0.151

Table 7.6: ***Regressions of Determinants of Ex-ante Lower Revenue Gain.***

This table presents the results of the OLS regression model for auctions' characteristics on revenue gain. Variable definitions are provided in Table 5.2. ***, **, * denote significance at the 1, 5, and 10 percent level, respectively. T-statistics in parentheses.

	Full Sample	**Bills**	**Bonds**	**Islamic**	**Conventional**
	(1)	(2)	(3)	(4)	(5)
Bidding	0.219	0.359***	0.237	0.476	-0.333
	(1.369)	(8.042)	(1.028)	(1.509)	(-1.530)
Bidders	-1.508***	-0.395***	-1.837***	-3.246***	-0.161
	(-4.069)	(-4.585)	(-3.340)	(-4.968)	(-0.367)
BTC	-0.149***	-0.049***	-0.224***	-0.187***	-0.085**
	(-3.966)	(-6.186)	(-3.793)	(-2.967)	(-2.039)
Spread	0.098*	-0.186**	0.100*	0.244**	0.027
	(1.884)	(-2.583)	(1.872)	(2.078)	(0.448)
Tail	0.008	-0.003	0.006	-0.005	0.015
	(0.724)	(-1.148)	(0.214)	(-0.325)	(0.974)
Volatility	-0.524***	-0.200***	-0.581***	-0.480**	-0.428***
	(-4.837)	(-4.256)	(-4.613)	(-2.282)	(-2.804)
Maturity	-1.038***	-0.517***	-1.000***	-1.540***	-1.218***
	(-6.970)	(-3.647)	(-5.161)	(-4.039)	(-6.959)
Constant	4.070***	0.195	5.011***	7.335***	3.895***
	(6.598)	(0.939)	(5.945)	(5.23)	(4.346)
Observations	1359	396	963	430	929
adj. R-sq	0.17	0.305	0.082	0.22	0.167

7.4 Determinants of Efficiency Gain

To test my hypotheses of the impact of auction competition and uncertainty on efficiency gain, I perform ordinary least squares regressions. Table 7.7 reports the results where the dependent variable is efficiency gain. As for regressions of revenue gain, the regressions for the subsample of treasury bills has the highest explanatory power.

The proxies for competition amongst bidders, i.e., *Bidders* and *BTC* are negatively and significant in the majority (8/10) of regression specifications. Although the coefficient of *Bidding* is significant, the direction is positive and not as expected. This finding is consistent with H3b in that increasing competition reduce efficiency gain when switching to a uniform-price auction format. This is consistent with Ausubel's et al.'s (2014) finding that that increasing investor demand reduces the gap in efficiency between uniform and discriminatory-price auctions.

Spread is positively and significantly associated with efficiency gain in specifications (1), (3) and (5), consistent with H4b that higher uncertainty favours the uniform-price auction format in terms of efficiency. The result is consistent with Lu & Yang (2003) in that bidders in a discriminatory-price auction bid more cautiously when market uncertainty is high to avoid the winner's curse. None of the results for the other proxies of uncertainty offers any support for H4b.

Maturity has a significantly positive relationship with efficiency gain in specification (2), (3) and (4). That is, a longer maturity encourages bidders to bid more efficient in uniform-price auctions, increasing efficiency gain.

For robustness, I rerun the tests using the ex-ante estimates of efficiency gain. The results are reported in Table 7.8. The results are nearly identical to those using ex-post

estimates of efficiency gains, discussed above. Hence, consistent with H3b, increasing competition among bidders reduces efficiency gains when switching to a uniform-price auction format.

Table 7.7: ***Regressions of Determinants of Ex-post Efficiency Gain.***

This table presents the results of the OLS regression model for auctions' characteristics on efficiency gain. Variable definitions are provided in Table 5.2. ***, **, * denote significance at the 1, 5, and 10 percent level, respectively. T-statistics in parentheses.

	Full Sample	**Bills**	**Bonds**	**Islamic**	**Conventional**
	(1)	(2)	(3)	(4)	(5)
Bidding	0.042***	0.144***	-0.006	0.016	0.050***
	(3.728)	(6.064)	(-0.451)	(0.78)	(3.157)
Bidders	-0.191***	-0.192***	-0.123***	-0.102**	-0.188***
	(-7.327)	(-4.533)	(-3.644)	(-2.519)	(-5.115)
BTC	-0.010***	-0.021***	-0.006	-0.006	-0.012***
	(-3.809)	(-5.976)	(-1.593)	(-1.072)	(-4.980)
Spread	0.009***	-0.043	0.007**	0.006	0.005**
	(3.338)	(-1.205)	(2.55)	(0.905)	(2.437)
Tail	0.01	-0.001	0.002	0.001	0.001
	(-0.031)	(-0.872)	(1.069)	(0.733)	(0.473)
Volatility	0.001	0.022	0.001	0.01	-0.005
	(0.203)	(1.341)	(0.141)	(0.605)	(-0.798)
Maturity	0.005	0.366***	0.036***	0.056**	-0.015
	(0.517)	(6.367)	(2.886)	(2.376)	(-1.120)
Constant	0.396***	-0.264***	0.370***	0.183**	0.402***
	(9.934)	(-3.104)	(6.986)	(2.252)	(6.975)
Observations	1359	396	963	430	929
adj. R-sq	0.083	0.247	0.082	0.069	0.152

Table 7.8: ***Robustness Regressions of Determinants of Ex-ante Efficiency Gain.***

This table presents the results of the OLS regression model for auctions' characteristics on efficiency gain. Variable definitions are provided in Table 5.2. ***, **, * denote significance at the 1, 5, and 10 percent level, respectively. T-statistics in parentheses.

	Full Sample	**Bills**	**Bonds**	**Islamic**	**Conventional**
	(1)	(2)	(3)	(4)	(5)
Bidding	0.017***	0.078***	-0.011	-0.001	0.021***
	(2.818)	(7.284)	(-1.551)	(-0.049)	-2.821
Bidders	-0.097***	-0.078***	-0.060***	-0.052**	-0.084***
	(-7.180)	(-4.071)	(-3.292)	(-2.331)	(-4.652)
BTC	-0.007***	-0.014***	-0.004***	-0.006**	-0.007***
	(-6.194)	(-8.251)	(-2.854)	(-2.575)	(-6.102)
Spread	0.006***	-0.050***	0.004***	0.004	0.003**
	(3.851)	(-2.751)	(3.033)	(1.206)	(2.304)
Tail	0.001	-0.001**	0.001	0.001	0.001
	(-0.675)	(-2.049)	(1.221)	(0.196)	(0.27)
Volatility	0.001	0.002	0.001	0.004	-0.003
	(-0.050)	(0.188)	(0.183)	(0.409)	(-0.961)
Maturity	-0.009	0.177***	0.009	0.021*	-0.023***
	(-1.634)	(5.66)	(1.331)	(1.811)	(-3.339)
Constant	0.266***	-0.138***	0.258***	0.171***	0.243***
	(11.456)	(-3.070)	(8.208)	(3.683)	(8.118)
Observations	1359	396	963	430	929
adj. R-sq	0.109	0.306	0.113	0.088	0.207

7.5 Chapter Summary

Treasury auction performance has been extensively studied in the US and other developed markets, but there is only sparse research on the determinants of auction performance when switching from a discriminatory-price auctions to a uniform-price auction format. My study fills this void in the literature.

The results, summarised in Table 7.8, are consistent with proposition H3a-b, that with increasing competition there is a decline in both revenue and efficiency gains when switching to a uniform-price auction format. Thus, increasing the competition level amongst bidders reduces the performance gap between discriminatory-price and uniform-price auctions. This is consistent with the prediction of auction theory that the number of bidders is related to the expected price (McAfee & McMillan, 1987), and supports the proposition that competition impacts on multi-unit auction performance (Vickrey, 1961; and Klemperer et al., 2003).

I find no evidence consistent with H4a-b, that higher uncertainty leads to higher revenue and efficiency gains when switching to a uniform-price auction format. Previous studies argue that dispersion of information influences auction performance in a sealed-bid first-price (FPA) auction (Engelbrecht-wiggans et al., 1983; Hendricks & Porter, 1989; Milgrom & Weber, 1982; Weverbergh, 1979; Robert Wilson, 1977), but much less is known about second-price auctions.

Table 7.9: ***Summary of Hypotheses and Results.***

Hypothesis		Test Variable	Expected Sign	Confirmed?	Comment
H1	Uniform-price auctions generate higher revenue.		+	Y	Significant for treasury bills only
H2	Uniform-price auctions generate higher efficiency.		+	Y	Significant for treasury bills only
H3a	Competition is negatively related to revenue gain.	*Bidding*	-	Y	
		Bidders	-	Y	
		BTC	-	N	
H3b	Competition is negatively related to efficiency gain.	*Bidding*	-	Y	
		Bidders	-	Y	
		BTC	-	N	
H4a	Uncertainty is positively related to revenue gain.	*Spread*	+	N	
		Tail	+	N	
		Volatility	+	N	
H4b	Uncertainty is positively related to efficiency gain.	*Spread*	+	N	
		Tail	+	N	
		Volatility	+	N	

Chapter 8

Summary and Conclusions

8.1 Introduction

Despite extensive research on the performance of treasury auctions in the US and other developed countries, there is scant literature on the same in emerging countries. My thesis fills this gap in the literature by examining the performance of a large sample of Indonesian treasury auctions. I compare the performance of discriminatory-price auctions with uniform-price auctions using ex-post and ex-ante scenarios. With the actual bidding data from discriminatory-price auctions, I invert the known residual supply function to perform a counterfactual uniform-price analysis. I employ a regression analysis to test several hypotheses on the determinants of revenue and efficiency gain when switching from a discriminatory-price auction to a uniform-price auction format. My sample consists of 297 conventional treasury bill auctions, 632 conventional treasury bond auctions, 99 Islamic treasury bill auctions, and 331 Islamic treasury bond auctions for the period between 2003 and 2017.

Section 8.2 summarises the key findings and contributions of my thesis. I outline the limitations of my study and provide possible future avenues for research in Section 8.3.

8.2 Findings and Contributions

The main findings and contributions of my thesis are summarised as follows. While not resolving any ambiguity in the literature, I providing substantive evidence that for Indonesian treasury bills (but not Indonesian treasury bonds), switching to a uniform-

price auction may significantly increase both revenue and efficiency (ex-post and ex-ante). This is consistent with the argument that in discriminatory-price auctions, bidders with less analytical capacities shade their bidding to avoid the existence of the winners' curse (Friedman, 1959; Brimmer, 1963; Rieber, 1964; and Goldstein, 1962). Therefore, the price level at which they bid is reduced, thereby decreasing auction revenue and efficiency. The finding is also consistent with my argument that in the Indonesian treasuries market, a collusive Nash equilibrium (in uniform auctions) is unlikely to exist. These finding are potentially very important to the MoF in Indonesia, in terms of increasing both auction revenue and efficiency.

Next, I provide new evidence on the impact of competition on the performance difference between discriminatory and uniform-price auctions. For both treasury bills and bonds, increasing the competition level amongst bidders in the auction reduces the performance gap between discriminatory and uniform-price auctions. Hence, competition has a significant impact on multi-unit auction performance (Vickrey, 1961; and Klemperer et al., 2003), consistent with the prediction of auction theory that the number of bidders is related to the expected price (McAfee & McMillan, 1987).

Finally, although inconsistent with my prediction, I find information uncertainty reduces gains in revenue, but has no impact on efficiency gains, when switching to a uniform-price auction format. Existing studies examine the dispersion of information on auction performance in a sealed-bid first-price (FPA) auction (Engelbrecht-wiggans et al., 1983; Hendricks & Porter, 1989; Milgrom & Weber, 1982; Weverbergh, 1979; Robert Wilson, 1977) but much less is understood about second-price auctions.

8.3 Limitations and Avenues for Future Research

There are several limitations in my research that open potential avenues for future research on treasury auctions. I have not accounted for the potential impact of a change in auction method on bidders' participation and bidding strategy. Uniform-price auctions may encourage more participation due to less bid preparation costs. In my empirical analysis, I have assumed that the number of bidders is stable and are known to each bidder. As with previous studies on treasury auctions, my sample is limited to treasury auctions domiciled in just one country, Indonesia. It might be worthwhile to compare auction performance across several countries with different auction settings so that the results may be generalisable to a wider market. It might also be worthwhile to estimate the impact of endogenous bidder participation when comparing the performance of the discriminatory-price auction to the uniform-price auction. The private independent value assumption that underlies my analysis may not be applicable to the Indonesian treasuries market. To improve the validity of the analysis, the consideration of value structure could be strengthened, including the paradigm of value affiliated. The non-parametric resampling approach is not feasible when the symmetry bidder assumption is broken down as the bidders' value estimations are likely to be correlated. It is not immediately clear how to treat the structure of general value because the development of an estimation approach with relevance to multi-unit divisible auctions is still lacking. Despite numerous attempts in the literature to describe the impact of risk behaviour in a single-unit auction environment, such consideration has not yet been attempted in multi-unit auctions. Thus, the model used in the thesis could be adapted to accommodate further exploration of bidders' risk attitude. Doing so is beyond the scope of my thesis and I will leave such investigation as another avenue for future research.

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Appendixes

Appendix 1: *Regressions of Determinants of Ex-post Upper Revenue Gain.*

This table presents the results of the OLS regression model for auctions' characteristics on revenue gain using bidding in IDR currency. Variable definitions are provided in Table 5.2. ***, **, * denote significance at the 1, 5, and 10 percent level, respectively. T-statistics in parentheses.

	Full Sample	**Islamic**		**Conventional**	
		Bills	**Bonds**	**Bills**	**Bonds**
	(1)	(2)	(3)	(4)	(5)
Bidding	0.333**	0.122	0.490	0.505***	-0.425
	(2.026)	(1.184)	(1.408)	(5.038)	(-1.301)
Bidders	-1.601***	-0.897***	-3.668***	-0.480**	-0.448
	(-4.042)	(-4.897)	(-4.543)	(-2.404)	(-0.563)
BTC	-0.161***	-0.003	-0.240**	-0.039**	-0.201***
	(-4.080)	(-0.170)	(-2.570)	(-2.421)	(-2.689)
Spread	0.097*	0.271	0.212*	-0.313**	0.045
	(1.818)	(1.605)	(1.760)	(-2.444)	(0.646)
Tail	0.010	0.001	-0.012	0.003	0.061
	(0.844)	(0.121)	(-0.341)	(0.566)	(1.018)
Volatility	-0.586***	-0.402**	-0.333	0.033	-0.444**
	(-5.195)	(-2.501)	(-1.349)	(0.320)	(-2.199)
Maturity	-1.022***	-5.750**	-1.794***	-0.089	-0.843***
	(-6.374)	(-1.988)	(-3.470)	(-0.433)	(-2.812)
Constant	0.735	3.341	4.222	-5.283***	8.634**
	(0.387)	(1.579)	(0.974)	(-4.562)	(2.241)
Observations	1359.000	99.000	331.000	297.000	632.000
adj. R-sq	0.163	0.243	0.178	0.163	0.037

Appendix 2: ***Regressions of Determinants of Ex-post Lower Revenue Gain.***

This table presents the results of the OLS regression model for auctions' characteristics on revenue gain using bidding in IDR currency. Variable definitions are provided in Table 5.2. ***, **, * denote significance at the 1, 5, and 10 percent level, respectively. T-statistics in parentheses.

	Full Sample	**Islamic**		**Conventional**	
		Bills	**Bonds**	**Bills**	**Bonds**
	(1)	(2)	(3)	(4)	(5)
Bidding	0.309*	0.029	0.476	0.488***	-0.448
	(1.898)	(0.326)	(1.381)	(5.211)	(-1.391)
Bidders	-1.498***	-0.585***	-3.628***	-0.377**	-0.287
	(-3.841)	(-3.119)	(-4.544)	(-2.059)	(-0.370)
BTC	-0.163***	0.010	-0.243***	-0.050***	-0.198***
	(-4.191)	(0.636)	(-2.661)	(-3.378)	(-2.673)
Spread	0.094*	0.211	0.212*	-0.289**	0.040
	(1.780)	(1.350)	(1.786)	(-2.491)	(0.582)
Tail	0.009	0.002	-0.012	0.003	0.059
	(0.793)	(0.478)	(-0.363)	(0.614)	(1.000)
Volatility	-0.588***	-0.343**	-0.342	0.011	-0.436**
	(-5.264)	(-2.300)	(-1.403)	(0.114)	(-2.171)
Maturity	-1.029***	-4.694	-1.834***	-0.377*	-0.856***
	(-6.525)	(-1.568)	(-3.604)	(-1.965)	(-2.900)
Constant	0.809	3.298	4.386	-5.149***	8.515**
	(0.431)	(1.614)	(1.020)	(-4.723)	(2.232)
Observations	1359.000	99.000	331.000	297.000	632.000
adj. R-sq	0.162	0.170	0.185	0.198	0.035

Appendix 3: ***Regressions of Determinants of Ex-post Efficiency Gain.***

This table presents the results of the OLS regression model for auctions' characteristics on efficiency gain using bidding in IDR currency. Variable definitions are provided in Table 5.2. ***, **, * denote significance at the 1, 5, and 10 percent level, respectively. T-statistics in parentheses.

	Full Sample	**Islamic**		**Conventional**	
		Bills	**Bonds**	**Bills**	**Bonds**
	(1)	(2)	(3)	(4)	(5)
Bidding	0.034***	0.018	-0.003	0.130***	0.002
	(3.290)	(1.212)	(-0.150)	(3.990)	(0.114)
Bidders	-0.184***	-0.094***	-0.084	-0.200***	-0.156***
	(-7.054)	(-4.378)	(-1.647)	(-3.108)	(-2.978)
BTC	-0.009***	-0.008**	-0.005	-0.016***	-0.008***
	(-3.599)	(-2.475)	(-0.662)	(-3.134)	(-3.647)
Spread	0.009***	-0.000	0.006	-0.025	0.005**
	(3.550)	(-0.030)	(0.857)	(-0.501)	(2.132)
Tail	-0.000	0.001	0.001	-0.001	0.004*
	(-0.096)	(0.797)	(0.265)	(-0.466)	(1.858)
Volatility	-0.002	-0.040**	0.012	0.063**	-0.014***
	(-0.308)	(-2.313)	(0.618)	(2.250)	(-2.722)
Maturity	0.004	-1.020**	0.050*	0.345***	0.026
	(0.394)	(-2.152)	(1.658)	(5.381)	(1.529)
Constant	0.098	0.486	0.257	-1.425***	0.444***
	(0.861)	(1.555)	(1.101)	(-3.768)	(2.701)
Observations	1359.000	99.000	331.000	297.000	632.000
adj. R-sq	0.081	0.193	0.045	0.184	0.095

Appendix 4: ***Regressions of Determinants of Ex-ante Upper Revenue Gain.***

This table presents the results of the OLS regression model for auctions' characteristics on revenue gain using bidding in IDR currency. Variable definitions are provided in Table 5.2. ***, **, * denote significance at the 1, 5, and 10 percent level, respectively. T-statistics in parentheses.

	Full Sample	**Islamic**		**Conventional**	
		Bills	**Bonds**	**Bills**	**Bonds**
	(1)	(2)	(3)	(4)	(5)
Bidding	0.259	0.066	0.499	0.330***	-0.452
	(1.632)	(0.984)	(1.427)	(5.253)	(-1.485)
Bidders	-1.673***	-0.634***	-3.969***	-0.364***	-0.208
	(-4.336)	(-4.454)	(-5.013)	(-3.020)	(-0.281)
BTC	-0.145***	0.003	-0.249***	-0.031***	-0.159**
	(-3.853)	(0.283)	(-2.843)	(-3.045)	(-2.197)
Spread	0.108**	0.150	0.275**	-0.225***	0.019
	(2.060)	(1.517)	(2.293)	(-2.774)	(0.280)
Tail	0.008	-0.000	-0.019	0.002	0.051
	(0.726)	(-0.109)	(-0.569)	(0.648)	(0.917)
Volatility	-0.534***	-0.292**	-0.329	-0.046	-0.422**
	(-4.885)	(-2.493)	(-1.382)	(-0.643)	(-2.165)
Maturity	-1.014***	-4.202*	-1.857***	-0.424***	-0.842***
	(-6.611)	(-1.696)	(-3.656)	(-3.137)	(-3.002)
Constant	1.800	2.727	4.859	-2.905***	8.155**
	(0.981)	(1.661)	(1.107)	(-3.901)	(2.253)
Observations	1359.000	99.000	331.000	297.000	632.000
adj. R-sq	0.172	0.269	0.199	0.223	0.033

Appendix 5: ***Regressions of Determinants of Ex-ante Lower Revenue Gain.***

This table presents the results of the OLS regression model for auctions' characteristics on revenue gain using bidding in IDR currency. Variable definitions are provided in Table 5.2. ***, **, * denote significance at the 1, 5, and 10 percent level, respectively. T-statistics in parentheses.

	Full Sample	**Islamic**		**Conventional**	
		Bills	**Bonds**	**Bills**	**Bonds**
	(1)	(2)	(3)	(4)	(5)
Bidding	0.250	0.013	0.419	0.357***	-0.544
	(1.333)	(0.208)	(1.045)	(5.266)	(-1.487)
Bidders	-1.590***	-0.456***	-3.882***	-0.278**	-0.146
	(-3.839)	(-3.331)	(-4.548)	(-2.474)	(-0.186)
BTC	-0.161***	0.008	-0.275***	-0.038***	-0.174**
	(-3.927)	(0.826)	(-2.937)	(-3.749)	(-2.196)
Spread	0.106*	0.111	0.295**	-0.219***	0.012
	(1.809)	(1.319)	(2.191)	(-2.904)	(0.159)
Tail	0.009	0.000	-0.020	0.002	0.057
	(0.706)	(0.144)	(-0.504)	(0.663)	(0.936)
Volatility	-0.567***	-0.254**	-0.360	-0.066	-0.429**
	(-4.945)	(-2.398)	(-1.473)	(-0.997)	(-2.068)
Maturity	-1.102***	-3.321	-2.111***	-0.601***	-0.938***
	(-6.872)	(-1.434)	(-4.010)	(-4.658)	(-3.156)
Constant	2.008	2.578*	6.382	-3.082***	8.933**
	(0.994)	(1.694)	(1.371)	(-4.047)	(2.185)
Observations	1359.000	99.000	331.000	297.000	632.000
adj. R-sq	0.169	0.218	0.202	0.268	0.032

Appendix 6: *Regressions of Determinants of Ex-ante Efficiency Gain*

This table presents the results of the OLS regression model for auctions' characteristics on efficiency gain using bidding in IDR currency. Variable definitions are provided in Table 5.2. ***, **, * denote significance at the 1, 5, and 10 percent level, respectively. T-statistics in parentheses.

	Full Sample	**Islamic**		**Conventional**	
		Bills	**Bonds**	**Bills**	**Bonds**
	(1)	(2)	(3)	(4)	(5)
Bidding	0.014**	0.008	-0.007	0.063***	-0.008
	(2.457)	(1.234)	(-0.605)	(4.345)	(-0.961)
Bidders	-0.094***	-0.041***	-0.053*	-0.065**	-0.062**
	(-6.938)	(-3.765)	(-1.901)	(-2.256)	(-2.329)
BTC	-0.007***	-0.004***	-0.006**	-0.011***	-0.003***
	(-6.046)	(-2.840)	(-2.118)	(-4.863)	(-2.992)
Spread	0.006***	0.002	0.005	-0.051**	0.003**
	(3.972)	(0.224)	(1.294)	(-2.155)	(2.048)
Tail	-0.000	0.000	0.000	-0.001	0.002*
	(-0.721)	(0.448)	(0.103)	(-0.863)	(1.923)
Volatility	-0.002	-0.017*	0.008	0.032**	-0.007***
	(-0.438)	(-1.817)	(0.798)	(2.071)	(-2.768)
Maturity	-0.009*	-0.751**	0.007	0.162***	0.011
	(-1.704)	(-2.462)	(0.469)	(4.876)	(1.369)
Constant	0.148**	0.333*	0.292**	-0.687***	0.310***
	(2.337)	(1.859)	(2.095)	(-4.027)	(3.728)
Observations	1359.000	99.000	331.000	297.000	632.000
adj. R-sq	0.107	0.234	0.055	0.226	0.091

Appendix 7: *Regressions of Determinants of Ex-post Upper Revenue Gain*

This table presents the results of the OLS regression model for auctions' characteristics on revenue gain. Variable definitions are provided in Table 5.2. ***, **, * denote significance at the 1, 5, and 10 percent level, respectively. T-statistics in parentheses.

	Full Sample									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Bidding	0.308*	-0.153	-0.170	-0.321***						
	(1.830)	(-1.375)	(-1.538)	(-2.984)						
Bidders	-1.500***				-0.928***	-0.939***	-1.085***			
	(-3.912)				(-3.738)	(-3.771)	(-4.332)			
BTC	-0.164***							-0.166***	-0.168***	-0.165***
	(-4.122)							(-4.173)	(-4.200)	(-4.292)
Spread	0.089*	0.065			0.064			0.075		
	(1.682)	(1.217)			(1.189)			(1.437)		
Tail	0.011		0.019			0.018			0.022*	
	(0.921)		(1.595)			(1.523)			(1.853)	
Volatility	-0.567***			-0.631***			-0.607***			-0.556***
	(-5.024)			(-5.489)			(-5.295)			(-4.756)
Maturity	-1.046***	-1.339***	-1.185***	-1.218***	-1.121***	-0.968***	-0.955***	-1.423***	-1.244***	-1.273***
	(-6.644)	(-11.554)	(-12.789)	(-14.620)	(-8.623)	(-8.717)	(-9.263)	(-12.115)	(-13.267)	(-14.875)
Constant	1359	1359	1359	1359	1359	1359	1359	1359	1359	1359
adj. R-sq	0.163	0.120	0.119	0.143	0.130	0.129	0.153	0.127	0.126	0.146

Appendix 8: *Regressions of Determinants of Ex-post Upper Revenue Gain*

This table presents the results of the OLS regression model for auctions' characteristics on revenue gain. Variable definitions are provided in Table 5.2. ***, **, * denote significance at the 1, 5, and 10 percent level, respectively. T-statistics in parentheses.

	Treasury Bills									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Bidding	0.543***	0.291***	0.300***	0.276***						
	(7.396)	(6.300)	(6.444)	(5.395)						
Bidders	-0.603***				0.039	0.034	0.030			
	(-4.260)				(0.358)	(0.308)	(0.279)			
BTC	-0.054***							-0.012	-0.019	-0.026**
	(-4.059)							(-0.885)	(-1.502)	(-2.073)
Spread	-0.227*	-0.388***			-0.408***			-0.389***		
	(-1.928)	(-3.618)			(-3.551)			(-3.462)		
Tail	-0.001		-0.008**			-0.007*			-0.006*	
	(-0.346)		(-2.080)			(-1.855)			(-1.711)	
Volatility	-0.062			-0.130**			-0.180***			-0.187***
	(-0.945)			(-1.987)			(-2.767)			(-2.841)
Maturity	-0.096	0.055	-0.307**	-0.376**	0.158	-0.211	-0.339*	0.142	-0.217	-0.365**
	(-0.458)	(0.307)	(-2.014)	(-2.247)	(0.822)	(-1.257)	(-1.880)	(0.752)	(-1.347)	(-2.079)
Constant	396	396	396	396	396	396	396	396	396	396
adj. R-sq	0.184	0.112	0.087	0.091	0.032	0.002	0.021	0.033	0.006	0.029

Appendix 9: *Regressions of Determinants of Ex-post Upper Revenue Gain*

This table presents the results of the OLS regression model for auctions' characteristics on revenue gain. Variable definitions are provided in Table 5.2. ***, **, * denote significance at the 1, 5, and 10 percent level, respectively. T-statistics in parentheses.

	Treasury Bonds									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Bidding	0.269	-0.274**	-0.289**	-0.466***						
	(1.110)	(-2.078)	(-2.206)	(-3.658)						
Bidders	-1.712***				-1.084***	-1.067***	-1.325***			
	(-2.994)				(-3.574)	(-3.493)	(-4.298)			
BTC	-0.242***							-0.257***	-0.255***	-0.249***
	(-3.914)							(-4.013)	(-3.982)	(-4.027)
Spread	0.089	0.057			0.062			0.073		
	(1.619)	(1.055)			(1.139)			(1.374)		
Tail	0.013		0.050			0.041			0.050	
	(0.426)		(1.543)			(1.282)			(1.588)	
Volatility	-0.646***			-0.704***			-0.680***			-0.584***
	(-4.938)			(-5.289)			(-5.092)			(-4.300)
Maturity	-0.976***	-1.013***	-0.861***	-0.910***	-0.890***	-0.737***	-0.755***	-1.158***	-0.975***	-1.031***
	(-4.771)	(-5.240)	(-5.036)	(-5.450)	(-4.454)	(-4.141)	(-4.294)	(-6.003)	(-5.659)	(-6.101)
Constant	963.000	963.000	963.000	963.000	963.000	963.000	963.000	963.000	963.000	963.000
adj. R-sq	0.076	0.025	0.026	0.054	0.035	0.035	0.063	0.036	0.036	0.055

Appendix 10: *Regressions of Determinants of Ex-post Upper Revenue Gain*

This table presents the results of the OLS regression model for auctions' characteristics on revenue gain. Variable definitions are provided in Table 5.2. ***, **, * denote significance at the 1, 5, and 10 percent level, respectively. T-statistics in parentheses.

	Islamic Treasury									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Bidding	0.510	-0.717***	-0.733***	-0.662***						
	(1.583)	(-3.062)	(-3.144)	(-2.860)						
Bidders	-3.134***				-2.787***	-2.820***	-2.677***			
	(-4.748)				(-5.638)	(-5.690)	(-5.473)			
BTC	-0.180***							-0.208***	-0.206***	-0.199***
	(-2.606)							(-2.810)	(-2.748)	(-2.778)
Spread	0.191	0.188			0.156			0.194*		
	(1.617)	(1.635)			(1.394)			(1.721)		
Tail	-0.000		0.028			0.012			0.025	
	(-0.024)		(1.612)			(0.711)			(1.440)	
Volatility	-0.455**			-0.560***			-0.388*			-0.600***
	(-2.073)			(-2.681)			(-1.897)			(-2.806)
Maturity	-1.531***	-2.063***	-1.555***	-1.469***	-1.689***	-1.289***	-1.232***	-1.840***	-1.315***	-1.244***
	(-3.903)	(-5.624)	(-6.177)	(-5.923)	(-5.487)	(-6.523)	(-6.564)	(-5.642)	(-6.591)	(-6.485)
Constant	430.000	430.000	430.000	430.000	430.000	430.000	430.000	430.000	430.000	430.000
adj. R-sq	0.213	0.129	0.121	0.135	0.194	0.187	0.194	0.124	0.114	0.132

Appendix 11: *Regressions of Determinants of Ex-post Upper Revenue Gain*

This table presents the results of the OLS regression model for auctions' characteristics on revenue gain. Variable definitions are provided in Table 5.2. ***, **, * denote significance at the 1, 5, and 10 percent level, respectively. T-statistics in parentheses.

	Conventional Treasury									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Bidding	-0.247	-0.536***	-0.523***	-0.442***						
	(-1.035)	(-3.166)	(-3.071)	(-2.596)						
Bidders	-0.396				-0.984***	-0.889**	-0.706**			
	(-0.818)				(-2.788)	(-2.571)	(-2.057)			
BTC	-0.113**							-0.132***	-0.133***	-0.139***
	(-2.559)							(-3.203)	(-3.225)	(-3.333)
Spread	0.053	0.075			0.088			0.060		
	(0.843)	(1.242)			(1.390)			(0.999)		
Tail	0.017		0.029			0.028			0.027	
	(1.005)		(1.645)			(1.584)			(1.555)	
Volatility	-0.445***			-0.473***			-0.473***			-0.528***
	(-2.805)			(-3.139)			(-3.171)			(-3.608)
Maturity	-1.186***	-1.326***	-1.156***	-1.212***	-1.067***	-0.908***	-1.014***	-1.376***	-1.232***	-1.282***
	(-6.184)	(-11.139)	(-11.020)	(-13.365)	(-6.869)	(-5.633)	(-6.990)	(-11.431)	(-11.615)	(-13.772)
Constant	929.000	929.000	929.000	929.000	929.000	929.000	929.000	929.000	929.000	929.000
adj. R-sq	0.152	0.141	0.141	0.150	0.139	0.138	0.147	0.136	0.137	0.149

Appendix 12: *Regressions of Determinants of Ex-ante Upper Revenue Gain*

This table presents the results of the OLS regression model for auctions' characteristics on revenue gain. Variable definitions are provided in Table 5.2. ***, **, * denote significance at the 1, 5, and 10 percent level, respectively. T-statistics in parentheses.

	Full Sample									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Bidding	0.236	-0.261**	-0.281***	-0.421***						
	(1.521)	(-2.421)	(-2.610)	(-4.026)						
Bidders	-1.589***				-1.141***	-1.156***	-1.287***			
	(-4.875)				(-4.679)	(-4.707)	(-5.217)			
BTC	-0.147***							-0.157***	-0.158***	-0.155***
	(-3.504)							(-4.103)	(-4.110)	(-4.173)
Spread	0.102**	0.078			0.079			0.092*		
	(2.554)	(1.494)			(1.517)			(1.786)		
Tail	0.008		0.018			0.017			0.020*	
	(0.558)		(1.596)			(1.476)			(1.807)	
Volatility	-0.520***			-0.583***			-0.546***			-0.484***
	(-5.409)			(-5.228)			(-4.907)			(-4.237)
Maturity	-1.034***	-1.352***	-1.179***	-1.210***	-1.083***	-0.906***	-0.895***	-1.429***	-1.225***	-1.253***
	(-7.396)	(-12.224)	(-13.316)	(-15.084)	(-8.732)	(-8.425)	(-8.901)	(-12.690)	(-13.648)	(-15.170)
Constant	1359.000	1359.000	1359.000	1359.000	1359.000	1359.000	1359.000	1359.000	1359.000	1359.000
adj. R-sq	0.171	0.129	0.128	0.150	0.143	0.142	0.162	0.133	0.131	0.147

Appendix 13: *Regressions of Determinants of Ex-ante Upper Revenue Gain*

This table presents the results of the OLS regression model for auctions' characteristics on revenue gain. Variable definitions are provided in Table 5.2. ***, **, * denote significance at the 1, 5, and 10 percent level, respectively. T-statistics in parentheses.

	Treasury Bills									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Bidding	0.383***	0.194***	0.202***	0.165***						
	(8.256)	(5.926)	(6.078)	(4.615)						
Bidders	-0.489***				-0.067	-0.071	-0.060			
	(-5.451)				(-0.923)	(-0.930)	(-0.838)			
BTC	-0.044***							-0.013	-0.019**	-0.027***
	(-5.206)							(-1.500)	(-2.304)	(-3.554)
Spread	-0.179**	-0.337***			-0.342***			-0.330***		
	(-2.324)	(-4.794)			(-4.414)			(-4.390)		
Tail	-0.002		-0.007**			-0.006**			-0.006**	
	(-0.876)		(-2.586)			(-2.119)			(-2.180)	
Volatility	-0.160***			-0.214***			-0.240***			-0.252***
	(-3.336)			(-4.374)			(-4.980)			(-5.226)
Maturity	-0.374**	-0.124	-0.441***	-0.588***	-0.033	-0.344***	-0.538***	-0.078	-0.387***	-0.598***
	(-2.581)	(-1.004)	(-4.086)	(-5.063)	(-0.249)	(-2.937)	(-4.390)	(-0.608)	(-3.453)	(-5.057)
Constant	396.000	396.000	396.000	396.000	396.000	396.000	396.000	396.000	396.000	396.000
adj. R-sq	0.277	0.130	0.096	0.152	0.063	0.023	0.104	0.065	0.029	0.120

Appendix 14: *Regressions of Determinants of Ex-ante Upper Revenue Gain*

This table presents the results of the OLS regression model for auctions' characteristics on revenue gain. Variable definitions are provided in Table 5.2. ***, **, * denote significance at the 1, 5, and 10 percent level, respectively. T-statistics in parentheses.

	Conventional Treasury									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Bidding	-0.319	-0.531***	-0.525***	-0.449***						
	(-1.453)	(-3.355)	(-3.292)	(-2.812)						
Bidders	-0.242				-0.910***	-0.845***	-0.672**			
	(-0.546)				(-2.774)	(-2.624)	(-2.097)			
BTC	-0.085**							-0.109***	-0.110***	-0.115***
	(-2.007)							(-2.718)	(-2.742)	(-2.862)
Spread	0.030	0.053			0.064			0.038		
	(0.493)	(0.914)			(1.052)			(0.651)		
Tail	0.016		0.026			0.024			0.023	
	(0.987)		(1.590)			(1.511)			(1.462)	
Volatility	-0.431***			-0.449***			-0.452***			-0.503***
	(-2.817)			(-3.086)			(-3.136)			(-3.565)
Maturity	-1.211***	-1.318***	-1.186***	-1.235***	-1.077***	-0.952***	-1.048***	-1.357***	-1.254***	-1.296***
	(-6.828)	(-11.665)	(-12.105)	(-14.438)	(-7.406)	(-6.397)	(-7.778)	(-11.873)	(-12.586)	(-14.676)
Constant	929.000	929.000	929.000	929.000	929.000	929.000	929.000	929.000	929.000	929.000
adj. R-sq	0.169	0.159	0.160	0.169	0.156	0.156	0.166	0.153	0.153	0.166

Appendix 15: *Regressions of Determinants of Ex-ante Upper Revenue Gain*

This table presents the results of the OLS regression model for auctions' characteristics on revenue gain. Variable definitions are provided in Table 5.2. ***, **, * denote significance at the 1, 5, and 10 percent level, respectively. T-statistics in parentheses.

	Treasury Bonds									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Bidding	0.241	-0.382***	-0.401***	-0.562***						
	(1.041)	(-2.987)	(-3.142)	(-4.526)						
Bidders	-1.895***				-1.329***	-1.320***	-1.549***			
	(-3.414)				(-4.466)	(-4.391)	(-5.112)			
BTC	-0.223***							-0.242***	-0.240***	-0.235***
	(-3.739)							(-3.883)	(-3.844)	(-3.860)
Spread	0.104*	0.070			0.078			0.090*		
	(1.929)	(1.329)			(1.470)			(1.730)		
Tail	0.007		0.046			0.035			0.046	
	(0.234)		(1.494)			(1.156)			(1.538)	
Volatility	-0.581***			-0.637***			-0.605***			-0.495***
	(-4.578)			(-4.922)			(-4.666)			(-3.718)
Maturity	-0.981***	-1.038***	-0.863***	-0.907***	-0.894***	-0.713***	-0.728***	-1.194***	-0.979***	-1.029***
	(-5.009)	(-5.654)	(-5.226)	(-5.589)	(-4.693)	(-4.125)	(-4.237)	(-6.483)	(-5.872)	(-6.253)
Constant	963.000	963.000	963.000	963.000	963.000	963.000	963.000	963.000	963.000	963.000
adj. R-sq	0.082	0.033	0.033	0.057	0.047	0.046	0.069	0.038	0.037	0.051

Appendix 16: *Regressions of Determinants of Ex-ante Upper Revenue Gain*

This table presents the results of the OLS regression model for auctions' characteristics on revenue gain. Variable definitions are provided in Table 5.2. ***, **, * denote significance at the 1, 5, and 10 percent level, respectively. T-statistics in parentheses.

	Islamic Treasury									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Bidding	0.545*	-0.710***	-0.721***	-0.667***						
	(1.705)	(-2.935)	(-3.031)	(-2.823)						
Bidders	-3.252***				-2.773***	-2.806***	-2.702***			
	(-5.252)				(-5.925)	(-6.007)	(-5.878)			
BTC	-0.153***							-0.167***	-0.158***	-0.152***
	(-2.813)							(-2.861)	(-2.763)	(-2.796)
Spread	0.258**	0.245**			0.221**			0.257**		
	(2.318)	(2.215)			(2.074)			(2.365)		
Tail	-0.003		0.024			0.012			0.020	
	(-0.225)		(1.585)			(0.787)			(1.357)	
Volatility	-0.398**			-0.481**			-0.322*			-0.510**
	(-1.987)			(-2.497)			(-1.748)			(-2.530)
Maturity	-1.540***	-2.061***	-1.416***	-1.343***	-1.721***	-1.163***	-1.117***	-1.811***	-1.138***	-1.071***
	(-4.039)	(-5.730)	(-5.712)	(-5.487)	(-5.906)	(-5.997)	(-6.039)	(-5.806)	(-5.902)	(-5.766)
Constant	430.000	430.000	430.000	430.000	430.000	430.000	430.000	430.000	430.000	430.000
adj. R-sq	0.219	0.120	0.103	0.115	0.197	0.182	0.188	0.113	0.093	0.108

Appendix 17: *Regressions of Determinants of Ex-post Lower Revenue Gain*

This table presents the results of the OLS regression model for auctions' characteristics on revenue gain. Variable definitions are provided in Table 5.2. ***, **, * denote significance at the 1, 5, and 10 percent level, respectively. T-statistics in parentheses.

	Full Sample									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Bidding	0.256*	-0.131	-0.144	-0.290***						
	(1.657)	(-1.197)	(-1.320)	(-2.682)						
Bidders	-1.322***				-0.805***	-0.809***	-0.956***			
	(-4.151)				(-3.466)	(-3.474)	(-4.068)			
BTC	-0.113***							-0.112***	-0.112***	-0.111***
	(-3.387)							(-3.037)	(-3.048)	(-3.245)
Spread	0.077**	0.052			0.054			0.060		
	(2.054)	(1.132)			(1.168)			(1.315)		
Tail	0.009		0.017			0.016			0.019*	
	(0.581)		(1.489)			(1.456)			(1.656)	
Volatility	-0.548***			-0.598***			-0.579***			-0.528***
	(-5.652)			(-5.318)			(-5.183)			(-4.623)
Maturity	-1.028***	-1.301***	-1.172***	-1.198***	-1.117***	-0.984***	-0.966***	-1.353***	-1.204***	-1.226***
	(-7.432)	(-11.913)	(-12.866)	(-14.575)	(-9.246)	(-9.196)	(-9.717)	(-12.191)	(-13.162)	(-14.692)
Constant	1359.000	1359.000	1359.000	1359.000	1359.000	1359.000	1359.000	1359.000	1359.000	1359.000
adj. R-sq	0.158	0.118	0.118	0.141	0.127	0.127	0.150	0.124	0.124	0.143

Appendix 18: *Regressions of Determinants of Ex-post Lower Revenue Gain*

This table presents the results of the OLS regression model for auctions' characteristics on revenue gain. Variable definitions are provided in Table 5.2. ***, **, * denote significance at the 1, 5, and 10 percent level, respectively. T-statistics in parentheses.

	Treasury Bills									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Bidding	0.526***	0.308***	0.319***	0.286***						
	(7.598)	(7.015)	(7.184)	(5.777)						
Bidders	-0.493***				0.106	0.105	0.103			
	(-3.712)				(1.005)	(0.976)	(0.989)			
BTC	-0.060***							-0.016	-0.023**	-0.031***
	(-5.109)							(-1.322)	(-2.029)	(-2.819)
Spread	-0.225**	-0.402***			-0.429***			-0.398***		
	(-2.070)	(-4.059)			(-4.021)			(-3.756)		
Tail	-0.002		-0.008**			-0.008**			-0.006*	
	(-0.701)		(-2.296)			(-2.126)			(-1.806)	
Volatility	-0.126*			-0.180***			-0.236***			-0.243***
	(-1.922)			(-2.754)			(-3.621)			(-3.676)
Maturity	-0.329	-0.068	-0.447***	-0.556***	0.024	-0.370**	-0.545***	0.014	-0.355**	-0.553***
	(-1.636)	(-0.394)	(-3.004)	(-3.393)	(0.132)	(-2.279)	(-3.130)	(0.075)	(-2.248)	(-3.234)
Constant	396.000	396.000	396.000	396.000	396.000	396.000	396.000	396.000	396.000	396.000
adj. R-sq	0.215	0.133	0.106	0.122	0.042	0.009	0.047	0.043	0.014	0.057

Appendix 19: *Regressions of Determinants of Ex-post Lower Revenue Gain*

This table presents the results of the OLS regression model for auctions' characteristics on revenue gain. Variable definitions are provided in Table 5.2. ***, **, * denote significance at the 1, 5, and 10 percent level, respectively. T-statistics in parentheses.

	Treasury Bonds									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Bidding	0.199	-0.242*	-0.251*	-0.422***						
	(0.857)	(-1.867)	(-1.944)	(-3.282)						
Bidders	-1.468***				-0.937***	-0.915***	-1.160***			
	(-2.868)				(-3.360)	(-3.261)	(-4.072)			
BTC	-0.136***							-0.140***	-0.138***	-0.135***
	(-2.714)							(-2.634)	(-2.597)	(-2.760)
Spread	0.076	0.047			0.054			0.059		
	(1.592)	(1.002)			(1.141)			(1.265)		
Tail	0.014		0.047			0.040			0.047	
	(0.477)		(1.505)			(1.301)			(1.531)	
Volatility	-0.618***			-0.658***			-0.639***			-0.549***
	(-4.849)			(-5.075)			(-4.927)			(-4.163)
Maturity	-0.961***	-1.016***	-0.885***	-0.929***	-0.917***	-0.780***	-0.797***	-1.106***	-0.951***	-1.004***
	(-4.912)	(-5.461)	(-5.247)	(-5.652)	(-4.802)	(-4.475)	(-4.642)	(-5.921)	(-5.601)	(-6.041)
Constant	963.000	963.000	963.000	963.000	963.000	963.000	963.000	963.000	963.000	963.000
adj. R-sq	0.068	0.025	0.026	0.052	0.034	0.034	0.060	0.031	0.031	0.050

Appendix 20: *Regressions of Determinants of Ex-post Lower Revenue Gain*

This table presents the results of the OLS regression model for auctions' characteristics on revenue gain. Variable definitions are provided in Table 5.2. ***, **, * denote significance at the 1, 5, and 10 percent level, respectively. T-statistics in parentheses.

	Islamic Treasury									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Bidding	0.534*	-0.622***	-0.634***	-0.581**						
	(1.676)	(-2.612)	(-2.678)	(-2.453)						
Bidders	-2.968***				-2.508***	-2.530***	-2.415***			
	(-4.778)				(-5.396)	(-5.440)	(-5.247)			
BTC	-0.157***							-0.168***	-0.162***	-0.156***
	(-2.591)							(-2.689)	(-2.678)	(-2.743)
Spread	0.192*	0.177			0.156			0.189*		
	(1.762)	(1.631)			(1.480)			(1.789)		
Tail	-0.001		0.023			0.012			0.020	
	(-0.064)		(1.471)			(0.768)			(1.311)	
Volatility	-0.402**			-0.495**			-0.352*			-0.519**
	(-1.968)			(-2.523)			(-1.861)			(-2.554)
Maturity	-1.498***	-1.976***	-1.498***	-1.417***	-1.679***	-1.278***	-1.223***	-1.774***	-1.265***	-1.195***
	(-3.934)	(-5.466)	(-6.083)	(-5.815)	(-5.663)	(-6.588)	(-6.606)	(-5.667)	(-6.618)	(-6.488)
Constant	430.000	430.000	430.000	430.000	430.000	430.000	430.000	430.000	430.000	430.000
adj. R-sq	0.211	0.124	0.116	0.129	0.188	0.181	0.188	0.123	0.113	0.128

Appendix 21: *Regressions of Determinants of Ex-post Lower Revenue Gain*

This table presents the results of the OLS regression model for auctions' characteristics on revenue gain. Variable definitions are provided in Table 5.2. ***, **, * denote significance at the 1, 5, and 10 percent level, respectively. T-statistics in parentheses.

	Conventional Treasury									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Bidding	-0.359	-0.503***	-0.497***	-0.419**						
	(-1.572)	(-3.016)	(-2.968)	(-2.499)						
Bidders	-0.095				-0.757**	-0.687**	-0.523*			
	(-0.219)				(-2.316)	(-2.144)	(-1.656)			
BTC	-0.061*							-0.075**	-0.075**	-0.080**
	(-1.799)							(-2.134)	(-2.122)	(-2.280)
Spread	0.034	0.054			0.062			0.042		
	(0.637)	(1.040)			(1.154)			(0.817)		
Tail	0.017		0.029*			0.027			0.025	
	(1.020)		(1.663)			(1.571)			(1.502)	
Volatility	-0.447***			-0.465***			-0.474***			-0.515***
	(-2.862)			(-3.091)			(-3.200)			(-3.530)
Maturity	-1.209***	-1.286***	-1.144***	-1.200***	-1.086***	-0.957***	-1.056***	-1.313***	-1.197***	-1.243***
	(-6.746)	(-11.379)	(-11.067)	(-13.384)	(-7.374)	(-6.194)	(-7.635)	(-11.569)	(-11.511)	(-13.647)
Constant	929.000	929.000	929.000	929.000	929.000	929.000	929.000	929.000	929.000	929.000
adj. R-sq	0.149	0.140	0.140	0.149	0.136	0.136	0.146	0.134	0.134	0.147

Appendix 22: *Regressions of Determinants of Ex-ante Lower Revenue Gain*

This table presents the results of the OLS regression model for auctions' characteristics on revenue gain. Variable definitions are provided in Table 5.2. ***, **, * denote significance at the 1, 5, and 10 percent level, respectively. T-statistics in parentheses.

	Full Sample									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Bidding	0.189	-0.238**	-0.252**	-0.388***						
	(1.271)	(-2.215)	(-2.352)	(-3.665)						
Bidders	-1.421***				-1.018***	-1.023***	-1.156***			
	(-4.633)				(-4.434)	(-4.433)	(-4.975)			
BTC	-0.095***							-0.098***	-0.098***	-0.096***
	(-2.943)							(-2.767)	(-2.759)	(-2.895)
Spread	0.086**	0.061			0.066			0.072		
	(2.366)	(1.318)			(1.413)			(1.544)		
Tail	0.007		0.016			0.015			0.017	
	(0.479)		(1.474)			(1.405)			(1.578)	
Volatility	-0.505***			-0.554***			-0.523***			-0.460***
	(-5.401)			(-5.068)			(-4.818)			(-4.111)
Maturity	-1.011***	-1.312***	-1.170***	-1.195***	-1.077***	-0.927***	-0.911***	-1.354***	-1.189***	-1.209***
	(-7.592)	(-12.427)	(-13.399)	(-15.047)	(-9.221)	(-8.913)	(-9.371)	(-12.546)	(-13.499)	(-14.931)
Constant	1359.000	1359.000	1359.000	1359.000	1359.000	1359.000	1359.000	1359.000	1359.000	1359.000
adj. R-sq	0.165	0.128	0.126	0.148	0.140	0.139	0.159	0.129	0.128	0.143

Appendix 23: *Regressions of Determinants of Ex-ante Lower Revenue Gain*

This table presents the results of the OLS regression model for auctions' characteristics on revenue gain. Variable definitions are provided in Table 5.2. ***, **, * denote significance at the 1, 5, and 10 percent level, respectively. T-statistics in parentheses.

	Treasury Bills									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Bidding	0.356***	0.198***	0.208***	0.165***						
	(7.960)	(6.229)	(6.426)	(4.759)						
Bidders	-0.389***				-0.020	-0.018	-0.009			
	(-4.530)				(-0.275)	(-0.243)	(-0.129)			
BTC	-0.047***							-0.016**	-0.022***	-0.031***
	(-6.265)							(-1.986)	(-2.890)	(-4.591)
Spread	-0.182**	-0.348***			-0.358***			-0.337***		
	(-2.548)	(-5.307)			(-5.010)			(-4.784)		
Tail	-0.003		-0.008***			-0.007**			-0.006**	
	(-1.424)		(-2.914)			(-2.523)			(-2.428)	
Volatility	-0.201***			-0.244***			-0.273***			-0.284***
	(-4.289)			(-5.062)			(-5.734)			(-5.998)
Maturity	-0.522***	-0.198	-0.530***	-0.697***	-0.117	-0.449***	-0.666***	-0.158	-0.476***	-0.714***
	(-3.705)	(-1.620)	(-4.863)	(-5.979)	(-0.909)	(-3.848)	(-5.474)	(-1.237)	(-4.218)	(-6.063)
Constant	396.000	396.000	396.000	396.000	396.000	396.000	396.000	396.000	396.000	396.000
adj. R-sq	0.305	0.147	0.113	0.186	0.074	0.033	0.138	0.080	0.044	0.161

Appendix 24: *Regressions of Determinants of Ex-ante Lower Revenue Gain*

This table presents the results of the OLS regression model for auctions' characteristics on revenue gain. Variable definitions are provided in Table 5.2. ***, **, * denote significance at the 1, 5, and 10 percent level, respectively. T-statistics in parentheses.

	Treasury Bonds									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Bidding	0.172	-0.347***	-0.359***	-0.514***						
	(0.767)	(-2.735)	(-2.827)	(-4.091)						
Bidders	-1.646***				-1.175***	-1.156***	-1.376***			
	(-3.282)				(-4.265)	(-4.167)	(-4.892)			
BTC	-0.116**							-0.122**	-0.119**	-0.117**
	(-2.423)							(-2.376)	(-2.331)	(-2.439)
Spread	0.087*	0.055			0.066			0.070		
	(1.794)	(1.187)			(1.393)			(1.498)		
Tail	0.009		0.043			0.035			0.043	
	(0.307)		(1.445)			(1.179)			(1.482)	
Volatility	-0.560***			-0.598***			-0.572***			-0.468***
	(-4.526)			(-4.741)			(-4.543)			(-3.613)
Maturity	-0.955***	-1.030***	-0.884***	-0.924***	-0.912***	-0.754***	-0.769***	-1.128***	-0.952***	-0.999***
	(-5.046)	(-5.771)	(-5.408)	(-5.762)	(-4.957)	(-4.444)	(-4.569)	(-6.257)	(-5.758)	(-6.135)
Constant	963.000	963.000	963.000	963.000	963.000	963.000	963.000	963.000	963.000	963.000
adj. R-sq	0.073	0.032	0.033	0.055	0.045	0.044	0.066	0.032	0.031	0.045

Appendix 25: *Regressions of Determinants of Ex-ante Lower Revenue Gain*

This table presents the results of the OLS regression model for auctions' characteristics on revenue gain. Variable definitions are provided in Table 5.2. ***, **, * denote significance at the 1, 5, and 10 percent level, respectively. T-statistics in parentheses.

	Islamic Treasury									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Bidding	0.522	-0.709***	-0.718***	-0.666***						
	(1.643)	(-2.953)	(-3.045)	(-2.836)						
Bidders	-3.176***				-2.731***	-2.763***	-2.654***			
	(-5.165)				(-5.882)	(-5.963)	(-5.819)			
BTC	-0.154***							-0.168***	-0.159***	-0.153***
	(-2.881)							(-2.921)	(-2.822)	(-2.862)
Spread	0.251**	0.236**			0.213**			0.248**		
	(2.274)	(2.154)			(2.012)			(2.304)		
Tail	-0.004		0.022			0.010			0.019	
	(-0.302)		(1.491)			(0.684)			(1.263)	
Volatility	-0.408**			-0.489**			-0.333*			-0.517***
	(-2.057)			(-2.560)			(-1.826)			(-2.589)
Maturity	-1.533***	-2.037***	-1.419***	-1.339***	-1.697***	-1.165***	-1.111***	-1.789***	-1.142***	-1.069***
	(-4.066)	(-5.731)	(-5.794)	(-5.538)	(-5.897)	(-6.082)	(-6.089)	(-5.804)	(-6.009)	(-5.836)
Constant	430.000	430.000	430.000	430.000	430.000	430.000	430.000	430.000	430.000	430.000
adj. R-sq	0.220	0.121	0.105	0.118	0.197	0.182	0.189	0.114	0.095	0.111

Appendix 26: *Regressions of Determinants of Ex-ante Lower Revenue Gain*

This table presents the results of the OLS regression model for auctions' characteristics on revenue gain. Variable definitions are provided in Table 5.2. ***, **, * denote significance at the 1, 5, and 10 percent level, respectively. T-statistics in parentheses.

	Conventional Treasury									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Bidding	0.172	-0.347***	-0.359***	-0.514***						
	(0.767)	(-2.735)	(-2.827)	(-4.091)						
Bidders	-1.646***				-1.175***	-1.156***	-1.376***			
	(-3.282)				(-4.265)	(-4.167)	(-4.892)			
BTC	-0.116**							-0.122**	-0.119**	-0.117**
	(-2.423)							(-2.376)	(-2.331)	(-2.439)
Spread	0.087*	0.055			0.066			0.070		
	(1.794)	(1.187)			(1.393)			(1.498)		
Tail	0.009		0.043			0.035			0.043	
	(0.307)		(1.445)			(1.179)			(1.482)	
Volatility	-0.560***			-0.598***			-0.572***			-0.468***
	(-4.526)			(-4.741)			(-4.543)			(-3.613)
Maturity	-0.955***	-1.030***	-0.884***	-0.924***	-0.912***	-0.754***	-0.769***	-1.128***	-0.952***	-0.999***
	(-5.046)	(-5.771)	(-5.408)	(-5.762)	(-4.957)	(-4.444)	(-4.569)	(-6.257)	(-5.758)	(-6.135)
Constant	963.000	963.000	963.000	963.000	963.000	963.000	963.000	963.000	963.000	963.000
adj. R-sq	0.073	0.032	0.033	0.055	0.045	0.044	0.066	0.032	0.031	0.045

Appendix 27: *Regressions of Determinants of Ex-post Efficiency Gain*

This table presents the results of the OLS regression model for auctions' characteristics on efficiency gain. Variable definitions are provided in Table 5.2. ***, **, * denote significance at the 1, 5, and 10 percent level, respectively. T-statistics in parentheses.

	Full Sample									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Bidding	0.037***	-0.033***	-0.034***	-0.034***						
	(3.239)	(-4.229)	(-4.327)	(-4.644)						
Bidders	-0.188***				-0.129***	-0.129***	-0.130***			
	(-7.752)				(-6.477)	(-6.473)	(-6.696)			
BTC	-0.007***							-0.007***	-0.006***	-0.006***
	(-2.861)							(-3.867)	(-3.778)	(-3.744)
Spread	0.007**	0.005*			0.006**			0.006**		
	(2.470)	(1.899)			(2.195)			(2.214)		
Tail	-0.000		0.000			0.000			0.000	
	(-0.256)		(0.176)			(0.051)			(0.183)	
Volatility	0.004			-0.002			-0.001			0.007
	(0.548)			(-0.242)			(-0.073)			(0.731)
Maturity	0.012	-0.029***	-0.020***	-0.020***	0.001	0.011	0.011	-0.031***	-0.020***	-0.021***
	(1.103)	(-4.178)	(-2.892)	(-3.359)	(0.126)	(1.167)	(1.257)	(-4.390)	(-2.801)	(-3.282)
Constant	1359.000	1359.000	1359.000	1359.000	1359.000	1359.000	1359.000	1359.000	1359.000	1359.000
adj. R-sq	0.063	0.021	0.019	0.019	0.055	0.052	0.052	0.011	0.008	0.008

Appendix 28: *Regressions of Determinants of Ex-post Efficiency Gain*

This table presents the results of the OLS regression model for auctions' characteristics on efficiency gain. Variable definitions are provided in Table 5.2. ***, **, * denote significance at the 1, 5, and 10 percent level, respectively. T-statistics in parentheses.

	Treasury Bills									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Bidding	0.137***	0.049***	0.052***	0.051***						
	(5.992)	(3.930)	(4.036)	(3.777)						
Bidders	-0.190***				-0.028	-0.024	-0.036			
	(-4.513)				(-1.114)	(-0.950)	(-1.465)			
BTC	-0.020***							-0.011***	-0.012***	-0.013***
	(-6.005)							(-3.691)	(-4.031)	(-4.519)
Spread	-0.041	-0.091***			-0.092***			-0.078**		
	(-1.152)	(-2.744)			(-2.679)			(-2.321)		
Tail	-0.001		-0.003***			-0.002***			-0.002***	
	(-1.065)		(-3.228)			(-2.793)			(-2.701)	
Volatility	0.022			0.003			-0.005			-0.012
	(1.331)			(0.166)			(-0.320)			(-0.723)
Maturity	0.364***	0.383***	0.291***	0.310***	0.409***	0.319***	0.332***	0.376***	0.298***	0.299***
	(6.368)	(7.528)	(6.112)	(6.056)	(7.734)	(6.069)	(6.007)	(7.273)	(5.975)	(5.637)
Constant	396.000	396.000	396.000	396.000	396.000	396.000	396.000	396.000	396.000	396.000
adj. R-sq	0.245	0.144	0.137	0.121	0.116	0.104	0.093	0.134	0.128	0.119

Appendix 29: *Regressions of Determinants of Ex-post Efficiency Gain*

This table presents the results of the OLS regression model for auctions' characteristics on efficiency gain. Variable definitions are provided in Table 5.2. ***, **, * denote significance at the 1, 5, and 10 percent level, respectively. T-statistics in parentheses.

	Treasury Bonds									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Bidding	-0.006	-0.054***	-0.055***	-0.056***						
	(-0.451)	(-6.777)	(-6.906)	(-7.229)						
Bidders	-0.123***				-0.137***	-0.136***	-0.138***			
	(-3.644)				(-6.736)	(-6.703)	(-6.853)			
BTC	-0.006							-0.008**	-0.008**	-0.008**
	(-1.593)							(-2.207)	(-2.135)	(-2.145)
Spread	0.007**	0.006**			0.007***			0.008***		
	(2.550)	(2.364)			(2.884)			(3.117)		
Tail	0.002		0.004**			0.003			0.004**	
	(1.069)		(2.007)			(1.486)			(2.110)	
Volatility	0.001			-0.002			0.002			0.011
	(0.141)			(-0.310)			(0.285)			(1.461)
Maturity	0.036***	0.024**	0.039***	0.036***	0.036***	0.053***	0.052***	0.012	0.032***	0.029***
	(2.886)	(2.040)	(3.548)	(3.329)	(2.983)	(4.556)	(4.456)	(1.052)	(2.904)	(2.725)
Constant	963.000	963.000	963.000	963.000	963.000	963.000	963.000	963.000	963.000	963.000
adj. R-sq	0.087	0.068	0.068	0.064	0.087	0.082	0.080	0.018	0.014	0.012

Appendix 30: *Regressions of Determinants of Ex-post Efficiency Gain*

This table presents the results of the OLS regression model for auctions' characteristics on efficiency gain. Variable definitions are provided in Table 5.2. ***, **, * denote significance at the 1, 5, and 10 percent level, respectively. T-statistics in parentheses.

	Islamic Treasury									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Bidding	0.016	-0.021	-0.022*	-0.021						
	(0.780)	(-1.598)	(-1.670)	(-1.607)						
Bidders	-0.102**				-0.082***	-0.083***	-0.089***			
	(-2.519)				(-2.974)	(-3.025)	(-3.061)			
BTC	-0.006							-0.007	-0.007	-0.006
	(-1.072)							(-1.336)	(-1.331)	(-1.264)
Spread	0.006	0.008			0.007			0.008		
	(0.905)	(1.273)			(1.132)			(1.308)		
Tail	0.001		0.002			0.001			0.002	
	(0.733)		(1.208)			(0.895)			(1.181)	
Volatility	0.010			0.006			0.012			0.005
	(0.605)			(0.419)			(0.770)			(0.331)
Maturity	0.056**	0.038*	0.061***	0.056***	0.049**	0.069***	0.063***	0.044**	0.068***	0.063***
	(2.376)	(1.801)	(3.699)	(3.414)	(2.573)	(4.902)	(4.681)	(2.301)	(4.777)	(4.535)
Constant	430.000	430.000	430.000	430.000	430.000	430.000	430.000	430.000	430.000	430.000
adj. R-sq	0.069	0.062	0.060	0.059	0.073	0.071	0.072	0.062	0.059	0.058

Appendix 31: *Regressions of Determinants of Ex-post Efficiency Gain*

This table presents the results of the OLS regression model for auctions' characteristics on efficiency gain. Variable definitions are provided in Table 5.2. ***, **, * denote significance at the 1, 5, and 10 percent level, respectively. T-statistics in parentheses.

	Conventional Treasury									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Bidding	0.050***	-0.026**	-0.025**	-0.023**						
	(3.157)	(-2.506)	(-2.450)	(-2.305)						
Bidders	-0.188***				-0.127***	-0.121***	-0.119***			
	(-5.115)				(-5.161)	(-4.959)	(-4.907)			
BTC	-0.012***							-0.010***	-0.010***	-0.010***
	(-4.980)							(-4.360)	(-4.408)	(-4.415)
Spread	0.005**	0.002			0.005**			0.001		
	(2.437)	(1.045)			(2.531)			(0.711)		
Tail	0.001		0.001			0.001			0.001	
	(0.473)		(0.702)			(0.905)			(0.673)	
Volatility	-0.005			-0.010*			-0.005			-0.014**
	(-0.798)			(-1.701)			(-0.898)			(-2.193)
Maturity	-0.015	-0.061***	-0.056***	-0.058***	-0.028**	-0.020*	-0.023**	-0.065***	-0.061***	-0.063***
	(-1.120)	(-8.459)	(-7.643)	(-8.932)	(-2.573)	(-1.740)	(-2.147)	(-9.090)	(-8.725)	(-10.104)
Constant	929.000	929.000	929.000	929.000	929.000	929.000	929.000	929.000	929.000	929.000
adj. R-sq	0.152	0.099	0.099	0.100	0.136	0.133	0.133	0.102	0.102	0.104

Appendix 32: *Regressions of Determinants of Ex-ante Efficiency Gain*

This table presents the results of the OLS regression model for auctions' characteristics on efficiency gain. Variable definitions are provided in Table 5.2. ***, **, * denote significance at the 1, 5, and 10 percent level, respectively. T-statistics in parentheses.

	Full Sample									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Bidding	0.017***	-0.021***	-0.022***	-0.023***						
	(2.974)	(-5.265)	(-5.474)	(-5.789)						
Bidders	-0.097***				-0.072***	-0.073***	-0.073***			
	(-8.258)				(-7.873)	(-7.883)	(-8.020)			
BTC	-0.007***							-0.008***	-0.008***	-0.007***
	(-4.646)							(-6.763)	(-6.644)	(-6.593)
Spread	0.006***	0.005***			0.005***			0.006***		
	(3.931)	(3.201)			(3.399)			(3.697)		
Tail	-0.000		0.000			-0.000			0.000	
	(-0.591)		(0.077)			(-0.154)			(0.227)	
Volatility	-0.000			-0.004			-0.002			0.002
	(-0.054)			(-0.992)			(-0.514)			(0.430)
Maturity	-0.009*	-0.027***	-0.019***	-0.019***	-0.010**	-0.002	-0.001	-0.031***	-0.021***	-0.021***
	(-1.729)	(-7.577)	(-5.624)	(-6.456)	(-2.221)	(-0.375)	(-0.323)	(-8.479)	(-5.984)	(-6.870)
Constant	1359.000	1359.000	1359.000	1359.000	1359.000	1359.000	1359.000	1359.000	1359.000	1359.000
adj. R-sq	0.109	0.054	0.047	0.048	0.093	0.085	0.086	0.047	0.037	0.037

Appendix 33: *Regressions of Determinants of Ex-ante Efficiency Gain*

This table presents the results of the OLS regression model for auctions' characteristics on efficiency gain. Variable definitions are provided in Table 5.2. ***, **, * denote significance at the 1, 5, and 10 percent level, respectively. T-statistics in parentheses.

	Treasury Bills									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Bidding	0.078***	0.034***	0.036***	0.034***						
	(7.284)	(5.106)	(5.048)	(4.371)						
Bidders	-0.078***				0.003	0.005	-0.003			
	(-4.071)				(0.223)	(0.323)	(-0.201)			
BTC	-0.014***							-0.009***	-0.010***	-0.011***
	(-8.251)							(-5.243)	(-6.039)	(-7.008)
Spread	-0.050***	-0.083***			-0.085***			-0.074***		
	(-2.751)	(-4.593)			(-4.656)			(-4.106)		
Tail	-0.001**		-0.002***			-0.002***			-0.002***	
	(-2.049)		(-4.133)			(-3.849)			(-3.526)	
Volatility	0.002			-0.007			-0.013			-0.017*
	(0.188)			(-0.686)			(-1.286)			(-1.693)
Maturity	0.177***	0.214***	0.133***	0.140***	0.226***	0.145***	0.147***	0.208***	0.137***	0.130***
	(5.660)	(7.410)	(5.169)	(5.056)	(7.658)	(5.149)	(4.998)	(7.352)	(5.234)	(4.641)
Constant	396.000	396.000	396.000	396.000	396.000	396.000	396.000	396.000	396.000	396.000
adj. R-sq	0.306	0.186	0.150	0.122	0.141	0.098	0.077	0.177	0.146	0.137

Appendix 34: *Regressions of Determinants of Ex-ante Efficiency Gain*

This table presents the results of the OLS regression model for auctions' characteristics on efficiency gain. Variable definitions are provided in Table 5.2. ***, **, * denote significance at the 1, 5, and 10 percent level, respectively. T-statistics in parentheses.

	Treasury Bonds									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Bidding	-0.011	-0.035***	-0.036***	-0.036***						
	(-1.551)	(-8.084)	(-8.277)	(-8.572)						
Bidders	-0.060***				-0.083***	-0.083***	-0.083***			
	(-3.292)				(-7.561)	(-7.456)	(-7.605)			
BTC	-0.004***							-0.006***	-0.006***	-0.006***
	(-2.854)							(-3.832)	(-3.707)	(-3.705)
Spread	0.004***	0.004***			0.005***			0.006***		
	(3.033)	(2.976)			(3.595)			(3.835)		
Tail	0.001		0.002**			0.002			0.002**	
	(1.221)		(2.212)			(1.622)			(2.329)	
Volatility	0.001			-0.001			0.002			0.008*
	(0.183)			(-0.255)			(0.595)			(1.919)
Maturity	0.009	0.003	0.013**	0.012**	0.010	0.022***	0.021***	-0.005	0.009	0.007
	(1.331)	(0.510)	(2.311)	(2.044)	(1.613)	(3.552)	(3.421)	(-0.737)	(1.418)	(1.206)
Constant	963.000	963.000	963.000	963.000	963.000	963.000	963.000	963.000	963.000	963.000
adj. R-sq	0.113	0.094	0.092	0.087	0.107	0.098	0.096	0.022	0.013	0.012

Appendix 35: *Regressions of Determinants of Ex-ante Efficiency Gain*

This table presents the results of the OLS regression model for auctions' characteristics on efficiency gain. Variable definitions are provided in Table 5.2. ***, **, * denote significance at the 1, 5, and 10 percent level, respectively. T-statistics in parentheses.

	Islamic Treasury									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Bidding	-0.001	-0.022***	-0.022***	-0.022***						
	(-0.049)	(-2.835)	(-2.891)	(-2.840)						
Bidders	-0.052**				-0.056***	-0.057***	-0.059***			
	(-2.331)				(-3.661)	(-3.722)	(-3.750)			
BTC	-0.006**							-0.007***	-0.007***	-0.007***
	(-2.575)							(-3.215)	(-3.138)	(-3.098)
Spread	0.004	0.005			0.004			0.005		
	(1.206)	(1.450)			(1.267)			(1.521)		
Tail	0.000		0.001			0.000			0.000	
	(0.196)		(0.718)			(0.159)			(0.617)	
Volatility	0.004			0.002			0.006			0.001
	(0.409)			(0.296)			(0.685)			(0.131)
Maturity	0.021*	0.014	0.028***	0.026***	0.026***	0.037***	0.035***	0.021**	0.035***	0.033***
	(1.811)	(1.385)	(3.449)	(3.304)	(2.679)	(5.420)	(5.419)	(2.161)	(5.096)	(5.052)
Constant	430.000	430.000	430.000	430.000	430.000	430.000	430.000	430.000	430.000	430.000
adj. R-sq	0.088	0.077	0.072	0.072	0.087	0.083	0.084	0.076	0.070	0.070

Appendix 36: *Regressions of Determinants of Ex-ante Efficiency Gain*

This table presents the results of the OLS regression model for auctions' characteristics on efficiency gain. Variable definitions are provided in Table 5.2. ***, **, * denote significance at the 1, 5, and 10 percent level, respectively. T-statistics in parentheses.

	Conventional Treasury									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Bidding	0.021***	-0.014***	-0.014**	-0.013**						
	(2.821)	(-2.661)	(-2.541)	(-2.408)						
Bidders	-0.084***				-0.060***	-0.056***	-0.054***			
	(-4.652)				(-4.660)	(-4.368)	(-4.320)			
BTC	-0.007***							-0.006***	-0.006***	-0.007***
	(-6.102)							(-5.801)	(-5.826)	(-5.851)
Spread	0.003**	0.002			0.003**			0.001		
	(2.304)	(1.316)			(2.390)			(1.021)		
Tail	0.000		0.000			0.000			0.000	
	(0.270)		(0.532)			(0.667)			(0.526)	
Volatility	-0.003			-0.006			-0.004			-0.007**
	(-0.961)			(-1.619)			(-1.028)			(-2.118)
Maturity	-0.023***	-0.043***	-0.040***	-0.041***	-0.028***	-0.023***	-0.025***	-0.046***	-0.043***	-0.044***
	(-3.339)	(-11.409)	(-10.359)	(-12.143)	(-4.771)	(-3.824)	(-4.426)	(-12.135)	(-11.563)	(-13.415)
Constant	929.000	929.000	929.000	929.000	929.000	929.000	929.000	929.000	929.000	929.000
adj. R-sq	0.207	0.165	0.165	0.166	0.191	0.188	0.189	0.172	0.172	0.174